

3

General Vector Spaces

SECTION 3.1 Introduction to General Vector Spaces

By the end of this section you will be able to

- understand what is meant by a vector space
- give examples of vector spaces
- prove properties of vector spaces

In the last chapter we used vectors to represent a quantity with ‘magnitude and direction’. In fact, this ‘magnitude–direction’ definition relates to a specific application of vectors, which is used in the fields of science and engineering. However, vectors have a much broader range of applications, and they are more generally defined as ‘elements of a vector space’. In this chapter we describe what is meant by a vector space and how it is mathematically defined.

We discuss the vector spaces that accommodate matrices, functions and polynomials. You may find this chapter more abstract than the previous chapters, because for Euclidean spaces $\mathbb{R}^2$ and $\mathbb{R}^3$ we were able to visualize and even plot the vectors. It is clearly impossible to draw a vector in four dimensions, but that doesn’t make the mathematics carried out in that vector space any less valid.

First, we define what is meant by a general vector space in terms of a set of axioms.



*What does the term **axiom** mean?*

An **axiom** is a statement which is self-evidently true and doesn’t have a proof.

We define vector space with 10 axioms, based on the two fundamental operations of linear algebra – vector addition and scalar multiplication.

3.1.1 Vector space

Let V be a non-empty set of elements called vectors. We define two operations on the set V – **vector addition** and **scalar multiplication**. Scalars are real numbers.

Let $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ be vectors in the set V . The set V is called a **vector space** if it satisfies the following 10 axioms.

1. The vector addition $\mathbf{u} + \mathbf{v}$ is also in the vector space V . Generally in mathematics we say that we have **closure** under vector addition if this property holds.
2. Commutative law: $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$.

3. Associative law: $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$.
4. Neutral element. There is a vector called the **zero** vector in V denoted by $\mathbf{O}$ which satisfies

$$\mathbf{u} + \mathbf{O} = \mathbf{u} \text{ for every vector } \mathbf{u} \text{ in } V$$

5. Additive inverse. For every vector $\mathbf{u}$ there is a vector $-\mathbf{u}$ (minus $\mathbf{u}$) which satisfies the following:

$$\mathbf{u} + (-\mathbf{u}) = \mathbf{O}$$

6. Let k be a real scalar then $k\mathbf{u}$ is also in V . We say that we have **closure** under scalar multiplication if this axiom is satisfied.
7. Associative law for scalar multiplication. Let k and c be real scalars then

$$k(c\mathbf{u}) = (kc)\mathbf{u}$$

8. Distributive law for vectors. Let k be a real scalar then

$$k(\mathbf{u} + \mathbf{v}) = k\mathbf{u} + k\mathbf{v}$$

9. Distributive law for scalars. Let k and c be real scalars then

$$(k + c)\mathbf{u} = k\mathbf{u} + c\mathbf{u}$$

10. Identity element. For every vector $\mathbf{u}$ in V we have

$$1\mathbf{u} = \mathbf{u}$$

We say that if the elements of the set V satisfy the above 10 axioms then V is called a vector space and the elements are known as vectors. This might seem like a long list to digest, so don't worry if it seems a little intimidating at this point. We will use these axioms frequently in the next few sections, and you will soon become familiar with them.

3.1.2 Examples of vector spaces



Can you think of any examples of vector spaces?

The Euclidean spaces of the last chapter - $V = \mathbb{R}^2, \mathbb{R}^3, \dots, \mathbb{R}^n$ - are *all* examples of vector spaces.



Are there any other examples of a vector space?

The set of matrices M_{22} that are *all* matrices of size 2 by 2 where matrix addition and scalar multiplication is defined as in chapter 1 form their own vector space.

Let $\mathbf{u} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $\mathbf{v} = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$ and $\mathbf{w} = \begin{pmatrix} i & j \\ k & l \end{pmatrix}$ be matrices in M_{22} .



What is the zero vector in M_{22} ?

The zero vector is the zero matrix of size 2 by 2 which is given by $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = \mathbf{O}$.

The rules of matrix algebra established in chapter 1 ensure that all 10 axioms are satisfied, defining M_{22} as an example of a vector space. You are asked to check this in Exercises 3.1.

We can show that the set M_{23} , which is the set of matrices of size 2 by 3, also forms a vector space. You are asked to do this in Exercises 3.1.

There also exists vector space which is neither Euclidean space nor formed by a set of matrices. For example, the set of polynomials denoted $P(t)$ whose elements take the form:

$$\mathbf{p}(t) = c_0 + c_1t + c_2t^2 + \cdots + c_nt^n$$

where $c_0, c_1, c_2, \dots$ and c_n are real numbers called the coefficients, form a vector space. The following are examples of polynomials

$$\mathbf{p}(t) = t^2 - 1, \mathbf{q}(t) = 1 + 2t + 7t^2 + 12t^3 - 3t^4 \text{ and } \mathbf{r}(t) = 7$$

The **degree** of a polynomial is the highest index (power) which has a non-zero coefficient, that is the maximum n for which $c_n \neq 0$.



What is the degree of $\mathbf{p}(t)$, $\mathbf{q}(t)$ and $\mathbf{r}(t)$?

$\mathbf{p}(t) = t^2 - 1$ is of degree 2, $\mathbf{q}(t) = 1 + 2t + 7t^2 + 12t^3 - 3t^4$ is of degree 4 and $\mathbf{r}(t) = 7$ is of degree 0. Note that the last polynomial $\mathbf{r}(t) = 7 = 7t^0$ is of degree 0 because 0 is the highest index with a non-zero coefficient (c_0).



How do we define the **zero polynomial?**

The zero polynomial, denoted by $\mathbf{O}$, is given by

$$\mathbf{O} = 0 + 0t + 0t^2 + \cdots + 0t^n$$

All the coefficient c 's are zero. We say that the zero polynomial has *no degree*.

Let $P(t)$ be the set of polynomials of degree less than or equal to n where n is a positive or zero integer. The zero polynomial $\mathbf{O}$ is also in the set $P(t)$.

We define the two operations of vector addition and scalar multiplication as:

1. Vector addition $\mathbf{p}(t) + \mathbf{q}(t)$ is the normal addition of polynomials.
2. Scalar multiplication $k\mathbf{p}(t)$ is the normal multiplication of a constant, k , with the polynomial, $\mathbf{p}(t)$.

In the next example we show that $P(t)$ is also a vector space.

Example 3.1

Show that $P(t)$, the set of polynomials of degree n or less, is indeed a vector space.

Solution

We need to check all 10 axioms with the elements or vectors (polynomials) of this space which have the general polynomial form:

$$\mathbf{p}(t) = c_0 + c_1t + c_2t^2 + \cdots + c_nt^n, \mathbf{q}(t) = d_0 + d_1t + d_2t^2 + \cdots + d_nt^n \text{ and} \\ \mathbf{r}(t) = e_0 + e_1t + e_2t^2 + \cdots + e_nt^n$$

(continued...)

1. Adding two vectors (two polynomials):

$$\begin{aligned}\mathbf{p}(t) + \mathbf{q}(t) &= \underbrace{c_0 + c_1t + c_2t^2 + \cdots + c_nt^n}_{=\mathbf{p}(t)} + \underbrace{d_0 + d_1t + d_2t^2 + \cdots + d_nt^n}_{=\mathbf{q}(t)} \\ &= (c_0 + d_0) + (c_1 + d_1)t + (c_2 + d_2)t^2 + \cdots + (c_n + d_n)t^n\end{aligned}$$

Hence $\mathbf{p}(t) + \mathbf{q}(t)$ is in $P(t)$ which means that we have *closure* under addition.

2. Commutative law:

We are required to show that $\mathbf{p}(t) + \mathbf{q}(t) = \mathbf{q}(t) + \mathbf{p}(t)$:

$$\begin{aligned}\mathbf{p}(t) + \mathbf{q}(t) &= c_0 + c_1t + c_2t^2 + \cdots + c_nt^n + d_0 + d_1t + d_2t^2 + \cdots + d_nt^n \\ &= \underbrace{d_0 + d_1t + d_2t^2 + \cdots + d_nt^n}_{=\mathbf{q}(t)} + \underbrace{c_0 + c_1t + c_2t^2 + \cdots + c_nt^n}_{=\mathbf{p}(t)} \\ &= \mathbf{q}(t) + \mathbf{p}(t)\end{aligned}$$

3. Associative law:

We need to show that $[\mathbf{p}(t) + \mathbf{q}(t)] + \mathbf{r}(t) = \mathbf{p}(t) + [\mathbf{q}(t) + \mathbf{r}(t)]$. Substituting the above polynomials into this yields:

$$\begin{aligned}[\mathbf{p}(t) + \mathbf{q}(t)] + \mathbf{r}(t) &= \left[\underbrace{c_0 + c_1t + c_2t^2 + \cdots + c_nt^n}_{=\mathbf{p}(t)} + \underbrace{d_0 + d_1t + d_2t^2 + \cdots + d_nt^n}_{=\mathbf{q}(t)} \right] \\ &\quad + \underbrace{e_0 + e_1t + e_2t^2 + \cdots + e_nt^n}_{=\mathbf{r}(t)} \\ &= c_0 + c_1t + c_2t^2 + \cdots + c_nt^n + d_0 + d_1t + d_2t^2 + \cdots + d_nt^n \\ &\quad + e_0 + e_1t + e_2t^2 + \cdots + e_nt^n \\ &= \underbrace{c_0 + c_1t + c_2t^2 + \cdots + c_nt^n}_{=\mathbf{p}(t)} \\ &\quad + \left(\underbrace{d_0 + d_1t + d_2t^2 + \cdots + d_nt^n}_{=\mathbf{q}(t)} + \underbrace{e_0 + e_1t + e_2t^2 + \cdots + e_nt^n}_{=\mathbf{r}(t)} \right) \\ &= \mathbf{p}(t) + [\mathbf{q}(t) + \mathbf{r}(t)]\end{aligned}$$

4. What is the zero vector in this case?

The zero polynomial $\mathbf{O}$ is the real number 0 and is in the set $P(t)$. Clearly

$$\mathbf{p}(t) + \mathbf{O} = \underbrace{c_0 + c_1t + c_2t^2 + \cdots + c_nt^n}_{=\mathbf{p}(t)} + 0 = \mathbf{p}(t)$$

5. Additive inverse. For the polynomial $\mathbf{p}(t)$ we have the inverse as $-\mathbf{p}(t)$ and

$$\begin{aligned}\mathbf{p}(t) - \mathbf{p}(t) &= c_0 + c_1t + c_2t^2 + \cdots + c_nt^n - (c_0 + c_1t + c_2t^2 + \cdots + c_nt^n) \\ &= c_0 + c_1t + c_2t^2 + \cdots + c_nt^n - c_0 - c_1t - c_2t^2 - \cdots - c_nt^n \\ &= (c_0 - c_0) + (c_1 - c_1)t + (c_2 - c_2)t^2 + \cdots + (c_n - c_n)t^n \\ &= 0 + 0 + 0 + \cdots + 0 = 0 = \mathbf{O}\end{aligned}$$

6. Let k be a real scalar then

$$\begin{aligned}k\mathbf{p}(t) &= k(c_0 + c_1t + c_2t^2 + \cdots + c_nt^n) \\ &= kc_0 + kc_1t + kc_2t^2 + \cdots + kc_nt^n\end{aligned}$$

Hence $k\mathbf{p}(t)$ is also in the space $P(t)$ because it is a polynomial of degree less than or equal to n .

7. Associative law for scalar multiplication. Let k_1 and k_2 be real scalars:

$$\begin{aligned}k_1(k_2\mathbf{p}(t)) &= k_1(k_2[c_0 + c_1t + c_2t^2 + \cdots + c_nt^n]) \\ &= k_1(k_2c_0 + k_2c_1t + k_2c_2t^2 + \cdots + k_2c_nt^n) \\ &= k_1k_2c_0 + k_1k_2c_1t + k_1k_2c_2t^2 + \cdots + k_1k_2c_nt^n \\ &= k_1k_2 \underbrace{(c_0 + c_1t + c_2t^2 + \cdots + c_nt^n)}_{=\mathbf{p}(t)} \quad [\text{factorizing}] \\ &= (k_1k_2)\mathbf{p}(t)\end{aligned}$$

8. Distributive law. Let k be a real scalar, then we have:

$$\begin{aligned}k[\mathbf{p}(t) + \mathbf{q}(t)] &= k \left[\underbrace{c_0 + c_1t + c_2t^2 + \cdots + c_nt^n}_{=\mathbf{p}(t)} + \underbrace{d_0 + d_1t + d_2t^2 + \cdots + d_nt^n}_{=\mathbf{q}(t)} \right] \\ &= kc_0 + kc_1t + kc_2t^2 + \cdots + kc_nt^n + kd_0 + kd_1t + kd_2t^2 + \cdots + kd_nt^n \\ &\equiv \underbrace{k(c_0 + c_1t + c_2t^2 + \cdots + c_nt^n)}_{=\mathbf{p}(t)} + \underbrace{k(d_0 + d_1t + d_2t^2 + \cdots + d_nt^n)}_{=\mathbf{q}(t)} \quad [\text{Factorising}] \\ &= k\mathbf{p}(t) + k\mathbf{q}(t)\end{aligned}$$

9. Distributive law for scalars. Let k_1 and k_2 be real scalars:

$$\begin{aligned}(k_1 + k_2)\mathbf{p}(t) &= (k_1 + k_2)(c_0 + c_1t + c_2t^2 + \cdots + c_nt^n) \\ &= k_1(c_0 + c_1t + \cdots + c_nt^n) + k_2(c_0 + c_1t + \cdots + c_nt^n) \\ &= k_1\mathbf{p}(t) + k_2\mathbf{p}(t)\end{aligned}$$

10. The identity element is the real number 1. We have

$$\begin{aligned}1\mathbf{p}(t) &= 1(c_0 + c_1t + c_2t^2 + \cdots + c_nt^n) \\ &= (1 \times c_0) + (1 \times c_1t) + (1 \times c_2t^2) + \cdots + (1 \times c_nt^n) \\ &= c_0 + c_1t + c_2t^2 + \cdots + c_nt^n = \mathbf{p}(t)\end{aligned}$$

All 10 axioms are satisfied, therefore the set $P(t)$ of polynomials of degree n or less is a vector space.

Example 3.2

Let $P(t)$ be the set of polynomials of degree equal to n where n is a positive integer. Show that this set $P(t)$ is not a vector space.

Remark Note the difference between Example 3.1 and this example. Here the polynomial is of degree n exactly. In Example 3.1 the degree of the polynomial was less than or equal to n .

Solution

If we check the first axiom, that is if we add *any* two vectors in the set $P(t)$, then their addition should also be in the set. Let

$$\mathbf{p}(t) = 4 + 5t^n \text{ and } \mathbf{q}(t) = 3 - 5t^n$$

We have $\mathbf{p}(t) + \mathbf{q}(t) = 4 + 5t^n + 3 - 5t^n = 7$. This result is not inside the set $P(t)$.

Why not?

Because $P(t)$ is the set of polynomials of degree n , but $\mathbf{p}(t) + \mathbf{q}(t) = 7$ is of degree 0, which means that it cannot be a member of this set $P(t)$.

Hence $P(t)$ is not a vector space because axiom 1 fails, that is we do not have closure under vector addition.

Example 3.3

Let V be the set of integers $\mathbb{Z}$. Let vector addition be defined as the normal addition of integers, and scalar multiplication by the usual multiplication of integers by a real scalar, which is any real number.

Show that this set is not a vector space with respect to this definition of vector addition and scalar multiplication.

Solution

The set of integers $\mathbb{Z}$ is not a vector space because when we multiply an integer by a real scalar, which is any real number, then the result may not be an integer.

For example, if we consider the integer 2 and multiply this by the scalar $1/3$ then the result is $\frac{1}{3}(2) = \frac{2}{3}$ which is not an integer. Our result after scalar multiplication is not in the set of integers $\mathbb{Z}$. Hence the set of integers fails axiom 6.

We do not have closure under scalar multiplication, therefore the set of integers $\mathbb{Z}$ is not a vector space with respect to vector addition and scalar multiplication as defined above.

In general if a set forms a vector space with the scalars being real numbers then we say we have a vector space over the real numbers.

Another example of vector space is that which contains real-valued functions. You should be familiar with functions.

Examples of functions are $f(x) = x$, $f(x) = x^2 + 1$, $f(x) = \sin(x)$ and $f(x) = e^x$.

Let a and b be real numbers where $a \leq b$, then the notation $[a, b]$ is called the interval and is graphically displayed as shown in Fig. 3.1.

We use square brackets to indicate that $[a, b]$ is inclusive of the end points a and b . ($]a, b[$ contains every value between the end points, but does not contain the end points a and b .)



Figure 3.1

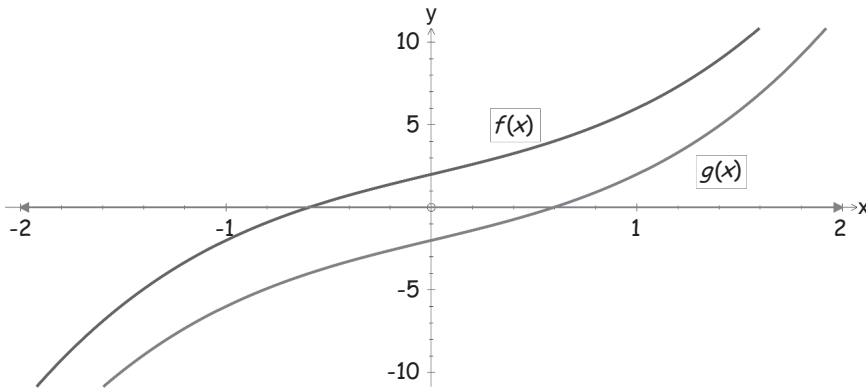


Figure 3.2

The functions in Fig. 3.2 are defined for the interval $[-2, 2]$.

Let $F[a, b]$ be the set of all functions defined in the interval $[a, b]$. Let $\mathbf{f}$ and $\mathbf{g}$ be functions in $F[a, b]$. Normally $\mathbf{f}$ and $\mathbf{g}$ (being vectors) would be in bold but since we are used to seeing functions without bold symbols we will sometimes write these as f and g respectively. We define:

(i) **Vector addition.** The sum $f + g$ is also in $F[a, b]$ and

$$(f + g)(x) = f(x) + g(x)$$

(ii) **Scalar multiplication.** Let k be a real scalar then for *all* x in $[a, b]$

$$(kf)(x) = kf(x)$$

The zero vector in $F[a, b]$ is the zero function $0(x)$ which is equal to zero for *all* x in the interval $[a, b]$, that is

$$0(x) = 0 \quad \text{for all } x \in [a, b]$$

For any function f there is a function $-f$ (minus f) for all x in the interval $[a, b]$ which satisfies

$$(-f)(x) = -f(x)$$

You are asked to prove that $F[a, b]$ is a vector space with respect to these definitions in Exercises 3.1.

Next we go on to prove some basic properties of general vector spaces, using the 10 axioms given in subsection 3.1.1.

3.1.3 Basic properties of general vector spaces

Proposition (3.1). Let V be a vector space and k be a real scalar. Then we have:

- (a) For any real scalar k we have $k\mathbf{O} = \mathbf{O}$ where $\mathbf{O}$ is the zero vector.
- (b) For the real number 0 and any vector $\mathbf{u}$ in V we have $0\mathbf{u} = \mathbf{O}$.
- (c) For the real number -1 and any vector $\mathbf{u}$ in V we have $(-1)\mathbf{u} = -\mathbf{u}$.
- (d) If $k\mathbf{u} = \mathbf{O}$, where k is a real scalar and $\mathbf{u}$ in V , then $k = 0$ or $\mathbf{u} = \mathbf{O}$.

We use the above 10 axioms given in subsection 3.1.1, to prove these properties.

(a) Proof.

Need to prove $k\mathbf{O} = \mathbf{O}$. By applying the above axiom 4, $\mathbf{u} + \mathbf{O} = \mathbf{u}$, we have

$$\begin{aligned} k\mathbf{O} &= k(\mathbf{O} + \mathbf{O}) \\ &= k\mathbf{O} + k\mathbf{O} \quad [\text{applying axiom 8 which is } k(\mathbf{u} + \mathbf{v}) = k\mathbf{u} + k\mathbf{v}] \end{aligned}$$

Adding $-k\mathbf{O}$ to both sides of this, $k\mathbf{O} = k\mathbf{O} + k\mathbf{O}$, gives

$$\begin{aligned} \underbrace{k\mathbf{O} + (-k\mathbf{O})}_{=\mathbf{O}} &= k\mathbf{O} + \underbrace{k\mathbf{O} + (-k\mathbf{O})}_{=\mathbf{O}} \quad [\text{By axiom 5 } \mathbf{u} + (-\mathbf{u}) = \mathbf{O}] \\ \mathbf{O} &= k\mathbf{O} \end{aligned}$$

Hence we have our required result, $k\mathbf{O} = \mathbf{O}$. ■

(b) Proof.

Required to prove $0\mathbf{u} = \mathbf{O}$. For real numbers $0 = 0 + 0$ we have

$$\begin{aligned} 0\mathbf{u} &= (0 + 0)\mathbf{u} \\ &= 0\mathbf{u} + 0\mathbf{u} \quad [\text{by axiom 9 } (k + c)\mathbf{u} = k\mathbf{u} + c\mathbf{u}] \end{aligned}$$

Adding $-0\mathbf{u}$ to both sides of this, $0\mathbf{u} = 0\mathbf{u} + 0\mathbf{u}$, gives

$$\begin{aligned} \underbrace{0\mathbf{u} + (-0\mathbf{u})}_{=\mathbf{O}} &= 0\mathbf{u} + \underbrace{0\mathbf{u} + (-0\mathbf{u})}_{=\mathbf{O}} \\ \mathbf{O} &= 0\mathbf{u} \end{aligned}$$

We have result (b), that is $0\mathbf{u} = \mathbf{O}$. ■

(c) Proof.

Required to prove $(-1)\mathbf{u} = -\mathbf{u}$. Adding $\mathbf{u}$ to $(-1)\mathbf{u}$ gives

$$\begin{aligned} \mathbf{u} + (-1)\mathbf{u} &= 1\mathbf{u} + (-1)\mathbf{u} \\ &= (1 - 1)\mathbf{u} = 0\mathbf{u} \quad \equiv \quad \mathbf{O} \\ &\quad \text{by part (b)} \end{aligned}$$

By axiom 5 we have $\mathbf{u} + (-\mathbf{u}) = \mathbf{O}$. Equating this with the above, $\mathbf{u} + (-1)\mathbf{u} = \mathbf{O}$, we have

$$\mathbf{u} + (-1)\mathbf{u} = \mathbf{u} + (-\mathbf{u})$$

Adding $-\mathbf{u}$ to both sides of this gives:

$$\begin{aligned} \underbrace{-\mathbf{u} + \mathbf{u}}_{=\mathbf{O}} + (-1)\mathbf{u} &= \underbrace{-\mathbf{u} + \mathbf{u}}_{=\mathbf{O}} + (-\mathbf{u}) \\ \mathbf{O} + (-1)\mathbf{u} &= \mathbf{O} + (-\mathbf{u}) \\ (-1)\mathbf{u} &= -\mathbf{u} \end{aligned}$$

Hence we have our result, $(-1)\mathbf{u} = -\mathbf{u}$. ■

(d) *Proof.*

Need to prove that $k\mathbf{u} = \mathbf{O}$ implies $k = 0$ or $\mathbf{u} = \mathbf{O}$.

If $k \neq 0$ (non-zero) then

$$\begin{aligned} k\mathbf{u} &= \mathbf{O} \\ \frac{1}{k}(k\mathbf{u}) &= \underbrace{\frac{1}{k}\mathbf{O}}_{=\mathbf{O}} \quad \left[\text{multiplying both sides by } \frac{1}{k} \right] \\ \left(\frac{1}{k} \right) \mathbf{u} &= \mathbf{O} \text{ gives } 1\mathbf{u} = \mathbf{O} \end{aligned}$$

Since $1\mathbf{u} = \mathbf{u}$, therefore $\mathbf{u} = \mathbf{O}$ which is our required result.

If $k = 0$ then clearly the result holds because $0\mathbf{u} = \mathbf{O}$ by part (b). ■

Proposition (3.2). Let V be a vector space. The zero vector $\mathbf{O}$ (neutral element) of V is unique.



What does this proposition mean?

There is only one zero vector in the vector space V . Remember, by axiom 4 there is a vector called the zero vector denoted by $\mathbf{O}$ which satisfies

$$\mathbf{u} + \mathbf{O} = \mathbf{u} \text{ for every vector } \mathbf{u}$$

Proof.

Suppose there is another vector, say $\mathbf{w}$, which also satisfies

$$\mathbf{u} + \mathbf{w} = \mathbf{u}$$



What do we need to prove?

Required to prove that $\mathbf{w} = \mathbf{O}$. Equating these two, $\mathbf{u} + \mathbf{O} = \mathbf{u}$ and $\mathbf{u} + \mathbf{w} = \mathbf{u}$, gives

$$\mathbf{u} + \mathbf{O} = \mathbf{u} + \mathbf{w}$$

Adding $-\mathbf{u}$ to both sides of this yields:

$$\begin{aligned} -\mathbf{u} + (\mathbf{u} + \mathbf{O}) &= -\mathbf{u} + (\mathbf{u} + \mathbf{w}) \\ \underbrace{(-\mathbf{u} + \mathbf{u})}_{=\mathbf{O}} + \mathbf{O} &= \underbrace{(-\mathbf{u} + \mathbf{u})}_{=\mathbf{O}} + \mathbf{w} \\ \mathbf{O} + \mathbf{O} &= \mathbf{O} + \mathbf{w} \text{ gives } \mathbf{O} = \mathbf{w} \end{aligned}$$

Hence $\mathbf{w} = \mathbf{O}$, which means that the zero vector is *unique*. ■

Proposition (3.3). Let V be a vector space and k be a real scalar. The vector $-\mathbf{u}$ which satisfies axiom 5 is unique:

$$\mathbf{u} + (-\mathbf{u}) = \mathbf{O}$$

Proof.

Suppose the vector $\mathbf{w}$ also satisfies this;

$$\mathbf{u} + \mathbf{w} = \mathbf{O}$$



What do we need to prove?

Required to prove that $\mathbf{w} = -\mathbf{u}$. Equating these two, $\mathbf{u} + (-\mathbf{u}) = \mathbf{O}$ and $\mathbf{u} + \mathbf{w} = \mathbf{O}$, gives

$$\mathbf{u} + (-\mathbf{u}) = \mathbf{u} + \mathbf{w}$$

Adding $-\mathbf{u}$ to both sides yields:

$$\begin{aligned} -\mathbf{u} + (\mathbf{u} + (-\mathbf{u})) &= -\mathbf{u} + (\mathbf{u} + \mathbf{w}) \\ \underbrace{(-\mathbf{u} + \mathbf{u})}_{=\mathbf{O}} + (-\mathbf{u}) &= \underbrace{(-\mathbf{u} + \mathbf{u})}_{=\mathbf{O}} + \mathbf{w} \\ \mathbf{O} + (-\mathbf{u}) &= \mathbf{O} + \mathbf{w} \Rightarrow -\mathbf{u} = \mathbf{w} \end{aligned}$$

Hence the vector $-\mathbf{u}$ is unique. ■



Summary

A vector space is formed by a non-empty set within which the operation of vector addition and scalar multiplication satisfies the above 10 axioms.

Examples of vector spaces include the sets of matrices, polynomials and functions.

Some properties of vector spaces are:

- (a) $k\mathbf{O} = \mathbf{O}$
- (b) $0\mathbf{u} = \mathbf{O}$
- (c) $(-1)\mathbf{u} = -\mathbf{u}$
- (d) If $k\mathbf{u} = \mathbf{O}$ then $k = 0$ or $\mathbf{u} = \mathbf{O}$

The zero vector $\mathbf{O}$ and the vector $-\mathbf{u}$ have the following properties

$$\mathbf{u} + (-\mathbf{u}) = \mathbf{O} \text{ and } \mathbf{u} + \mathbf{O} = \mathbf{u}$$

and are unique.



EXERCISES 3.1

(Brief solutions at end of book. Full solutions available at <http://www.oup.co.uk/companion/singh>.)

Throughout this exercise assume the usual operations of vector addition and scalar multiplication as defined in early chapters or in section 3.1.

1. Show that $\mathbb{R}^2$ is a vector space.
2. Show that $\mathbb{R}^3$ is a vector space.
3. Show that the set M_{22} of 2 by 2 matrices is a vector space.
4. Show that the set M_{23} of 2 by 3 matrices is a vector space.
5. Show that the set $F[a, b]$ which is the set of all functions defined on the interval $[a, b]$ is a vector space.
6. Show that the following set of matrices does not form a vector space:

$$\begin{pmatrix} 1 & 0 \\ 0 & a \end{pmatrix}$$

7. Show that matrices $\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$ form a vector space.
8. Show that the set of 2 by 2 non-invertible matrices do not form a vector space.
9. Let P_2 be the set of polynomials of degree 2 or less and

$$\mathbf{p}(x) = ax^2 + bx + c \text{ and } \mathbf{q}(x) = dx^2 + ex + f$$

be members of P_2 .

Show that the set P_2 is a vector space with respect to the usual vector addition and scalar multiplication.

10. Consider the set $\mathbb{R}^2$ with the normal vector addition, but scalar multiplication defined by

$$k \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ kb \end{pmatrix}$$

Show that the set $\mathbb{R}^2$ with this scalar multiplication definition is not a vector space.

11. Let V be the set $\begin{pmatrix} a \\ b \end{pmatrix}$ in $\mathbb{R}^2$ where $a \geq 0$ and $b \geq 0$. Show that with respect to normal vector addition and scalar multiplication, the given set is not a vector space.
12. Let V be the set of rationals (fractions) $\mathbb{Q}$. Let vector addition be defined as the normal addition of rationals and scalar multiplication by the usual multiplication of rationals by a scalar, which is any real number.
Show that this set is not a vector space with respect to this definition of vector addition and scalar multiplication.
13. Let V be the set of real numbers $\mathbb{R}$. Show that this set is a vector space with respect to the usual definition of vector addition and scalar multiplication.
14. Let V be the set of positive real numbers $\mathbb{R}^+$. Show that this set is not a vector space with respect to the usual definition of vector addition and scalar multiplication.
15. Prove that if the vector $\mathbf{u}$ is in the vector space V then the vector $-\mathbf{u}$ is also in V .
16. Let V be a vector space and $\mathbf{u}$ be a vector in V . Prove that

$$(-k)\mathbf{u} = -(k\mathbf{u}) = k(-\mathbf{u})$$

17. Let V be a vector space and $\mathbf{u}$ and $\mathbf{w}$ be vectors in V . Let $k \neq 0$ be a scalar. Prove that if $k\mathbf{u} = k\mathbf{w}$ then $\mathbf{u} = \mathbf{w}$.
18. Let V be a vector space with a non-zero vector $\mathbf{u}$ ($\neq \mathbf{0}$) and scalars k_1 and k_2 . Prove that if $k_1\mathbf{u} = k_2\mathbf{u}$ then $k_1 = k_2$.
19. Let $\mathbf{u}$ and $\mathbf{w}$ be vectors in a vector space V . Prove that

$$-(\mathbf{u} + \mathbf{w}) = -\mathbf{u} - \mathbf{w}$$

20. Let $\mathbf{u}$ be a vector in a vector space V . Prove that

$$\underbrace{\mathbf{u} + \mathbf{u} + \mathbf{u} + \cdots + \mathbf{u}}_{n \text{ copies}} = n\mathbf{u}$$

SECTION 3.2 Subspace of a Vector Space

By the end of this section you will be able to

- prove some properties of subspaces
- understand what is meant by a linear combination and spanning set

In section 3.1 we discussed the whole vector space V . In this section we show that parts of the whole vector space also form a vector space in their own right. We will show that a non-empty set *within* V , which is *closed* under the basic operations of vector addition and scalar multiplication, is also a legitimate vector space.

3.2.1 Examples of vector subspaces

Let V be a vector space and S be a non-empty subset of V . If the set S satisfies all 10 axioms of a vector space with respect to the same vector addition and scalar multiplication as V then S is also a vector space. We say S is a subspace of V .

Definition (3.4). A non-empty subset S of a vector space V is called a **subspace** of V if it is also a vector space with respect to the same vector addition and scalar multiplication as V .

We illustrate this in Fig. 3.3.

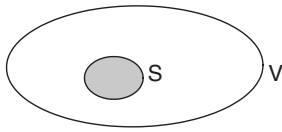


Figure 3.3

Note the difference between subspace and subset. A **subset** is merely a specific set of elements chosen from V . A subset must also satisfy the 10 axioms of vector space to be called a **subspace**.

Example 3.4

Let V be the set $\mathbb{R}^2$ and vector addition and scalar multiplication be defined as normal. Let the set S be the single element $\mathbf{O}$ which in this case is the origin. Show that S is a subspace of V .

Solution

We know from the last section that $\mathbb{R}^2$ is a vector space and also that it contains the origin $\mathbf{O}$. This means that the given set S containing the single element $\mathbf{O}$ is a non-empty subset of V .

Vector addition and scalar multiplication in this set is defined as

$$\mathbf{O} + \mathbf{O} = \mathbf{O} \text{ and } k\mathbf{O} = \mathbf{O}, \text{ where } k \text{ is a scalar.}$$

By checking *all* 10 axioms of the last section, we can show that the set S is also a vector space. You are asked to do this in Exercises 3.2.

Hence by definition (3.4) we conclude that S is a subspace of $V = \mathbb{R}^2$.

We can represent this graphically in Fig. 3.4.

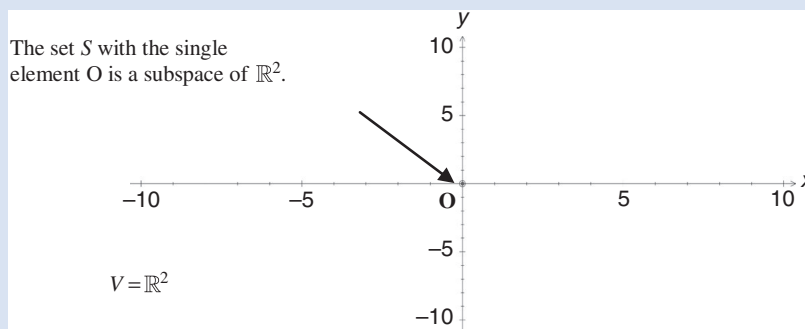


Figure 3.4

A set with a *single* element is called a **singleton set**.

More generally, we will use the following proposition to check if a given subset qualifies as a subspace.

(You will need to look again at the 10 axioms to understand the proof below.)

Proposition (3.5). Let S be a non-empty subset of a vector space V . Then S is subspace of $V \Leftrightarrow$:

- (a) If $\mathbf{u}$ and $\mathbf{v}$ are vectors in the set S then the vector addition $\mathbf{u} + \mathbf{v}$ is also in S .
- (b) If $\mathbf{u}$ is a vector in S then for every scalar k we have, $k\mathbf{u}$ is also in S .

Note that this proposition means that we must have *closure* under both vector addition and scalar multiplication. This means that S is a subspace of $V \Leftrightarrow$ both the following are satisfied, as shown in Fig. 3.5.

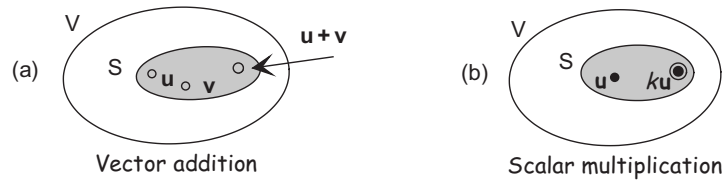


Figure 3.5

? *How do we prove this proposition?*

($\Rightarrow$). We first assume that S is a subspace of the vector space V , and from this we deduce conditions (a) and (b) [show that we have closure under vector addition and scalar multiplication].

($\Leftarrow$). Then we assume conditions (a) and (b) are satisfied, and from this we deduce that the set S is a subspace of the vector space V .

Proof.

Let $\mathbf{u}$ and $\mathbf{v}$ be vectors in the set S .

($\Rightarrow$). Let S be a subspace of the vector space V . By the above definition (3.4) we have closure under vector addition and scalar multiplication because the set S is itself a vector space. [Remember, axioms 1 and 6 state that we have closure under vector addition and scalar multiplication]. Hence conditions (a) and (b) hold.

($\Leftarrow$). Assume conditions (a) and (b) are satisfied, that is we have closure under vector addition and scalar multiplication.

Required to prove that all 10 axioms of the last section are satisfied.

We have closure, therefore axioms 1 and 6 are satisfied. Axioms 2, 3, 7, 8, 9 and 10 are satisfied because these axioms are true for all vectors in the vector space V and vectors $\mathbf{u}$ and $\mathbf{v}$ are vectors in the vector space V .

For the set S we need to prove axioms 4 and 5 which are:

- 4. There is a zero vector $\mathbf{O}$ in V which satisfies

$$\mathbf{u} + \mathbf{O} = \mathbf{u} \quad \text{for every vector } \mathbf{u} \text{ in } V$$

5. For every vector $\mathbf{u}$ there is a vector $-\mathbf{u}$ which satisfies the following:

$$\mathbf{u} + (-\mathbf{u}) = \mathbf{O}$$

We have to show (Fig. 3.6) that the zero vector, $\mathbf{O}$, and $-\mathbf{u}$ are also in S for any $\mathbf{u}$ in S .

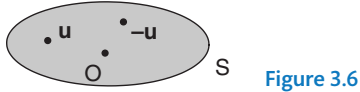


Figure 3.6

We are assuming closure under scalar multiplication (part (b)), which means that if $\mathbf{u}$ is in S then $k\mathbf{u}$ is also in S for any real scalar k . Substituting $k = 0$ gives

$$0\mathbf{u} = \mathbf{O} \text{ is also in } S$$

Substituting $k = -1$ gives

$$(-1)\mathbf{u} = -\mathbf{u} \text{ is also in } S$$

Hence the zero vector, $\mathbf{O}$, and $-\mathbf{u}$ are in the set S .

Now that we have shown all 10 axioms of a vector space are satisfied, we can conclude that S is a vector space, therefore it is a subspace of V .



What does Proposition (3.5) mean?

Proposition (3.5) makes life a lot easier because we only need to check closure under vector addition and scalar multiplication to show that a given set is a subspace. No need to check all 10 axioms.

In everyday English, closure refers to the act of closing or sealing, such as the lid for a container. It has the same meaning here: $\mathbf{u} + \mathbf{v}$ and $k\mathbf{u}$ are closed inside the subspace as shown in Fig. 3.7.

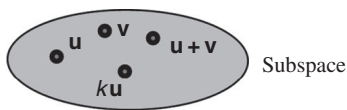


Figure 3.7

If vectors $\mathbf{u}$ and $\mathbf{v}$ are in subspace S then $\mathbf{u} + \mathbf{v}$ and $k\mathbf{u}$ cannot escape from S .

Example 3.5

Let V be the set $\mathbb{R}^2$ and vector addition and scalar multiplication be defined as normal. Let S be the set of vectors of the form $\begin{pmatrix} 0 \\ y \end{pmatrix}$. Show that S is a subspace of V .

(continued...)

Solution

We only need to check conditions (a) and (b) of Proposition (3.5). These are the closure conditions under vector addition and scalar multiplication. We need to show:

(a) If $\mathbf{u}$ and $\mathbf{v}$ are vectors in the set S then the vector addition $\mathbf{u} + \mathbf{v}$ is also in S .

(b) If $\mathbf{u}$ is a vector in S and k is any real scalar then $k\mathbf{u}$ is also in S .

Let $\mathbf{u} = \begin{pmatrix} 0 \\ a \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} 0 \\ b \end{pmatrix}$ be vectors in S . Then

$$\mathbf{u} + \mathbf{v} = \begin{pmatrix} 0 \\ a \end{pmatrix} + \begin{pmatrix} 0 \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ a + b \end{pmatrix} \quad \left[\begin{array}{l} \text{closure under} \\ \text{vector addition} \end{array} \right]$$

which is in the set S and

$$k\mathbf{u} = k \begin{pmatrix} 0 \\ a \end{pmatrix} = \begin{pmatrix} 0 \\ ka \end{pmatrix} \quad \left[\begin{array}{l} \text{closure under} \\ \text{scalar multiplication} \end{array} \right]$$

which is in S as well.

Conditions (a) and (b) are satisfied, therefore the given set S is a subspace of the vector space $\mathbb{R}^2$. Note that the set S is the y axis in the xy plane as shown in Fig. 3.8.

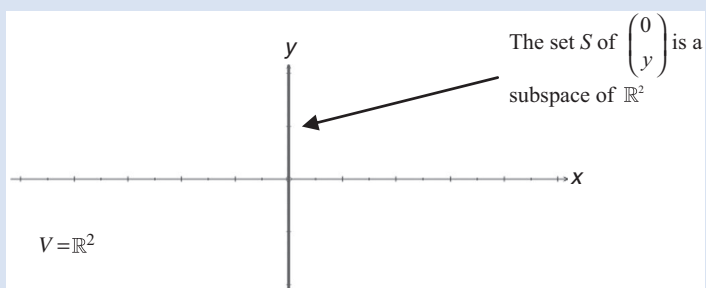


Figure 3.8

Example 3.6

Let S be the subset of vectors of the form $\begin{pmatrix} x \\ y \end{pmatrix}$ where $x \geq 0$ in the vector space $\mathbb{R}^2$. Show that S is not a subspace of $\mathbb{R}^2$.

Solution

How do we show that S is not a subspace of $\mathbb{R}^2$?

If we can show that we do not have closure under vector addition or scalar multiplication of vectors in S , then we can conclude that S is not a subspace. Consider the scalar $k = -1$ and the vector $\mathbf{u} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$; then clearly $\mathbf{u}$ is in the set S , but the scalar multiplication

$$k\mathbf{u} = (-1) \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -1 \\ -2 \end{pmatrix}$$

is not in S .

Why not?

Because the first entry in the vector $\begin{pmatrix} -1 \\ -2 \end{pmatrix}$ is -1 which is less than 0 and the set S only contains vectors of the form $\begin{pmatrix} x \\ y \end{pmatrix}$ where the first entry $x \geq 0$ (greater than or equal to zero).

Hence S is not a subspace of $\mathbb{R}^2$ (Fig. 3.9).

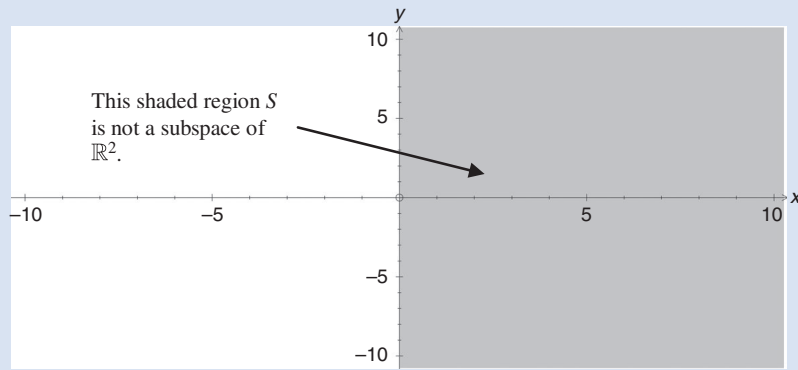


Figure 3.9

Clearly S is a subset but not a subspace of $\mathbb{R}^2$.

Note that by Example 3.5 the vertical axis $x = 0$ is a subspace of $\mathbb{R}^2$ but the whole region to the right of $x \geq 0$ is not.

Example 3.7

Let M_{22} be the set of 2 by 2 matrices. From the last section we know that this set is a vector space. Let S be a subset of M_{22} , containing matrices of the form $\begin{pmatrix} 1 & b \\ c & d \end{pmatrix}$.

Show that S is not a subspace of V .

Solution

We need to show that one of closure conditions (a) or (b) of Proposition (3.5) fails.

Let $\mathbf{u} = \begin{pmatrix} 1 & b \\ c & d \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ be vectors in the subset S . Then

$$\mathbf{u} + \mathbf{v} = \begin{pmatrix} 1 & b \\ c & d \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1+1 & b+0 \\ c+0 & d+0 \end{pmatrix} = \begin{pmatrix} 2 & b \\ c & d \end{pmatrix}$$

Hence $\mathbf{u} + \mathbf{v}$ is not a member of the set S because the elements in S are of the form $\begin{pmatrix} 1 & b \\ c & d \end{pmatrix}$ but $\mathbf{u} + \mathbf{v} = \begin{pmatrix} 2 & b \\ c & d \end{pmatrix}$. The first entry in the matrix $\mathbf{u} + \mathbf{v}$ needs to be 1, not 2, to qualify as an element of the subset S . Therefore S is not a subspace of the vector space M_{22} .

3.2.2 Revision of linear combination

Linear combination combines the two fundamental operations of linear algebra – vector addition and scalar multiplication.

In the last chapter we introduced linear combination in $\mathbb{R}^n$. For example, we had

$$\mathbf{u} = k_1\mathbf{e}_1 + k_2\mathbf{e}_2 + \cdots + k_n\mathbf{e}_n \quad (k\text{'s are scalars})$$

which is a **linear combination** of the standard unit vectors $\mathbf{e}_1, \mathbf{e}_2, \dots$ and $\mathbf{e}_n$.

Similarly for general vector spaces we define linear combination as:

Definition (3.6). Let $\mathbf{v}_1, \mathbf{v}_2, \dots$ and $\mathbf{v}_n$ be vectors in a vector space. If a vector $\mathbf{x}$ can be expressed as

$$\mathbf{x} = k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 + \cdots + k_n\mathbf{v}_n \quad (\text{where } k\text{'s are scalars})$$

then we say $\mathbf{x}$ is a **linear combination** of the vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots$ and $\mathbf{v}_n$.

Example 3.8

Let P_2 be the set of all polynomials of degree less than or equal to 2.

Let $\mathbf{v}_1 = t^2 - 1$, $\mathbf{v}_2 = t^2 + 3t - 5$ and $\mathbf{v}_3 = t$ be vectors in P_2 .

Show that the quadratic polynomial

$$\mathbf{x} = 7t^2 - 15$$

is a linear combination of $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.

Solution

How do we show $\mathbf{x}$ is a linear combination of vectors $\mathbf{v}_1, \mathbf{v}_2$ and $\mathbf{v}_3$?

We need to find the values of the scalars k_1, k_2 and k_3 which satisfy

$$k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 = \mathbf{x} \quad (*)$$

How can we determine these scalars?

By substituting $\mathbf{v}_1 = t^2 - 1$, $\mathbf{v}_2 = t^2 + 3t - 5$, $\mathbf{v}_3 = t$ and $\mathbf{x} = 7t^2 - 15$ into (*):

$$\begin{aligned} k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 &= k_1(t^2 - 1) + k_2(t^2 + 3t - 5) + k_3t \\ &= k_1t^2 - k_1 + k_2t^2 + 3k_2t - 5k_2 + k_3t && \text{[expanding]} \\ &= (k_1 + k_2)t^2 + (3k_2 + k_3)t - (k_1 + 5k_2) && \text{[factorizing]} \\ &= 7t^2 - 15 && \text{[remember } \mathbf{x} = 7t^2 - 15] \end{aligned}$$

By equating coefficients of the last two lines

$$(k_1 + k_2)t^2 + (3k_2 + k_3)t - (k_1 + 5k_2) = 7t^2 - 15$$

gives

$$\begin{aligned} k_1 + k_2 &= 7 && \text{[equating } t^2] \\ 3k_2 + k_3 &= 0 && \text{[equating } t] \\ k_1 + 5k_2 &= 15 && \text{[equating constants]} \end{aligned}$$

Solving these equations gives the values of the scalars: $k_1 = 5$, $k_2 = 2$ and $k_3 = -6$.
Substituting these into $k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 = \mathbf{x}$:

$$5\mathbf{v}_1 + 2\mathbf{v}_2 - 6\mathbf{v}_3 = \mathbf{x}$$

This means that adding 5 lots of $\mathbf{v}_1$, 2 lots of $\mathbf{v}_2$ and -6 lots of $\mathbf{v}_3$ gives the vector $\mathbf{x}$:

$$5(t^2 - 1) + 2(t^2 + 3t - 5) - 6t = 7t^2 - 15$$

You may like to check the algebra.

We conclude that $\mathbf{x}$ is a linear combination of $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.

The next proposition allows us to check that S is a subspace, by carrying out the test for scalar multiplication and vector addition in a single calculation.

Proposition (3.7). A non-empty subset S containing vectors $\mathbf{u}$ and $\mathbf{v}$ is a subspace of a vector space $V \Leftrightarrow$ any linear combination $k\mathbf{u} + c\mathbf{v}$ is also in S (k and c are scalars).



How do we prove this proposition?

Since it is a ' $\Leftrightarrow$ ' statement we need to prove it both ways.

Proof.

($\Rightarrow$). Let S be a subspace of V , then S is a vector space. If $\mathbf{u}$ and $\mathbf{v}$ are in S then $k\mathbf{u} + c\mathbf{v}$ is also in S because the vector space S is closed under scalar multiplication and vector addition.

($\Leftarrow$). Assume $k\mathbf{u} + c\mathbf{v}$ is in S .

Substituting $k = c = 1$ into $k\mathbf{u} + c\mathbf{v}$ we have $\mathbf{u} + \mathbf{v}$ is also in S . Similarly for $c = 0$ we have $k\mathbf{u} + c\mathbf{v} = k\mathbf{u}$ is in S .

Hence we have closure under vector addition and scalar multiplication. By Proposition (3.5): S is subspace of $V \Leftrightarrow S$ is closed under vector addition and scalar multiplication.

We conclude that S is a subspace. ■

Proposition (3.7) is another test for a subspace. Hence a subspace of a vector space V is a non-empty subset S of V , such that for all vectors $\mathbf{u}$ and $\mathbf{v}$ in S and all scalars k and c we have $k\mathbf{u} + c\mathbf{v}$ is also in S .

Example 3.9

Let S be the subset of vectors of the form $(x \ y \ 0)^T$ in the vector space $\mathbb{R}^3$. Show that S is a subspace of $\mathbb{R}^3$.

(continued...)

Solution

How do we show that S is a subspace of $\mathbb{R}^3$?

We can use the above Proposition (3.7), which means we need to show that any linear combination $k\mathbf{u} + c\mathbf{v}$ is in S for any vectors $\mathbf{u}$ and $\mathbf{v}$ in S .

Let $\mathbf{u} = (a \ b \ 0)^T$ and $\mathbf{v} = (c \ d \ 0)^T$ be in S . Then for real scalars k_1 and k_2 we have

$$\begin{aligned} k_1\mathbf{u} + k_2\mathbf{v} &= k_1 \begin{pmatrix} a \\ b \\ 0 \end{pmatrix} + k_2 \begin{pmatrix} c \\ d \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} k_1 a \\ k_1 b \\ 0 \end{pmatrix} + \begin{pmatrix} k_2 c \\ k_2 d \\ 0 \end{pmatrix} = \begin{pmatrix} k_1 a + k_2 c \\ k_1 b + k_2 d \\ 0 \end{pmatrix} \end{aligned}$$

Hence $k_1\mathbf{u} + k_2\mathbf{v}$ is also in S .

By the above Proposition (3.7):

S is subspace of $V \Leftrightarrow$ any linear combination $k\mathbf{u} + c\mathbf{v}$ is also in S .

We conclude that the given set S is a subspace of the vector space $\mathbb{R}^3$.

You might find it easier to use this test (3.7) rather than (3.5) because you only need to recall that the linear combination is closed in S .

Note that the given set S in the above example describes the xy plane in three-dimensional space $\mathbb{R}^3$ as shown in Fig. 3.10:

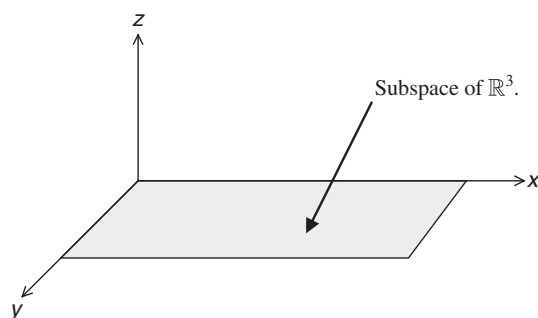


Figure 3.10

Sometimes we write $\begin{pmatrix} x \\ y \\ 0 \end{pmatrix}$ as the set $S = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mid z = 0 \right\}$ where the vertical line in the set means ‘such that’. That is, S contains the set of vectors in $\mathbb{R}^3$ such that the last entry is zero.

3.2.3 Revision of spanning sets

Definition (3.8). If every vector in V can be produced by a linear combination of vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots$ and $\mathbf{v}_n$ then these vectors **span** or **generate** the vector space V . We write this as $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$.

For example, the standard unit vectors $\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ span $\mathbb{R}^2$ because a linear combination of these vectors $\mathbf{e}_1$ and $\mathbf{e}_2$ can produce any vector in $\mathbb{R}^2$.

For example, the vector $\mathbf{u} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ spans the line $y = x$ in $\mathbb{R}^2$ as Fig. 3.11(a) shows.

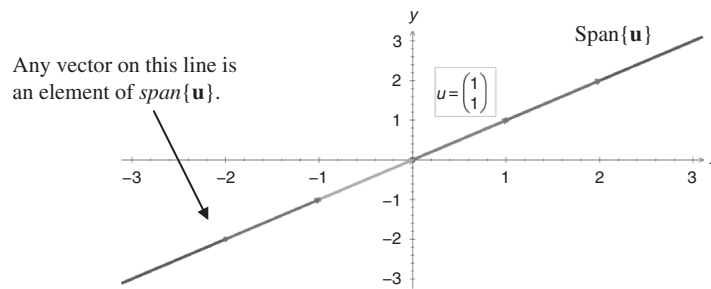


Figure 3.11(a)

Proposition (3.9). Let S be a non-empty subset of a vector space V . The set $\text{span}\{S\}$ is a subspace of the vector space V .

Proof – Exercises 3.2. ■

For example, in the above Fig. 3.11(a) $\text{span}\{\mathbf{u}\}$ is a subspace of $\mathbb{R}^2$. This is illustrated in Fig. 3.11(b).

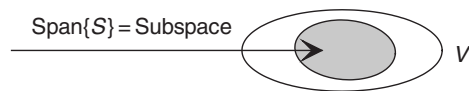


Figure 3.11(b)

Example 3.10

Let P_3 be the vector space containing the set of polynomials of degree 3 or less. Show that the set $\{1, t, t^2, t^3\}$ spans $\{P_3\}$.

Solution

Let $at^3 + bt^2 + ct + d$ be a general polynomial in the space P_3 . Then we need to show that there are scalars k_1, k_2, k_3 and k_4 which satisfy the following:

$$k_1 t^3 + k_2 t^2 + k_3 t + k_4 = at^3 + bt^2 + ct + d$$

Equating coefficients of t^3, t^2, t and constant gives $k_1 = a, k_2 = b, k_3 = c$ and $k_4 = d$ respectively.

We have found scalars, $k_1 = a, k_2 = b, k_3 = c$ and $k_4 = d$, therefore the linear combination of $\{1, t, t^2, t^3\}$ can generate any vector in P_3 which means $\{1, t, t^2, t^3\}$ spans P_3 .

Example 3.11

Let P_2 be the set of polynomials of degree 2 or less. Let the following vectors be in P_2 :

$$\mathbf{v}_1 = t^2 - t - 1, \quad \mathbf{v}_2 = 6t^2 + 3t - 3 \text{ and } \mathbf{v}_3 = t^2 + 5t + 1$$

Determine whether the vector $\mathbf{u} = 3t^2 + 2t + 1$ is spanned by $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.

Solution

How do we check that the vector $\mathbf{u}$ is spanned by $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$?

We need to see if scalars, k_1 , k_2 and k_3 exist which satisfy the following linear combination:

$$k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 = \mathbf{u}$$

Substituting $\mathbf{v}_1 = t^2 - t - 1$, $\mathbf{v}_2 = 6t^2 + 3t - 3$, $\mathbf{v}_3 = t^2 + 5t + 1$ and $\mathbf{u} = 3t^2 + 2t + 1$ into this:

$$\begin{aligned} k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 &= k_1(t^2 - t - 1) + k_2(6t^2 + 3t - 3) + k_3(t^2 + 5t + 1) \\ &= (k_1 + 6k_2 + k_3)t^2 + (-k_1 + 3k_2 + 5k_3)t + (-k_1 - 3k_2 + k_3) \quad [\text{rearranging}] \\ &= 3t^2 + 2t + 1 \end{aligned}$$

Equating the coefficients of the last two lines gives:

$$\begin{aligned} k_1 + 6k_2 + k_3 &= 3 && (\text{equating } t^2) \\ -k_1 + 3k_2 + 5k_3 &= 2 && (\text{equating } t) \\ -k_1 - 3k_2 + k_3 &= 1 && (\text{equating constants}) \end{aligned}$$

What are we trying to find?

We need to find values of the scalars k_1 , k_2 and k_3 which satisfy the above linear system. Writing the above system as an augmented matrix gives:

$$\begin{array}{l} R_1 \\ R_2 \\ R_3 \end{array} \left(\begin{array}{ccc|c} 1 & 6 & 1 & 3 \\ -1 & 3 & 5 & 2 \\ -1 & -3 & 1 & 1 \end{array} \right)$$

Executing row operations, $R_2 + R_1$ and $R_3 + R_1$, gives:

$$\begin{array}{l} R_1 \\ R_2^* = R_2 + R_1 \\ R_3^* = R_3 + R_1 \end{array} \left(\begin{array}{ccc|c} 1 & 6 & 1 & 3 \\ 0 & 9 & 6 & 5 \\ 0 & 3 & 2 & 4 \end{array} \right)$$

Carrying out the row operation $3R_3^* - R_2^*$:

$$\begin{array}{l} R_1 \\ R_2^* \\ 3R_3^* - R_2^* \end{array} \begin{array}{c} k_1 \quad k_2 \quad k_3 \\ \left(\begin{array}{ccc|c} 1 & 6 & 1 & 3 \\ 0 & 9 & 6 & 5 \\ 0 & 0 & 0 & 7 \end{array} \right) \end{array}$$

From the bottom row we have $0k_1 + 0k_2 + 0k_3 = 0 = 7$, which means that the linear system is inconsistent, therefore we have no solution. There are no scalars (k 's) which satisfy the above linear system. Hence the vector $\mathbf{u}$ does not exist in the $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$. This means that we cannot make $\mathbf{u}$ out of a linear combination of $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.

By Proposition (3.9) we have $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is a subspace of P_2 in the above example but $\mathbf{u} = 3t^2 + 2t + 1$ is not in the $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$. We can illustrate this as shown in Fig. 3.12.

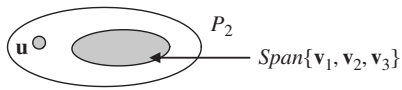


Figure 3.12

Example 3.12

Let $F[-\pi, \pi]$ be the vector space of continuous functions defined in the interval $[-\pi, \pi]$. Let the functions $\mathbf{f} = \cos^2(x)$ and $\mathbf{g} = \sin^2(x)$. Determine whether 2 is in $\text{span}\{\mathbf{f}, \mathbf{g}\}$.

Solution

We need to find scalars k and c such that

$$k\mathbf{f} + c\mathbf{g} = k\cos^2(x) + c\sin^2(x) = 2$$

From trigonometry we have the fundamental identity $\cos^2(x) + \sin^2(x) = 1$. Multiplying this by 2 gives us our result $2\cos^2(x) + 2\sin^2(x) = 2$.

Hence with $k = c = 2$ we have $k\mathbf{f} + c\mathbf{g} = 2$, therefore 2 is in the $\text{span}\{\mathbf{f}, \mathbf{g}\}$.

Example 3.13

Let M_{22} be the vector space of 2 by 2 matrices. Consider the matrices

$$\mathbf{A} = \begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 0 & 2 \\ 0 & -1 \end{pmatrix} \text{ and } \mathbf{C} = \begin{pmatrix} 0 & 1 \\ 5 & 2 \end{pmatrix}$$

Determine whether the matrix $\mathbf{D} = \begin{pmatrix} 1 & 2 \\ -4 & -2 \end{pmatrix}$ is within the $\text{span}\{\mathbf{A}, \mathbf{B}, \mathbf{C}\}$.

Solution

We need to solve the linear combination

$$k_1\mathbf{A} + k_2\mathbf{B} + k_3\mathbf{C} = \mathbf{D}$$

for scalars k_1 , k_2 and k_3 :

$$\begin{aligned} k_1\mathbf{A} + k_2\mathbf{B} + k_3\mathbf{C} &= k_1 \begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix} + k_2 \begin{pmatrix} 0 & 2 \\ 0 & -1 \end{pmatrix} + k_3 \begin{pmatrix} 0 & 1 \\ 5 & 2 \end{pmatrix} \\ &= \begin{pmatrix} k_1 & -k_1 \\ k_1 & 2k_1 \end{pmatrix} + \begin{pmatrix} 0 & 2k_2 \\ 0 & -k_2 \end{pmatrix} + \begin{pmatrix} 0 & k_3 \\ 5k_3 & 2k_3 \end{pmatrix} \\ &= \begin{pmatrix} k_1 & -k_1 + 2k_2 + k_3 \\ k_1 + 5k_3 & 2k_1 - k_2 + 2k_3 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ -4 & -2 \end{pmatrix} \quad [\text{matrix } \mathbf{D}] \end{aligned}$$

Equating entries of the matrix gives

$$k_1 = 1, \quad -k_1 + 2k_2 + k_3 = 2, \quad k_1 + 5k_3 = -4 \quad \text{and} \quad 2k_1 - k_2 + 2k_3 = -2$$

We have $k_1 = 1$, but how do we find k_2 and k_3 ?

Substituting $k_1 = 1$ into the third equation $k_1 + 5k_3 = -4$ yields

$$1 + 5k_3 = -4 \text{ gives } k_3 = -1$$

(continued...)

Substituting $k_1 = 1$ and $k_3 = -1$ into the last equation $2k_1 - k_2 + 2k_3 = -2$:

$$2(1) - k_2 + 2(-1) = -2$$

$$-k_2 = -2 \text{ which gives } k_2 = 2$$

Hence we have found scalars, $k_1 = 1$, $k_2 = 2$ and $k_3 = -1$, which satisfy

$$k_1\mathbf{A} + k_2\mathbf{B} + k_3\mathbf{C} = \mathbf{A} + 2\mathbf{B} - \mathbf{C} = \mathbf{D}$$

Therefore $\mathbf{D}$ belongs to $\text{span}\{\mathbf{A}, \mathbf{B}, \mathbf{C}\}$ because $\mathbf{D}$ can be made by a linear combination of $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$.



Summary

Proposition (3.5). The set S is a subspace of a vector space $\Leftrightarrow$:

- (a) S is closed under vector addition.
- (b) S is closed under scalar multiplication.

(3.6) If a vector $\mathbf{x}$ can be expressed as

$$\mathbf{x} = k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 + \cdots + k_n\mathbf{v}_n \text{ (where } k\text{'s are scalars)}$$

then we say $\mathbf{x}$ is a *linear combination* of the vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots$ and $\mathbf{v}_n$.

(3.8) If *every* vector in V can be produced by a linear combination of vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots$ and $\mathbf{v}_n$ then we say these vectors *span* or *generate* the vector space V .



EXERCISES 3.2

(Brief solutions at end of book. Full solutions available at <http://www.oup.co.uk/companion/singh>.)

Throughout this exercise assume your scalars to be real numbers.

1. Consider Example 3.4. Show that S is a vector space with respect to the usual vector addition and scalar multiplication of $\mathbb{R}^2$.
2. Let S be the set of vectors $\begin{pmatrix} a \\ 0 \end{pmatrix}$ in the vector space $\mathbb{R}^2$. Show that S is a subspace of $\mathbb{R}^2$.
3. Let S be the set of vectors $\begin{pmatrix} a \\ a \end{pmatrix}$ in the vector space $\mathbb{R}^2$. Show that S is a subspace of $\mathbb{R}^2$.
4. Let S be the set of vectors $(0 \ 0 \ c)^T$ in the vector space $\mathbb{R}^3$. Show that S is a subspace of $\mathbb{R}^3$.

5. Let S be the set of vectors $(1 \ b \ c \ d)^T$ in the vector space $\mathbb{R}^4$. Show that S is not a subspace of $\mathbb{R}^4$.
6. Let S be the set of vectors $(a \ b \ c)^T$ where $a + b + c = 0$ in the vector space $\mathbb{R}^3$. Determine whether S is a subspace of $\mathbb{R}^3$.
7. Let $S = \left\{ \begin{pmatrix} a \\ b \end{pmatrix} \mid a \text{ and } b \text{ are integers} \right\}$ be a subset of $\mathbb{R}^2$. Show that S is not a subspace of $\mathbb{R}^2$.
8. Let $S = \left\{ \begin{pmatrix} a \\ b \\ c \end{pmatrix} \mid a, b \text{ and } c \text{ are rational numbers} \right\}$ be a subset of $\mathbb{R}^3$. Show that S is not a subspace of $\mathbb{R}^3$.
9. Let M_{22} be the set of matrices of size 2 by 2. Let S be the subset of matrices of the form $\begin{pmatrix} a & 1 \\ c & d \end{pmatrix}$. Show that S is not a subspace of M_{22} .
10. Let M_{22} be the set of matrices of size 2 by 2. Let

$$S = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mid a, b, c, d \text{ are all integers} \right\}$$

Show that S is not a subspace of M_{22} .

11. Let S be the set of symmetric matrices (these are matrices $\mathbf{A}$ such that $\mathbf{A}^T = \mathbf{A}$, that is $\mathbf{A}$ transposed is equal to $\mathbf{A}$). Let V be the set of all matrices. Show that S is a subspace of V .
12. Let $\mathbf{v}_1 = t^2 - 1$, $\mathbf{v}_2 = t + 1$ and $\mathbf{v}_3 = 2t^2 + t - 1$ be vectors in P_2 where P_2 is the set of polynomials of degree 2 or less. Show that $\mathbf{x} = 7t^2 + 8t + 1$ is a linear combination of $\mathbf{v}_1$, $\mathbf{v}_2$ and $\mathbf{v}_3$.
13. Let $\mathbf{p}_1 = t^2 + 2t - 1$, $\mathbf{p}_2 = 2t + 1$ and $\mathbf{p}_3 = 5t^2 + 2t - 3$ be vectors in P_2 . Show that the following are linear combinations of these vectors $\mathbf{p}_1$, $\mathbf{p}_2$ and $\mathbf{p}_3$:
 (a) $\mathbf{x} = 4t^2 - 2t - 3$ (b) $\mathbf{x} = -2t^2 - 2$ (c) $\mathbf{x} = 6$
14. Let V be the set of real valued continuous functions. Then V is a vector space. Let $\mathbf{v}_1 = \sin^2(t)$ and $\mathbf{v}_2 = \cos^2(t)$. Show that the following are linear combinations of $\mathbf{v}_1$ and $\mathbf{v}_2$.
 (a) $\mathbf{x} = 1$ (b) $\mathbf{x} = \pi$ (c) $\mathbf{x} = \cos(2t)$
15. Let S be a subspace of $\mathbb{R}^3$ given by $\begin{pmatrix} a \\ b \\ 0 \end{pmatrix}$. Let $\mathbf{u} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} -1 \\ 5 \\ 0 \end{pmatrix}$. Show that the vectors $\mathbf{u}$ and $\mathbf{v}$ span S .
16. Let P_2 be the set of polynomials of degree 2 or less. Let

$$\mathbf{p}_1 = t^2 + 3, \mathbf{p}_2 = 2t^2 + 5t + 6 \text{ and } \mathbf{p}_3 = 5t$$

Determine whether the vector $\mathbf{x} = t^2 + 3$ is in the span $\{\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3\}$.

17. Let M_{22} be the set of 2 by 2 matrices. Consider the matrices

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} -1 & 2 \\ 5 & 7 \end{pmatrix} \text{ and } \mathbf{C} = \begin{pmatrix} 2 & 6 \\ 8 & 0 \end{pmatrix}$$

Determine whether the matrix $\mathbf{D} = \begin{pmatrix} 7 & -3 \\ -14 & -26 \end{pmatrix}$ belongs to $\text{span}\{\mathbf{A}, \mathbf{B}, \mathbf{C}\}$.

18. Let $F[0, 2\pi]$ be the vector space of continuous functions. Let the functions $\mathbf{f} = \cos(2x)$ and $\mathbf{g} = \sin(2x)$. Determine whether the following are in $\text{span}\{\mathbf{f}, \mathbf{g}\}$:

- (a) 0 (b) $\sin(2x)$ (c) $\cos^2(x) - \sin^2(x)$ (d) 1

19. Determine whether $x + 1$ and $(x + 1)^2$ span P_2 .

20. Prove Proposition (3.9).

21. Let S be a subspace of a vector space V . Show that if T is a subset of S it may not be a subspace of V .

22. Let S and T be subspaces of a vectors space V . Prove that

- (a) $S \cap T$ (b) $S \cup T$

are subspaces of V .

[$S \cap T$ is the **intersection** of S and T and is the set of elements which lie in both S and T . This means if $\mathbf{u} \in S \cap T$ then $\mathbf{u} \in S$ and $\mathbf{u} \in T$.

$S \cup T$ is the **union** of S and T and is the set all the elements in S or T .]

SECTION 3.3 Linear Independence and Basis

By the end of this section you will be able to

- test vectors for linear independence
- prove properties about linear independence
- determine whether given vectors are a basis for a vector space

We discussed the terms linear independence and basis during chapter 2, using examples in Euclidean space $\mathbb{R}^n$. In this section we expand these definitions to general vector spaces V .

From previous sections we know that the set of matrices, polynomials and functions form a vector space which means that they behave the same way although they are different objects.

A basis is a sort of coordinate system for vector space which we measure a vector against.

3.3.1 Linear independence

We extend the definition of linear independence to a general vector space V .

Definition (3.10). We say vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots$ and $\mathbf{v}_n$ in V are **linearly independent** $\Leftrightarrow$ the *only* real scalars $k_1, k_2, k_3, \dots$ and k_n which satisfy:

$$k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 + \dots + k_n\mathbf{v}_n = \mathbf{O} \quad \text{are} \quad k_1 = k_2 = k_3 = \dots = k_n = 0$$



What does this mean?

The only solution to the linear combination $k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 + \dots + k_n\mathbf{v}_n = \mathbf{O}$ requires all the scalars $k_1, k_2, k_3, \dots$ and k_n to be equal to zero. In other words, you cannot make any one of the vectors $\mathbf{v}_j$ say, by a linear combination of the others.

Example 3.14

Let M_{22} be the vector space of size 2 by 2 matrices. Consider the matrices:

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \text{ and } \mathbf{B} = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}$$

Show that the matrices $\mathbf{A}$ and $\mathbf{B}$ are linearly independent.

Solution

How do we show that matrices $\mathbf{A}$ and $\mathbf{B}$ are linearly independent?

Required to show that $k\mathbf{A} + c\mathbf{B} = \mathbf{O}$ is true only when the scalars $k = c = 0$.

Let k and c be scalars, then by applying the above definition (3.10) to these matrices with the zero matrix $\mathbf{O} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ we have

$$\begin{aligned} k\mathbf{A} + c\mathbf{B} &= k \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + c \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix} \\ &= \begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix} + \begin{pmatrix} 0 & 2c \\ 2c & 0 \end{pmatrix} = \begin{pmatrix} k & 2c \\ 2c & k \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = \mathbf{O} \end{aligned}$$

Equating entries gives $k = 0$ and $2c = 0$. Since both (all) our scalars, $k = 0$ and $c = 0$, are zero, then by definition (3.10) we conclude that the given matrices $\mathbf{A}$ and $\mathbf{B}$ are linearly independent.

Linear independence means that we cannot make any one vector out of a combination of the others.



What does this mean in relation to Example 3.14?

It means that matrix $\mathbf{A}$ is not a multiple of matrix $\mathbf{B}$.

Example 3.15

Let V be the vector space of continuous functions defined on the real line.

Let $\mathbf{u} = \cos(x)$, $\mathbf{v} = \sin(x)$ and $\mathbf{w} = 2$ be vectors in V .

Show that the vectors $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ are linearly independent.

(continued...)

Solution

By the above definition (3.10) we need to consider the linear combination

$$k_1 \mathbf{u} + k_2 \mathbf{v} + k_3 \mathbf{w} = k_1 \cos(x) + k_2 \sin(x) + k_3(2) = 0 \quad (*)$$

where k_1 , k_2 and k_3 are scalars.

What do we need to prove?

We need to show that the given vectors $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ are linearly independent.

How?

Required to show that all the scalars are equal to zero, that is $k_1 = k_2 = k_3 = 0$.

We need three equations, so we substitute three different values of x into (*). We choose values that help to simplify the equations.

Substituting $x = 0$ (to eliminate the sine term) into (*) gives

$$k_1 \underbrace{\cos(0)}_{=1} + k_2 \underbrace{\sin(0)}_{=0} + k_3(2) = k_1 + 2k_3 = 0 \quad (1)$$

Next we substitute $x = \pi$ into (*):

$$k_1 \underbrace{\cos(\pi)}_{=-1} + k_2 \underbrace{\sin(\pi)}_{=0} + k_3(2) = -k_1 + 2k_3 = 0 \quad (2)$$

Substituting $x = \pi/2$ into equation (*):

$$k_1 \underbrace{\cos\left(\frac{\pi}{2}\right)}_{=0} + k_2 \underbrace{\sin\left(\frac{\pi}{2}\right)}_{=1} + k_3(2) = k_2 + 2k_3 = 0 \quad (3)$$

We can write these three equations - (1), (2) and (3) - in an augmented matrix and solve:

$$\begin{array}{l} R_1 \\ R_2 \\ R_3 \end{array} \begin{pmatrix} k_1 & k_2 & k_3 \\ 1 & 0 & 2 \\ -1 & 0 & 2 \\ 0 & 1 & 2 \end{pmatrix} \begin{array}{c} \longrightarrow \\ \end{array} \begin{array}{l} R_1 \\ R_2 + R_1 \\ R_3 \end{array} \begin{pmatrix} k_1 & k_2 & k_3 \\ 1 & 0 & 2 \\ 0 & 0 & 4 \\ 0 & 1 & 2 \end{pmatrix}$$

By inspection we can see that $k_1 = k_2 = k_3 = 0$.

Since the only solution is $k_1 = k_2 = k_3 = 0$, the given vectors $\mathbf{u} = \cos(x)$, $\mathbf{v} = \sin(x)$ and $\mathbf{w} = 2$ are linearly independent.



In the above Example 3.15, how do we know which values of x to substitute?

Since the linear combination

$$k_1 \mathbf{u} + k_2 \mathbf{v} + k_3 \mathbf{w} = k_1 \cos(x) + k_2 \sin(x) + k_3(2) = 0$$

is valid for any real value of x , we can try any real number. Normally we choose x values which simplify our arithmetic.

3.3.2 Linear dependence



What does linear dependence mean?

Definition (3.11). The vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots$ and $\mathbf{v}_n$ in a vector space V are *linearly dependent* $\Leftrightarrow$ the scalars $k_1, k_2, k_3, \dots$ and k_n are *not all zero* and satisfy

$$k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 + \dots + k_n\mathbf{v}_n = \mathbf{O}$$

Example 3.16

Let V be the vector space of continuous functions defined on the real line.

Let $\mathbf{u} = \cos^2(x)$, $\mathbf{v} = \sin^2(x)$ and $\mathbf{w} = 2$ be vectors in V .

Show that the vectors $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ are linearly dependent.

Solution

From our knowledge of trigonometry, we have the fundamental trigonometric identity

$$\cos^2(x) + \sin^2(x) = 1$$

Multiplying each side by 2 gives

$$2\cos^2(x) + 2\sin^2(x) = 2$$

$$2\mathbf{u} + 2\mathbf{v} = \mathbf{w} \quad [\text{because } \mathbf{u} = \cos^2(x), \mathbf{v} = \sin^2(x) \text{ and } \mathbf{w} = 2]$$

$2\mathbf{u} + 2\mathbf{v} = \mathbf{w}$ implies that we can make vector $\mathbf{w}$ out of a linear combination of vectors $\mathbf{u}$ and $\mathbf{v}$, (2 lots of $\mathbf{u}$ plus 2 lots of $\mathbf{v}$), therefore vectors $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ are linearly dependent.

Example 3.17

Let P_2 be the vector space of polynomials of degree 2 or less. Decide whether the following vectors in P_2 are linearly independent or dependent.

(a) $\mathbf{p} = 6t^2 + 8t + 2$ and $\mathbf{q} = 3t^2 + 4t + 1$

(b) $\mathbf{p} = 2t^2 + 3t + 2$ and $\mathbf{q} = t^2 + t + 1$

(c) $\mathbf{p} = t^2 + 3t - 1$, $\mathbf{q} = 2t^2 + 7t + 5$ and $\mathbf{r} = 7$

Solution

(a) What do you notice about the first two given vectors, $\mathbf{p} = 6t^2 + 8t + 2$ and $\mathbf{q} = 3t^2 + 4t + 1$?

The vector $\mathbf{p}$ is double vector $\mathbf{q}$, that is $\mathbf{p} = 2\mathbf{q}$ or $\mathbf{p} - 2\mathbf{q} = \mathbf{O}$.

What does this mean?

$\mathbf{p}$ and $\mathbf{q}$ are linearly dependent because we can make vector $\mathbf{p}$ by doubling $\mathbf{q}$.

Notice that the non-zero scalars are 1 and -2 because $\mathbf{p} - 2\mathbf{q} = (1)\mathbf{p} + (-2)\mathbf{q} = \mathbf{O}$.

(b) In this case the vector $\mathbf{p} = 2t^2 + 3t + 2$ is not a multiple of $\mathbf{q} = t^2 + t + 1$, which means that they are linearly independent.

(continued...)

(c) With three vectors it is much harder to spot any relationship between them, so we have to carry out the general procedure for solving simultaneous equations.

Let k_1 , k_2 and k_3 be scalars. We need to find values for k_1 , k_2 and k_3 , so that they satisfy

$$k_1\mathbf{p} + k_2\mathbf{q} + k_3\mathbf{r} = \mathbf{O} \quad (*)$$

Substituting our given vectors $\mathbf{p} = t^2 + 3t - 1$, $\mathbf{q} = 2t^2 + 7t + 5$, $\mathbf{r} = 7$ and $\mathbf{O} = 0$ into (*):

$$\begin{aligned} k_1\mathbf{p} + k_2\mathbf{q} + k_3\mathbf{r} &= k_1(t^2 + 3t - 1) + k_2(2t^2 + 7t + 5) + k_3(7) \\ &= k_1t^2 + 3k_1t - k_1 + 2k_2t^2 + 7k_2t + 5k_2 + 7k_3 \\ &\equiv (k_1 + 2k_2)t^2 + (3k_1 + 7k_2)t + (-k_1 + 5k_2 + 7k_3) \\ &\quad \text{collecting like terms} \\ &= 0 \end{aligned}$$

Equating the coefficients of the last two lines gives

$$\begin{array}{ll} t^2 : & k_1 + 2k_2 = 0 \\ t : & 3k_1 + 7k_2 = 0 \\ \text{const} : & -k_1 + 5k_2 + 7k_3 = 0 \end{array}$$

The only solution to these equations is $k_1 = k_2 = k_3 = 0$.

We conclude that the given vectors $\mathbf{p}$, $\mathbf{q}$ and $\mathbf{r}$ are linearly independent.

3.3.3 Properties of linear dependence and independence

Some of the properties associated with the linear independence and dependence of vectors in Euclidean space $\mathbb{R}^n$ were proved during chapter 2 in subsection 2.3.4. We can extend these results to the general vector space V . The proofs are very similar to the Euclidean space proofs that you were asked to show in chapter 2.

Proposition (3.12). Two vectors $\mathbf{u}$ and $\mathbf{v}$ in a vector space V are linearly **dependent** $\Leftrightarrow$ one of the vectors is a multiple of the other.

Proof.

($\Rightarrow$). Assume vectors $\mathbf{u}$ and $\mathbf{v}$ are linearly dependent.



What do we need to prove?

We need to prove one vector is a multiple of the other. The vectors $\mathbf{u}$ and $\mathbf{v}$ are linearly dependent, therefore by the above definition (3.11) there must be non-zero scalars k or c such that

$$k\mathbf{u} + c\mathbf{v} = \mathbf{O}$$

Suppose $k \neq 0$ then $k\mathbf{u} = -c\mathbf{v}$ which implies $\mathbf{u} = -\frac{c}{k}\mathbf{v}$. We have $\mathbf{u} = -\frac{c}{k}\mathbf{v}$, therefore the vector $\mathbf{u}$ is a multiple of the vector $\mathbf{v}$.

($\Leftarrow$). Now we prove it the other way. Assume that the vector $\mathbf{u}$ is a multiple of vector $\mathbf{v}$ which means that we can write this as

$$\mathbf{u} = m\mathbf{v} \quad (m \text{ is a scalar})$$

Taking $-m\mathbf{v}$ from both sides, we have $\mathbf{u} - m\mathbf{v} = \mathbf{0}$. Since this linear combination (1) $\mathbf{u} - m\mathbf{v} = \mathbf{0}$ gives the zero vector with non-zero scalars, then by definition (3.11) we conclude that vectors $\mathbf{u}$ and $\mathbf{v}$ are linearly dependent. ■

Proposition (3.13). Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$ be a set of linearly dependent vectors in a vector space V . Then the set of vectors $T = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n, \mathbf{v}_{n+1}\}$ are also linearly dependent in V .



What does this proposition mean?

It means that if a set of n vectors are linearly dependent then the same n vectors plus another vector are also linearly dependent – the addition of more vectors does *not* change dependence.

Proof.

Since we are given that the vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$ are linearly dependent, we can write these as

$$k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 + \dots + k_n\mathbf{v}_n = \mathbf{0} \quad (*)$$

where *all* the k 's are not zero. Consider the linear combination

$$k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + \dots + k_n\mathbf{v}_n + \underbrace{k_{n+1}\mathbf{v}_{n+1}}_{\text{extra vector}}$$

If we take the case where $k_{n+1} = 0$ then we have

$$\underbrace{k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + \dots + k_n\mathbf{v}_n}_{=\mathbf{0} \text{ by } (*)} + \underbrace{k_{n+1}\mathbf{v}_{n+1}}_{\text{extra vector}} = \mathbf{0}$$

By (*) all the scalars are not zero in the last equation, therefore by definition (3.11) we conclude that the set of vectors $T = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n, \mathbf{v}_{n+1}\}$ are linearly dependent. ■

Proposition (3.14). The vectors in the set $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$ are linearly dependent $\Leftrightarrow$ one of these vectors, say $\mathbf{v}_k$, is a linear combination of the preceding vectors, that is

$$\mathbf{v}_k = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 + \dots + c_{k-1}\mathbf{v}_{k-1}$$

 What does this proposition mean?

This means that if the vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots$ and $\mathbf{v}_n$ are linearly dependent, then one of these vectors can be written in terms of the other vectors. This proposition also goes the other way, that is if one of the vectors can be written as a linear combination of the preceding vectors then these vectors are linearly dependent.

Proof.

($\Rightarrow$). Assume the vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$ are linearly dependent. We need to show that we can write the vector $\mathbf{v}_k$ as a linear combination of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{k-1}$.

By the above definition (3.11):

Dependence: If $k_1, k_2, \dots, k_n$ are not all zero then $k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + \dots + k_n\mathbf{v}_n = \mathbf{0}$

We can write the vectors in S as

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 + \dots + c_n\mathbf{v}_n = \mathbf{0}$$

where *all* the scalars c 's are *not* zero. Let k be the largest subscript for which $c_k \neq 0$ [not zero] then

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 + \dots + c_{k-1}\mathbf{v}_{k-1} + c_k\mathbf{v}_k = \mathbf{0}$$

Carrying out vector algebra

$$\begin{aligned} c_k\mathbf{v}_k &= -c_1\mathbf{v}_1 - c_2\mathbf{v}_2 - c_3\mathbf{v}_3 - \dots - c_{k-1}\mathbf{v}_{k-1} && \text{[transposing]} \\ \mathbf{v}_k &= -\frac{c_1}{c_k}\mathbf{v}_1 - \frac{c_2}{c_k}\mathbf{v}_2 - \frac{c_3}{c_k}\mathbf{v}_3 - \dots - \frac{c_{k-1}}{c_k}\mathbf{v}_{k-1} && \text{[dividing by } c_k \neq 0\text{]} \\ &= d_1\mathbf{v}_1 + d_2\mathbf{v}_2 + d_3\mathbf{v}_3 + \dots + d_{k-1}\mathbf{v}_{k-1} \text{ where } d_j = -\frac{c_j}{c_k} \end{aligned}$$

As can be seen, the vector $\mathbf{v}_k$ can be written as a linear combination of the preceding vectors.

($\Leftarrow$). Now we go the other way. Assume that we can write the vector $\mathbf{v}_k$ as a linear combination of the preceding vectors. This means that we have

$$\mathbf{v}_k = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 + \dots + c_{k-1}\mathbf{v}_{k-1} \quad (c\text{'s are scalars})$$

Rearranging this gives

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 + \dots + c_{k-1}\mathbf{v}_{k-1} - \mathbf{v}_k = \mathbf{0}$$

Hence the vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_{k-1}$ and $\mathbf{v}_k$ are linearly dependent.

 Why?

Because we can write the last line as

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 + \dots + c_{k-1}\mathbf{v}_{k-1} + (-1)\mathbf{v}_k = \mathbf{0}$$

There is at least one non-zero scalar, -1 .

By the previous Proposition (3.13):

If $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ are dependent then $T = \{\mathbf{v}_1, \dots, \mathbf{v}_n, \mathbf{v}_{n+1}\}$ are also dependent.

We conclude that the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k, \mathbf{v}_{k+1}, \dots$ and $\mathbf{v}_n$ (adding more vectors does not change dependency) are linearly dependent. This completes our proof. ■

Note that the above proposition also implies that if none of the vectors in a set can be written as a linear combination of the preceding vectors then the set is linearly independent. Hence to prove linear independence it is enough to show that none of the vectors is a linear combination of the preceding vectors.

Example 3.18

Let P_n be the vector space of polynomials of degree n or less.

Show that the set of vectors $\{1, t, t^2, t^3, \dots, t^n\}$ are linearly independent.

Solution

Note that none of polynomials can be written as a linear combination of the preceding polynomials. For example, we cannot write the polynomial such as t^k as a linear combination of $\{1, t, t^2, t^3, \dots, t^{k-1}\}$ with scalars c 's:

$$c_0 1 + c_1 t + c_2 t^2 + c_3 t^3 + \dots + c_{k-1} t^{k-1} \neq t^k \quad [\text{not equal}]$$

Since none of the polynomials is a linear combination of the preceding polynomials we conclude that the given set $\{1, t, t^2, t^3, \dots, t^n\}$ is linearly independent.

3.3.4 Basis vectors

This section extends the definition of basis vectors to the general vector space V .

During chapter 2 we covered basis in section 2.4.2, but only for the Euclidean space $\mathbb{R}^n$.

In this section we discuss the nature of a basis in a general vector space V .

You can think of a basis as the *axes* of a coordinate system that describes a vector space. This means that every vector in a vector space V can be written in terms of the coordinate system which is represented by the basis vectors. You can decompose each vector into its basis vectors.

Definition (3.15). Consider the n vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots$ and $\mathbf{v}_n$ in the vector space V . These vectors form a **basis** of $V \Leftrightarrow$

- (i) $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots$ and $\mathbf{v}_n$ span V
- (ii) $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots$ and $\mathbf{v}_n$ are linearly independent

We can write these n vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots$ and $\mathbf{v}_n$ as a set

$$B = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$$

These are generally called the **basis vectors**.

? What does the term ' $\{v_1, v_2, v_3, \dots, v_n\}$ spans V ' mean?

If every vector in V can be produced by a linear combination of vectors $\{v_1, v_2, v_3, \dots, v_n\}$ then we say that these vectors **span** or **generate** the vector space V . This is generally proved by considering an arbitrary vector in V and then showing that it is a linear combination of vectors $v_1, v_2, v_3, \dots$ and v_n .

Next we show how to prove that a set of given vectors form a basis for a general vector space. The proofs are similar to the ones in the last chapter but we cannot visualize the vectors because we are dealing with general vector spaces.

Example 3.19

Let P_n be the vector space of polynomials of degree n or less.

Show that the set of vectors $S = \{1, t, t^2, \dots, t^n\}$ is a basis for P_n .

Solution

How do we prove this result?

We need to show two things for S to qualify as a basis:

1. The set of vectors S must span P_n .
2. The set of vectors S must be linearly independent.

Span: Consider an arbitrary vector $c_0 + c_1t + c_2t^2 + c_3t^3 + \dots + c_nt^n$ in P_n . This can be written as a linear combination of $\{1, t, t^2, t^3, \dots, t^n\}$ as follows:

$$k_0 + k_1t + k_2t^2 + k_3t^3 + \dots + k_nt^n = c_0 + c_1t + c_2t^2 + c_3t^3 + \dots + c_nt^n$$

where $k_j = c_j$ for $j = 1, 2, 3, \dots$ and n . Hence the vectors $\{1, t, t^2, t^3, \dots, t^n\}$ span P_n .

Linearly independent: We have already shown this in Example 3.18.

Both conditions (1 and 2) are satisfied, therefore the given vectors $S = \{1, t, t^2, \dots, t^n\}$ form a basis for P_n .

These basis vectors $\{1, t, t^2, t^3, \dots, t^n\}$ are generally known as the natural or standard basis for P_n . This is not the only basis for P_n ; there are an infinite number of them. For example, you can multiply each element in this $\{1, t, t^2, t^3, \dots, t^n\}$ by a non-zero number, and the set would then be another basis for P_n .

We can write any vector in terms of these basis vectors. This is similar to writing vectors in $\mathbb{R}^2$ in terms of $e_1 = (1 \ 0)^T$ representing the x axis and $e_2 = (0 \ 1)^T$ representing the y axis.

Example 3.20

Let M_{22} be the vector space containing matrices of size 2 by 2. Show that the following matrices are not a basis for M_{22} :

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \text{ and } B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Solution

How do we show this result?

We need to show either that the matrices **A** and **B** do not span M_{22} or that they are linearly dependent.

To show that matrices **A** and **B** do not span M_{22} , we only need to select a matrix in M_{22} and prove that a linear combination of **A** and **B** do not generate the selected matrix.

Consider the matrix $\mathbf{C} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$. Let k_1 and k_2 be scalars then

$$\begin{aligned} k_1\mathbf{A} + k_2\mathbf{B} &= k_1 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + k_2 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} k_1 & 0 \\ 0 & k_1 \end{pmatrix} + \begin{pmatrix} 0 & k_2 \\ k_2 & 0 \end{pmatrix} = \begin{pmatrix} k_1 & k_2 \\ k_2 & k_1 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \end{aligned}$$

By equating entries we have

$$k_1 = 1, k_2 = 2, k_2 = 3 \text{ and } k_1 = 4$$

This is inconsistent because we have two different values for k_1 , 1 and 4, which means that we cannot obtain the matrix $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$. Hence the given matrices $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ do not span M_{22} because we cannot generate the matrix **C**, therefore these matrices cannot form a basis for M_{22} . The scalar k_2 also has two different values; 2 and 3.

[Note that matrices **A** and **B** are linearly independent but that is not enough to be a basis.]

In fact, the given matrices only produce matrices of the form $\begin{pmatrix} a & b \\ b & a \end{pmatrix}$, which means that we cannot obtain the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ where $a \neq d$ or $b \neq c$.

In the study of Fourier series, we discuss periodic (one which repeats itself) continuous functions. For example, the following is a periodic continuous function:

$$y = \frac{2}{\pi} \sin(x) + \frac{2}{3\pi} \sin(3x) + \frac{2}{5\pi} \sin(5x)$$

Figure 3.13 illustrates that this function repeats every 2π intervals.

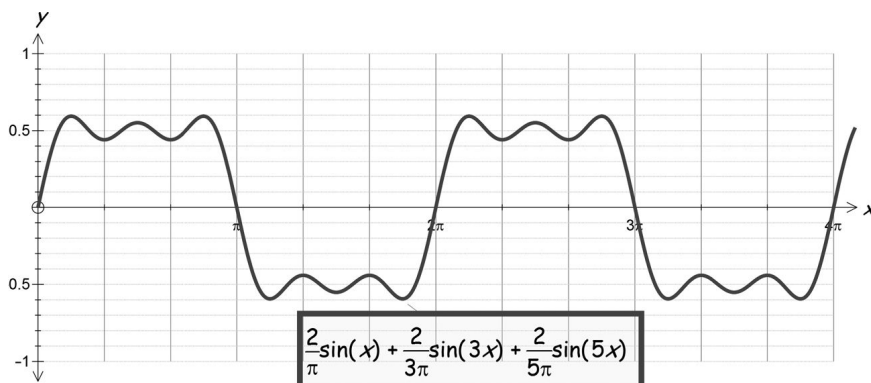


Figure 3.13

As you can see, this periodic function y is made up of a linear combination of $\sin(x)$, $\sin(3x)$ and $\sin(5x)$. Any function in the vector space of $F[0, 2\pi]$ of the form

$$y = k_1 \sin(x) + k_2 \sin(3x) + k_3 \sin(5x)$$

has a basis $\{\sin(x), \sin(3x), \sin(5x)\}$. Clearly these vectors span the space and you are asked to show linear independence of these in Exercises 3.3.

We can represent any periodic continuous function as a linear combination of sines and cosines. In general, a Fourier series will have the form

$$y = k_0 + k_1 \cos(x) + k_2 \cos(2x) + \cdots + c_1 \sin(x) + c_2 \sin(2x) + \cdots$$

where k 's and c 's are scalars. A basis for this general function is

$$\{1, \cos(x), \cos(2x), \cos(3x), \dots, \sin(x), \sin(2x), \sin(3x), \dots\}$$

We need 1 in the basis because this will create the constant term k_0 in the above. Hence we have an infinite number of basis vectors.

3.3.5 Uniqueness

Choosing a basis for a vector space is essentially choosing a coordinate system which allows each vector to be expressed uniquely.

Proposition (3.16). Let the set $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a basis for a vector space V . Every vector in V can be expressed *uniquely* as a linear combination of the basis vectors.



What does this proposition mean?

Any vector in V written as the linear combination of basis vector has unique scalars.

Proof.

Consider an arbitrary vector $\mathbf{w}$ in V . Since $B = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$ is a basis for the vector space V , there exist scalars $k_1, k_2, k_3, \dots$ and k_n such that

$$k_1 \mathbf{v}_1 + k_2 \mathbf{v}_2 + k_3 \mathbf{v}_3 + \cdots + k_n \mathbf{v}_n = \mathbf{w} \quad (\dagger)$$

Suppose we can write this vector $\mathbf{w}$ as another linear combination

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + c_3 \mathbf{v}_3 + \cdots + c_n \mathbf{v}_n = \mathbf{w} \quad (\dagger\dagger)$$

where c 's are scalars.



What do we need to prove?

Required to prove that the two sets of scalars are equal: $k_1 = c_1, k_2 = c_2, \dots$ and $k_n = c_n$.

Subtracting the two $\mathbf{w}$ vectors, $(\dagger) - (\dagger\dagger)$, gives

$$\begin{aligned} k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 + \cdots + k_n\mathbf{v}_n - c_1\mathbf{v}_1 - c_2\mathbf{v}_2 - c_3\mathbf{v}_3 - \cdots - c_n\mathbf{v}_n &= \mathbf{w} - \mathbf{w} \\ (k_1 - c_1)\mathbf{v}_1 + (k_2 - c_2)\mathbf{v}_2 + (k_3 - c_3)\mathbf{v}_3 + \cdots + (k_n - c_n)\mathbf{v}_n &= \mathbf{O} \end{aligned} \quad (*)$$

Since $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots$ and $\mathbf{v}_n$ are basis vectors, they are linearly independent, which means that all the bracketed terms (scalars) in $(*)$ must be equal to zero (Definition (3.10)). We have

$$\begin{aligned} k_1 - c_1 = 0, k_2 - c_2 = 0, k_3 - c_3 = 0, \dots \text{ and } k_n - c_n = 0 \\ k_1 = c_1, k_2 = c_2, k_3 = c_3, \dots \text{ and } k_n = c_n \end{aligned}$$

Hence we have proved that the basis vector representation of any vector is unique. ■



Summary

Consider the linear combination:

$$k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 + \cdots + k_n\mathbf{v}_n = \mathbf{O}$$

If the only solution is $k_1 = k_2 = k_3 = \cdots = k_n = 0$ then vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ are linearly independent, otherwise they are dependent.

Definition (3.15). Vectors $B = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$ of the vector space V form a **basis** of $V \Leftrightarrow$

- (i) $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$ span V
- (ii) $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$ are linearly independent



EXERCISES 3.3

(Brief solutions at end of book. Full solutions available at <http://www.oup.co.uk/companion/singh>.)

1. Let M_{22} be the vector space of size 2 by 2 matrices. Decide whether the following matrices are linearly independent or dependent:

$$(a) \mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \text{ and } \mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (b) \mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \text{ and } \mathbf{B} = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$$

$$(c) \mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \text{ and } \mathbf{B} = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} \quad (d) \mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \text{ and } \mathbf{B} = \begin{pmatrix} 2/5 & 4/5 \\ 6/5 & 8/5 \end{pmatrix}$$

On a course in differential equations it is important that you can test the following functions in questions 2 and 3 for linear independence.

2. Let V be the vector space of continuous functions defined on the real line. Test the following vectors for linear independence:
 - (a) $\mathbf{f} = (x+1)^2$ and $\mathbf{g} = x^2 + 2x + 1$
 - (b) $\mathbf{f} = 2$ and $\mathbf{g} = x^2$
 - (c) $\mathbf{f} = 1$ and $\mathbf{g} = e^x$
 - (d) $\mathbf{f} = \cos(x)$ and $\mathbf{g} = \sin(x)$
 - (e) $\mathbf{f} = \sin(x)$ and $\mathbf{g} = \sin(2x)$
3. Let V be the vector space of continuous functions defined on the real line. Test the following vectors for linear independence:
 - (a) $\mathbf{f} = \cos^2(x)$, $\mathbf{g} = \sin^2(x)$ and $\mathbf{h} = 5$
 - (b) $\mathbf{f} = \cos(2x)$, $\mathbf{g} = \sin^2(x)$ and $\mathbf{h} = \cos^2(x)$
 - (c) $\mathbf{f} = 1$, $\mathbf{g} = x$ and $\mathbf{h} = x^2$
 - (d) $\mathbf{f} = \sin(2x)$, $\mathbf{g} = \sin(x)\cos(x)$ and $\mathbf{h} = \cos(x)$
 - (e) $\mathbf{f} = e^x \sin(2x)$, $\mathbf{g} = e^x \sin(x)\cos(x)$ and $\mathbf{h} = e^x \cos(x)$
 - (f) $\mathbf{f} = 1$, $\mathbf{g} = e^x$ and $\mathbf{h} = e^{-x}$
 - (g) $\mathbf{f} = e^x$, $\mathbf{g} = e^{2x}$ and $\mathbf{h} = e^{3x}$
4. Let $\mathbf{f} = \sin(x)$, $\mathbf{g} = \sin(3x)$ and $\mathbf{h} = \sin(5x)$ be in the vector space of continuous functions $F[0, 2\pi]$. Show that these vectors are linearly independent.
5. Let M_{22} be the vector space of size 2 by 2 matrices. Show that the following matrices are a basis for M_{22} :

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \mathbf{C} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \text{ and } \mathbf{D} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

[These are generally called the standard or natural basis for M_{22}]

We denote the vector space of polynomials of degree ≤ 2 by P_2 which is discussed in the next two questions.

6. Show that $\{1, t-1, (t-1)^2\}$ forms a basis for the vector space P_2 . Write the vector $\mathbf{p} = t^2 + 1$ in terms of these basis vectors.
7. Show that $\{1, t^2 - 2t, 5(t-1)^2\}$ does *not* form a basis for the vector space P_2 .
8. Prove that if a set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$ is linearly independent then the set $\{k\mathbf{v}_1, k\mathbf{v}_2, k\mathbf{v}_3, \dots, k\mathbf{v}_n\}$, where k is a non-zero scalar, is also linearly independent.
9. Prove that any non-zero vector $\mathbf{v}$ on its own is a linearly independent vector in a vector space V .
10. Prove that the zero vector, $\mathbf{O}$, on its own is a linearly dependent vector in a vector space V .
11. Consider the set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$. Prove that if any two vectors $\mathbf{v}_j = \mathbf{v}_m$ where $j \neq m$ in the set are equal then the set is linearly dependent.
12. Consider the set of linearly independent vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$. Prove that any non-empty subset of this is also linearly independent.
13. Consider the set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$ which spans a vector space V . Let $\mathbf{w}$ be a vector in V but not in the set S . Prove that $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n, \mathbf{w}\}$ spans V but is linearly dependent.

14. Consider the set of vectors $B = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$ in a vector space V . Prove that if the set B is a basis for V and $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_m\}$ is a set of linearly independent vectors in V then $m \leq n$.
15. *Prove that if $B_1 = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$ and $B_2 = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \dots, \mathbf{u}_m\}$ are bases for a vector space V then $n = m$. [That is every basis of a vector space has the same number of vectors.]
16. Prove that if the largest number of linearly independent vectors in a vector space V is n then any n linearly independent vectors form a basis for V .
17. *Let S be a subspace of a vector space V . Prove that if S and V have the same basis then $S = V$.

SECTION 3.4 Dimension

By the end of this section you will be able to

- understand what is meant by the dimension of a vector space
- determine the dimension of a vector space and subspace
- prove properties of finite dimensional vector spaces

We have used the word ‘dimension’ without really defining it. Our physical interpretation of a vector space of $\mathbb{R}^n$ is that it has n dimensions.

 *How can we find the dimensions of other vector spaces such as the set of matrices M_{mn} or polynomials P_n ?*

In this section we give the precise definition of dimension so that we can give a numerical value to the dimension of any vector space. This is a challenging section. You may need to look back at the definition of a basis of a vector space to understand the proofs later on in this section.

3.4.1 Introduction to dimension

 *What does the word ‘dimension’ mean in everyday language?*

Dimension normally refers to the size of an object in a particular direction. For example, the height and diameter of a cylinder are the dimensions of the cylinder.

 *What does the term dimension of a vector space mean?*

The definition of dimension is the number of basis vectors or axes needed to describe the given vector space V .

Definition (3.17). The number of vectors in a basis of a non-zero vector space V is the **dimension** of the space.

We will show later in this section that the number of vectors in a basis depends on two things: linear independence and spanning.

We will prove that if we have too many vectors in the basis set, they become linearly dependent and if there are too few vectors then they will not span the whole vector space.

Example 3.21

What is the dimension of $\mathbb{R}^2$ and $\mathbb{R}^3$?

Evaluate the dimension of $\mathbb{R}^n$.

Solution

From the previous chapter we know that $\mathbf{e}_1 = (1 \ 0)^T$ and $\mathbf{e}_2 = (0 \ 1)^T$ (unit vectors on the x and y axes respectively) form a basis for $\mathbb{R}^2$.

What is the dimension of $\mathbb{R}^2$?

Two vectors, $\mathbf{e}_1$ and $\mathbf{e}_2$, are needed for the basis of $\mathbb{R}^2$, therefore dimension of $\mathbb{R}^2$ is 2.

What is the dimension of $\mathbb{R}^3$?

Similarly, we have the unit vectors $\mathbf{e}_1 = (1 \ 0 \ 0)^T$, $\mathbf{e}_2 = (0 \ 1 \ 0)^T$ and $\mathbf{e}_3 = (0 \ 0 \ 1)^T$ in the directions of x , y and z axes respectively, is the natural basis for $\mathbb{R}^3$, therefore the dimension of $\mathbb{R}^3$ is 3 because we have three vectors in the basis of $\mathbb{R}^3$.

What is the dimension of $\mathbb{R}^n$?

From the last section we have that the n vectors $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \dots, \mathbf{e}_n\}$ form a natural basis for $\mathbb{R}^n$, so the dimension of $\mathbb{R}^n$ is n .

The number of vectors in the basis gives the dimension of the vector space V . This is often denoted by $\dim(V)$.

$$\dim(\mathbb{R}^2) = 2, \quad \dim(\mathbb{R}^3) = 3 \text{ and } \dim(\mathbb{R}^n) = n$$



Why does the above definition (3.17) say **non-zero** space?

Because the zero vector space $\{\mathbf{0}\}$ is linearly dependent, there are no vectors in the basis of $\{\mathbf{0}\}$, which means that it has dimension 0. (The zero vector does not need any axes.)

3.4.2 Finite dimensional vector space



What does the term *finite dimensional vector space* mean?

It is a vector space V which has a finite number of vectors in its basis.

Definition (3.18). In general, if a finite number of vectors form a basis for a vector space V then we say V is **finite dimensional**. Otherwise, the vector space V is known as infinite dimensional.

If the vector space V consists only of the zero vector then it is also finite dimensional.



Can you think of any finite dimensional vector spaces?

The Euclidean spaces – $\mathbb{R}^2$, $\mathbb{R}^3$, $\mathbb{R}^4$, $\dots$, $\mathbb{R}^n$.

 *Are there any other examples of finite dimensional vector spaces?*

The set P_2 of polynomials of degree 2 or less for example, or the set of all 2 by 2 matrices M_{22} . (These were covered in the previous section.)

Definition (3.19). In general, if n vectors $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$ form a basis for a vector space V then we say that V is n -**dimensional**.

 *What is $\dim(M_{22})$ equal to?*

The standard basis for M_{22} (matrices of size 2 by 2) from the Exercises 3.3 question 5 is

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \mathbf{C} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \text{ and } \mathbf{D} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

Therefore $\dim(M_{22}) = 4$ because we have four matrices in $\{\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}\}$ which form a basis for M_{22} .

 *What is $\dim(P_2)$ equal to?*

Remember that the standard basis for P_2 (the set of all polynomials of degree 2 or less) is the set $\{1, t, t^2\}$, which means $\dim(P_2) = 3$ since the basis consists of three vectors.

Table 3.1 shows some vector spaces and their dimensions.

Table 3.1 Some vector spaces and their dimensions

Vector space	Dimension
$\mathbb{R}^n$	n
P_n	$n + 1$
M_{mn}	mn

The dimension does not need to be the number of vectors in the standard basis; it can be any basis for the given vector space because the number of vectors is the same, as the following theorem shows.

Theorem (3.20). Every basis for a finite dimensional vector space has the *same number* of vectors.

Proof.

We proved this in question 15 of Exercises 3.3. ■

Since every basis has the *same* number of vectors in a finite dimensional vector space, definition (3.17) means that:

$$\text{number of vectors in any basis} = \text{dimension of vector space}$$

Theorem (3.20) is important in linear algebra and definitely worth learning.

There are, of course, many infinite dimensional vector spaces, such as the set of *all* polynomials P (or P_∞). However, we confine ourselves to finite dimensional vector spaces for the remainder of this chapter.

3.4.3 Subspaces revisited

We discussed subspaces in section 3.2 of this chapter. A subspace is a non-empty subset S in a vector space V , which forms a vector space in its own right, with respect to the same vector addition and scalar multiplication as its parent set V . This is illustrated in Fig. 3.14.

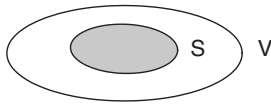


Figure 3.14



What is the dimension of a subspace of V ?

That depends on the number of vectors in the basis of the subspace.

Example 3.22

Let $\mathbb{R}^2$ be a vector space and S be the set of vectors $\begin{pmatrix} 0 \\ a \end{pmatrix}$. In Example 3.5 we showed that this is a subspace of $\mathbb{R}^2$. Find a basis for the subspace S and determine $\dim(S)$.

Solution

Every vector of S is of the form $\begin{pmatrix} 0 \\ a \end{pmatrix}$, which we can express in terms of $\mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ as

$$\begin{pmatrix} 0 \\ a \end{pmatrix} = a \begin{pmatrix} 0 \\ 1 \end{pmatrix} = a\mathbf{e}_2$$

This vector $\mathbf{e}_2$ forms a basis for S , therefore $\dim(S) = 1$. This is shown graphically in Fig. 3.15.

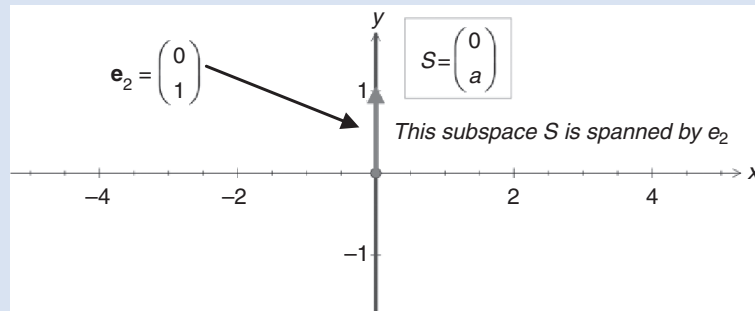


Figure 3.15

Figure 3.15 shows that the subspace S is the vertical axis (y axis) and $\mathbf{e}_2$ generates any vector in S , along this line.

Example 3.23

Let $\mathbf{u} = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}$, $\mathbf{v} = \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix}$ and $\mathbf{w} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ span a subspace S of $\mathbb{R}^3$.

What is the dimension of S ?

Solution

What do you notice about the first two vectors $\mathbf{u}$ and $\mathbf{v}$?

They are linearly dependent because $\mathbf{u} = -2\mathbf{v}$. Note that the vector $\mathbf{w}$ is linearly independent of $\mathbf{u}$ and $\mathbf{v}$.

How many vectors are in the basis of the subspace S ?

Two vectors $\mathbf{u}$ and $\mathbf{w}$ (or $\mathbf{v}$ and $\mathbf{w}$).

What is $\dim(S)$ equal to?

Two because we only have two vectors in the basis of S , that is $\dim(S) = 2$. This means that the given vectors $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ span a subspace S which is a plane in $\mathbb{R}^3$ as demonstrated in Fig. 3.16.

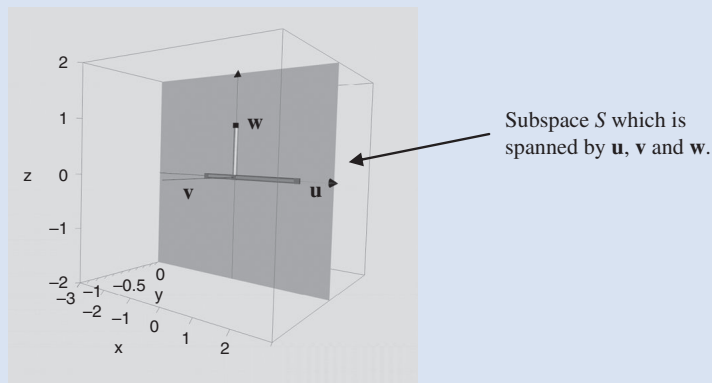


Figure 3.16

Note: a general misconception is that the dimension of a space is how many entries are required to specify a vector in that space. This is clearly wrong because in Example 3.23 above we require three entries to specify a vector in the subspace S but it has dimension 2. This means the space S in Fig. 3.16 can be spanned by two basis vectors or two axes rather than the three given vectors.

Example 3.24

Let S be a set of vectors in P_2 which are of the form

$$ax^2 + (a+b)x + (a+b) \text{ where } a \text{ and } b \text{ are scalars}$$

Then S is a subspace of P_2 . Find a basis for S and determine $\dim(S)$.

Solution

We can write vectors in S as

$$ax^2 + (a+b)x + (a+b) = ax^2 + ax + bx + (a+b) \quad [\text{expanding}]$$

$$= a(x^2 + x + 1) + b(x + 1) \quad [\text{factorizing}]$$

(continued...)

Hence the vectors $x^2 + x + 1$ and $x + 1$ span S , because we have demonstrated that the original set can be generated by these two vectors

What else do we need to show for these vectors to be a basis for S ?

We need to prove that they are linearly independent. Since $x^2 + x + 1$ and $x + 1$ are not multiples of each other, they are linearly independent.

Hence the two vectors $\{x^2 + x + 1, x + 1\}$ form a basis for S , therefore $\dim(S) = 2$.

3.4.4 Properties of finite dimensional vector spaces

In this section we show some important properties of bases and dimension. This is a demanding section because the proofs of propositions are lengthy.

Lemma (3.21). Let V be a finite n -dimensional vector space. We have the following:

- (a) Let $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$ be a set of linearly independent vectors. Then $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$ where $m > n$ (m is greater than n) is linearly dependent.
- (b) If the n vectors $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \dots, \mathbf{u}_n\}$ span V then $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \dots, \mathbf{u}_m\}$ where $m < n$ does not span V .

 *What does part (a) mean?*

In n -dimensional vector space, if you add additional vectors to n linearly independent vectors then the set becomes linearly dependent.

 *How do we prove this result?*

By using proof by contradiction.

Proof of (a).

The number of basis vectors in V is n . Suppose that $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$ are linearly independent. Then we must have $m \leq n$ (m is less than or equal to n).

 *Why?*

Because by question 14 of Exercises 3.3 we know that the number of vectors in a linearly independent set must be less than or equal to $n, m \leq n$ (the number of basis vectors).

However, we are given that $m > n$ (m is greater than n), therefore our supposition that $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$ is a linearly independent set of vectors must be wrong, so this set is linearly dependent. ■

 *What does part (b) mean?*

If n vectors span V where V is an n -dimensional vector space, then less than n vectors cannot span V . Again, we use proof by contradiction.

Proof of (b).

Suppose $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \dots, \mathbf{u}_m\}$ where $m < n$ span V , that is

$$\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \dots, \mathbf{u}_m\} = V \quad (*)$$

The dimension of the left hand side, $\dim(\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \dots, \mathbf{u}_m\}) \leq m$ but the dimension of V is n . This is a contradiction because we are given $m < n$ and the dimension of both sides of (*) must be equal. Hence $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m\}$ where $m < n$ cannot span V . ■

Theorem (3.22). Let V be a finite n -dimensional vector space. We have the following:

- (a) Any linearly independent set of n vectors $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$ form a basis for V .
- (b) Any spanning set of n vectors $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \dots, \mathbf{u}_n\}$ forms a basis for V .



How do we prove these results?

By using the definition of basis as described in the last section, that is:

Definition (3.15) A set of vectors is a basis for $V \Leftrightarrow$ it is linearly independent and spans V .

Proof of (a).

We are given that the vectors $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$ are linearly independent so we only need to show that these vectors also span V . Suppose there is a vector $\mathbf{w}$ in V such that

$$\mathbf{w} = k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + k_3\mathbf{v}_3 + \dots + k_n\mathbf{v}_n + k_{n+1}\mathbf{v}_{n+1} \quad (*)$$

where k 's are scalars. [We are supposing that the given vectors do not span V that is why we have added an extra vector $\mathbf{v}_{n+1}$].

By the above Lemma (3.21) part (a):

In a n dimension space, vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$ where $m > n$ are linearly dependent.

The set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n, \mathbf{v}_{n+1}\}$ is linearly dependent so we can write the vector $\mathbf{v}_{n+1}$ in terms of its preceding vectors, that is

$$\mathbf{v}_{n+1} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 + \dots + c_n\mathbf{v}_n \text{ where } c\text{'s are scalars}$$

Substituting this into (*) gives

$$\begin{aligned} \mathbf{w} &= k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + \dots + k_n\mathbf{v}_n + k_{n+1}\mathbf{v}_{n+1} \\ &= k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + \dots + k_n\mathbf{v}_n + k_{n+1}(c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_n\mathbf{v}_n) \\ &= (k_1 + k_{n+1}c_1)\mathbf{v}_1 + (k_2 + k_{n+1}c_2)\mathbf{v}_2 + \dots + (k_n + k_{n+1}c_n)\mathbf{v}_n \end{aligned}$$

Thus the vector $\mathbf{w}$ can be written as a linear combination of the given linearly independent vectors $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$. Thus $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$ spans V , therefore it forms a basis for V . ■

Proof of (b).

We are given that $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \dots, \mathbf{u}_n\}$ spans the vector space V . Let $\mathbf{w}$ be an arbitrary vector of V , then we can write

$$\mathbf{w} = k_1\mathbf{u}_1 + k_2\mathbf{u}_2 + \dots + k_{n-1}\mathbf{u}_{n-1} + k_n\mathbf{u}_n \quad (\dagger)$$

where k 's are scalars because $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \dots, \mathbf{u}_n\}$ spans V .



What do we need to show for this set of vectors to be a basis for V ?

Required to prove that the set of vectors under consideration $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \dots, \mathbf{u}_n\}$ is linearly independent. Suppose $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \dots, \mathbf{u}_n\}$ is linearly dependent then we can write the vector $\mathbf{u}_n$ in terms of its preceding vectors, that is

$$\mathbf{u}_n = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + c_3\mathbf{u}_3 + \dots + c_{n-1}\mathbf{u}_{n-1} \text{ where } c\text{'s are scalars}$$

Substituting this into the above (†) gives

$$\begin{aligned} \mathbf{w} &= k_1\mathbf{u}_1 + k_2\mathbf{u}_2 + \dots + k_{n-1}\mathbf{u}_{n-1} + k_n\mathbf{u}_n \\ &= k_1\mathbf{u}_1 + k_2\mathbf{u}_2 + \dots + k_{n-1}\mathbf{u}_{n-1} + k_n(c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \dots + c_{n-1}\mathbf{u}_{n-1}) \\ &= (k_1 + k_nc_1)\mathbf{u}_1 + (k_2 + k_nc_2)\mathbf{u}_2 + \dots + (k_{n-1} + k_nc_{n-1})\mathbf{u}_{n-1} \end{aligned}$$

This shows that the above $n - 1$ vectors $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \dots, \mathbf{u}_{n-1}\}$ span V . This is impossible because, by the above Lemma (3.21), part (b):

If n vectors $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ span V then $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m\}$ where $m < n$ does *not* span V .

Fewer than n vectors *cannot* span V . Thus $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \dots, \mathbf{u}_n\}$ is linearly independent which means it is a basis for the given vector space V . ■

Theorem (3.22) says two things:

1. The *maximum independent* set for an n -dimensional vector space is n vectors. If you add any more vectors then it becomes linearly dependent. The basis is a *maximum linearly independent* set.
2. The *minimum spanning* set for an n -dimensional vector space is n vectors. If you remove any of the vectors of the spanning set then it no longer spans V . The basis is a *minimum spanning* set.

Two Definitions of Basis

1. A basis for a vector space is the *largest independent* set of vectors.
2. A basis for a vector space is the *smallest spanning* set of vectors.

This means if you have n linearly independent vectors in an n -dimensional vector space V then it is a basis for V . You do not have to check that it spans V .

Also, if you have n spanning vectors in an n -dimensional vector space V then it is a basis for V . You do not have to check that the set is linearly independent.

A basis (axes) is the most efficient set of vectors used to describe a given vector space.

Let us apply the above theorem and see how it simplifies matters.

Example 3.25

Show that the following vectors form a basis for P_n :

$$S = \{1, t + 1, (t + 1)^2, (t + 1)^3, \dots, (t + 1)^n\}$$

Solution

We have $\dim(P_n) = n + 1$, and there are $n + 1$ given vectors.

We need to show that the set of vectors in S are linearly independent or span the vector space P_n . Note that *none* of the polynomials in the set S can be written as a linear combination of the preceding polynomials in the set. This means that we *cannot* write $(t + 1)^m$ where $m \leq n$ in terms of the vectors before we get to $(t + 1)^m$.

What can you conclude about this set?

Since we cannot write any of the polynomials (vectors) in terms of the preceding polynomials (vectors), the set S must be linearly independent. Again, by the above Theorem (3.22) part (a), we conclude that the vectors in the set S are a basis for P_n .

Example 3.26

Let M_{22} be the vector space of all 2 by 2 matrices. Show that the following matrices do not form a basis for M_{22} :

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix} \text{ and } \mathbf{C} = \begin{pmatrix} 2 & 1 \\ 1 & 5 \end{pmatrix}$$

Solution

What is the dimension of M_{22} ?

Recall from Table 3.1 in section 3.4.2 that $\dim(M_{22}) = 2 \times 2 = 4$. In the above list, we have three matrices so they cannot form a basis for M_{22} .

Why not?

Because $\dim(M_{22}) = 4$, and by Definition (3.17) we need exactly four matrices for a basis.

Example 3.27

Let $F[0, 2\pi]$ be a vector space of periodic continuous functions which is spanned by

$$S = \{\sin(x) \cos(x), \sin(2x), \cos(2x)\}$$

Show that these vectors do not form a basis for $F[0, 2\pi]$. Also find a basis for this space.

Solution

From trigonometry we have the identity

$$\sin(2x) = 2 \sin(x) \cos(x)$$

Hence the first two vectors in S are dependent, therefore the set S cannot be a basis.

A basis can be established if we remove one of the vectors; $\sin(x) \cos(x)$ or $\sin(2x)$. A basis for the given space is $\{\sin(2x), \cos(2x)\}$.

**Summary**

(3.17). The number of vectors in a basis of a non-zero space V is the **dimension** of the space.

(3.20). Every basis for a finite dimensional vector space has the *same number* of vectors.

(3.22). Let V be a finite n -dimensional vector space. We have the following:

(a) Any linearly *independent* set of n vectors forms a basis for V .

(b) Any *spanning* set of n vectors forms a basis for V .

**EXERCISES 3.4**

(Brief solutions at end of book. Full solutions available at <http://www.oup.co.uk/companion/singh>.)

1. Find the dimension of the following vector spaces and exhibit a basis for each space:

- (a) $\mathbb{R}^5$ (b) $\mathbb{R}^7$ (c) $\mathbb{R}^{11}$ (d) $\mathbb{R}^{13}$ (e) M_{33}
 (f) M_{44} (g) M_{23} (h) P_3 (i) P_5 (j) $\mathbf{O}$

2. Let $\mathbb{R}^2$ be a vector space and S be the subspace given by the set of vectors $\begin{pmatrix} a \\ 0 \end{pmatrix}$. Find a basis for S and evaluate $\dim(S)$.

3. Let $\mathbf{u} = (0 \ 0 \ 3)^T$, $\mathbf{v} = (-5 \ 0 \ 0)^T$ and $\mathbf{w} = (0 \ 0 \ 9)^T$ span a subspace S of $\mathbb{R}^3$. Determine the dimension of S .

4. Let S be the set of vectors in P_2 which are of the form

$$at^2 + b$$

Then S is a subspace of V . Find a basis for S and determine $\dim(S)$.

5. Consider the subspace S which consists of matrices of the form $\begin{pmatrix} a & b \\ b & c \end{pmatrix}$ (this is the set of symmetric matrices) of the vector space M_{22} . Find a basis for S and evaluate the dimension of S .

6. Find the dimension of the following vector spaces and subspaces:

- (a) M_{mn} (matrices of size m by n).
 (b) The subspace S of P_3 given by the cubic polynomials $S = \{at^3 + bt^2 + c\}$.
 (c) Let S be the set of vectors in P_3 which are of the form

$$S = \{at^3 + bt^2 + ct + d\}$$

7. Show that the following vectors are a basis for the corresponding vector spaces:

- (a) $\mathbf{u} = \begin{pmatrix} 1 \\ 5 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ for $\mathbb{R}^2$.
 (b) $\mathbf{u} = \begin{pmatrix} 0 \\ 3 \\ 4 \end{pmatrix}$, $\mathbf{v} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ and $\mathbf{w} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ for $\mathbb{R}^3$.

- (c) $\{t^n, t^{n-1}, \dots, t, 1\}$ for the vector space P_n .
- (d) $\left\{ \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \begin{pmatrix} 0 & 3 \\ 0 & 2 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 5 \end{pmatrix} \right\}$ for M_{22} .
8. Explain why the following vectors are not a basis for the corresponding vector spaces:
- (a) $\{1+t, 1+t^2, 1+2t+t^2, 1+2t\}$ for P_2 .
- (b) $\{1+t, 1+2t+t^2\}$ for P_2 .
- (c) $\{(1+t)^2, 1+t^2, 2+4t+2t^2\}$ for P_2 .
- (d) $\left\{ \mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \mathbf{C} = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}, \mathbf{D} = \begin{pmatrix} 0 & 0 \\ 0 & 5 \end{pmatrix} \right\}$ for M_{22} .
- (e) $\left\{ \mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 2 & 0 \\ 2 & 2 \end{pmatrix}, \mathbf{C} = \begin{pmatrix} 1 & 0 \\ 3 & 6 \end{pmatrix}, \mathbf{D} = \begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix} \right\}$ for M_{22} .
9. Show that the following set of vectors is a basis for $\mathbb{R}^5$:

$$S = \left\{ \begin{pmatrix} 1 \\ 2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ -5 \\ 2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -3 \\ 5 \\ 2 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 3 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \right\}$$

10. Let S be a subspace of an n -dimensional vector space V . Prove that $\dim(S) \leq n$.
11. Let S be a subspace of an n -dimensional vector space V . Prove that if $\dim(S) = n$ then $S = V$.
12. Let n be the dimension of a finite dimensional vector space. Prove that n is a positive integer or zero.
13. Let V be a one-dimensional vector space and vector $\mathbf{v} \neq \mathbf{0}$ be in V . Prove that $\mathbf{v}$ is a basis for V .
14. Let V be an n -dimensional vector space and $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$ be a set of vectors in V such that **none** of the vectors is a linear combination of the preceding vectors. Prove that the set S forms a basis for V .
15. Let P be the vector space of all polynomials. Prove that P is infinite dimensional.

SECTION 3.5 Properties of a Matrix

By the end of this section you will be able to

- understand what is meant by the row and column space of a matrix
- determine a basis for the row space of a matrix
- find the rank of a matrix

In this section we examine the rows and columns of a matrix and define what is meant by rank. These are important concepts in solving linear equations which will be discussed in the next section.

You will need to know some of the work of chapter 1, such as elementary row operations and the definition of row echelon form. In this section we will not carry out the row operations but use MATLAB to place a given matrix into row echelon form. Of course, you don't have to use MATLAB; any appropriate software will do or you can use hand calculations.

3.5.1 Row and column vectors

? *What are the 'row vectors of a matrix'?*

The row vectors of a matrix are the entries in the rows of a given matrix. For example, the row vectors of $\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$ are $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$ because the first row of matrix $\mathbf{A}$ has the entries 1, 2 and 3 and the second row has entries 4, 5 and 6.

? *What are the column vectors of $\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$?*

The matrix $\mathbf{A}$ has three columns vectors:

$$\begin{pmatrix} 1 \\ 4 \end{pmatrix}, \begin{pmatrix} 2 \\ 5 \end{pmatrix} \text{ and } \begin{pmatrix} 3 \\ 6 \end{pmatrix}$$

We generalize this by considering the m by n matrix:

$$\mathbf{A} = \begin{matrix} & \text{Col 1} & \text{Col 2} & \cdots & \text{Col } n \\ \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} & \text{Row 1} \\ & & & & \text{Row 2} \\ & & & & \vdots \\ & & & & \text{Row } m \end{matrix}$$

? *What are the row vectors of this matrix $\mathbf{A}$?*

The **row vectors** of $\mathbf{A}$ denoted $\mathbf{r}_1, \mathbf{r}_2, \dots$ and $\mathbf{r}_m$ are given by:

$$\mathbf{r}_1 = \begin{pmatrix} a_{11} \\ a_{12} \\ \vdots \\ a_{1n} \end{pmatrix}, \quad \mathbf{r}_2 = \begin{pmatrix} a_{21} \\ a_{22} \\ \vdots \\ a_{2n} \end{pmatrix}, \dots \text{ and } \mathbf{r}_m = \begin{pmatrix} a_{m1} \\ a_{m2} \\ \vdots \\ a_{mn} \end{pmatrix}$$

? *What are the column vectors of the matrix $\mathbf{A}$?*

The **column vectors** of $\mathbf{A}$ denoted $\mathbf{c}_1, \mathbf{c}_2, \dots$ and $\mathbf{c}_n$ are given by:

$$\mathbf{c}_1 = \begin{pmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{pmatrix}, \quad \mathbf{c}_2 = \begin{pmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{pmatrix}, \dots \text{ and } \mathbf{c}_n = \begin{pmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{pmatrix}$$

For example, the row and column vectors of $\mathbf{B} = \begin{pmatrix} -3 & 6 \\ -5 & 2 \\ -2 & 7 \end{pmatrix}$ are $\mathbf{r}_1 = \begin{pmatrix} -3 \\ 6 \end{pmatrix}$, $\mathbf{r}_2 = \begin{pmatrix} -5 \\ 2 \end{pmatrix}$, $\mathbf{r}_3 = \begin{pmatrix} -2 \\ 7 \end{pmatrix}$ and $\mathbf{c}_1 = \begin{pmatrix} -3 \\ -5 \\ -2 \end{pmatrix}$, $\mathbf{c}_2 = \begin{pmatrix} 6 \\ 2 \\ 7 \end{pmatrix}$ respectively.

3.5.2 Row and column space

The row space of a matrix $\mathbf{A}$ is the space spanned by the row vectors of $\mathbf{A}$. Remember, the space spanned by vectors means the space containing all the linear combinations of these vectors.

 What is the row space of the above matrix $\mathbf{B}$?

It is the space, S , spanned by the vectors $\mathbf{r}_1 = \begin{pmatrix} -3 \\ 6 \end{pmatrix}$, $\mathbf{r}_2 = \begin{pmatrix} -5 \\ 2 \end{pmatrix}$ and $\mathbf{r}_3 = \begin{pmatrix} -2 \\ 7 \end{pmatrix}$.

Any linear combination of these vectors, $\mathbf{r}_1$, $\mathbf{r}_2$ and $\mathbf{r}_3$, is in the row space S . Hence

$$\text{Row Space } S = \text{span}\{\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3\} = k_1 \begin{pmatrix} -3 \\ 6 \end{pmatrix} + k_2 \begin{pmatrix} -5 \\ 2 \end{pmatrix} + k_3 \begin{pmatrix} -2 \\ 7 \end{pmatrix}$$

where k_1 , k_2 and k_3 are scalars. Each row vector has two entries of real numbers and we can show that S is a subspace of $\mathbb{R}^2$.

The row space S is the set of vectors $\mathbf{u}$ such that $\mathbf{u} = k_1\mathbf{r}_1 + k_2\mathbf{r}_2 + k_3\mathbf{r}_3$. This row space S , spanned by $\mathbf{r}_1$, $\mathbf{r}_2$ and $\mathbf{r}_3$, is the vector space given by

$$S = \{\mathbf{u} \mid \mathbf{u} = k_1\mathbf{r}_1 + k_2\mathbf{r}_2 + k_3\mathbf{r}_3\}$$

In this case, these vectors span the whole of $\mathbb{R}^2$ because no two vectors in S are multiples of each other (they are linearly independent). The row space of the above matrix $\mathbf{B}$ occupies $\mathbb{R}^2$. Similarly the column space of a general matrix $\mathbf{A}$ is the space spanned by the column vectors of $\mathbf{A}$. We formally define the row and column space as follows:

Definition (3.23). Let $\mathbf{A}$ be any matrix. Then

- (a) The **row space** of the matrix $\mathbf{A}$ is the space spanned by the row vectors of matrix $\mathbf{A}$.
- (b) The **column space** of the matrix $\mathbf{A}$ is the space spanned by the column vectors of matrix $\mathbf{A}$.

 What is the column space of the above matrix $\mathbf{B}$?

It is the space spanned by the vectors $\mathbf{c}_1 = \begin{pmatrix} -3 \\ -5 \\ -2 \end{pmatrix}$ and $\mathbf{c}_2 = \begin{pmatrix} 6 \\ 2 \\ 7 \end{pmatrix}$. Any linear combination of these vectors, $\mathbf{c}_1$ and $\mathbf{c}_2$, is in the column space of matrix $\mathbf{B}$.

 What sort of space is spanned by these vectors?

Each vector has three entries of real numbers and we can show that the column space is a subspace of $\mathbb{R}^3$. The vector space spanned by vectors $\mathbf{c}_1$ and $\mathbf{c}_2$ is actually a plane in $\mathbb{R}^3$ (Fig. 3.17).

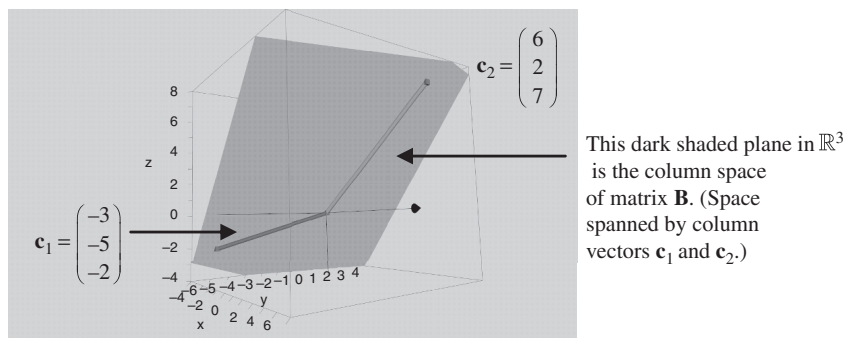


Figure 3.17

In general, we have the following proposition:

Proposition (3.24). Let $\mathbf{A}$ be a m by n matrix, that is we have

$$\mathbf{A} = \left(\begin{array}{c} \overbrace{\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}}^{n \text{ columns}} \end{array} \right) \left. \vphantom{\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}} \right\} m \text{ rows}$$

- (a) The **row space** of matrix $\mathbf{A}$ is a subspace of $\mathbb{R}^n$.
- (b) The **column space** of matrix $\mathbf{A}$ is a subspace of $\mathbb{R}^m$.

Proof – See Exercises 3.5. ■

For example, the row and column space of matrix $\mathbf{A} = \begin{pmatrix} 1 & 4 \\ 5 & 9 \end{pmatrix}$ is a subspace of $\mathbb{R}^2$.

The row space of matrix $\mathbf{B} = \begin{pmatrix} -1 & 3 & 9 \\ -5 & 2 & 6 \end{pmatrix}$ is a subspace of $\mathbb{R}^3$ but the column space is a subspace of $\mathbb{R}^2$.

3.5.3 Basis of a row space



Why do we want to find a basis for the row space of a matrix?

Recall that a basis is formed by the least number of vectors, or axes, required to describe the space. Our matrix may contain zero rows, or rows that are linearly dependent. By using row operations in the row space to reduce our matrix to its simplest form (or basis), we often end up with a much more efficient matrix, which is *row equivalent* to the original.

? What does row equivalent matrix mean?

Two matrices are *row equivalent* if one is obtained from the other by elementary row operations. (This was discussed in chapter 1.) These operations are:

1. Multiplying a row by a non-zero constant.
2. Adding or subtracting a multiple of one row from another.
3. Interchanging rows.

Consider the matrix **A** with rows **a**₁ and **a**₂, converted using row operations, to the matrix **R**:

$$\begin{matrix} \mathbf{a}_1 \\ \mathbf{a}_2 \end{matrix} \begin{pmatrix} 1 & 2 & 3 \\ 4 & 8 & 12 \end{pmatrix} = \mathbf{A} \quad \Rightarrow \quad \begin{matrix} \mathbf{r}_1 = \mathbf{a}_1 \\ \mathbf{r}_2 = \mathbf{a}_2 - 4\mathbf{a}_1 \end{matrix} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \end{pmatrix} = \mathbf{R}$$

Note that the rows of matrix **R**, **r**₁ and **r**₂, are simply a linear combination of rows **a**₁ and **a**₂:

$$\mathbf{r}_1 = \mathbf{a}_1 \text{ and } \mathbf{r}_2 = \mathbf{a}_2 - 4\mathbf{a}_1$$

The rows of matrix **R** are a linear combination of the rows of matrix **A**. This means that the row vectors of matrix **R** lie in the row space of matrix **A**.

? What can you predict about the row space of row equivalent matrices?

The space they occupy is equal.

For example, the row space *S* of the above matrix **A** is given by the vectors **v** in *S* such that:

$$\mathbf{v} = k_1\mathbf{a}_1 + k_2\mathbf{a}_2 = k_1 \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + k_2 \begin{pmatrix} 4 \\ 8 \\ 12 \end{pmatrix} = k_1 \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + 3k_2 \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = c \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = c\mathbf{r}_1$$

$$\text{where } c = k_1 + 3k_2$$

Hence all the vectors **v** are in the row space *S* of matrix **R**. In fact, you can span the row space of matrix **A** with just one vector $(1 \ 2 \ 3)^T$ which is the non-zero row in matrix **R**. Actually the row space *S* created by matrix **A** is the same as the row space created by matrix **R**. We have

$$\text{Row space of } \mathbf{A} = \text{Row space of } \mathbf{R}$$

Proposition (3.25). If matrices **A** and **R** are row equivalent then their *row spaces* are equal.

? How do we prove this result?

By showing that

1. The row space of **A** is in row space of **R**.
2. The row space of **R** is in row space of **A**.

If both these conditions are satisfied then the row spaces of matrices **A** and **R** must be equal.

Proof.

Let $\mathbf{A}$ and $\mathbf{R}$ be row equivalent m by n matrices. Let the row vectors of $\mathbf{A}$ be $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \dots, \mathbf{a}_m$ and the row vectors of $\mathbf{R}$ be $\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3, \dots, \mathbf{r}_m$. This means that we have

$$\mathbf{A} = \begin{pmatrix} \mathbf{a}_1 \\ \vdots \\ \mathbf{a}_m \end{pmatrix} \quad \Rightarrow \quad \mathbf{R} = \begin{pmatrix} \mathbf{r}_1 \\ \vdots \\ \mathbf{r}_m \end{pmatrix}$$

1) Matrices $\mathbf{A}$ and $\mathbf{R}$ are row equivalent, therefore the $\mathbf{r}$ row vectors are obtained from the $\mathbf{a}$ row vectors by elementary row operations. This means that every $\mathbf{r}$ row vector is a linear combination of the $\mathbf{a}$ row vectors. Therefore the row space of matrix $\mathbf{A}$ lies in the row space of matrix $\mathbf{R}$.

2) Similarly, by considering the above argument the other way, we have that the row space of matrix $\mathbf{R}$ lies in the row space of matrix $\mathbf{A}$.

Hence the row space of matrices $\mathbf{A}$ and $\mathbf{R}$ are equal. ■

Remember, we can use elementary row operations to put a matrix into row echelon form. Additionally, we can apply this to find a basis for the row space of a matrix as the following proposition states:

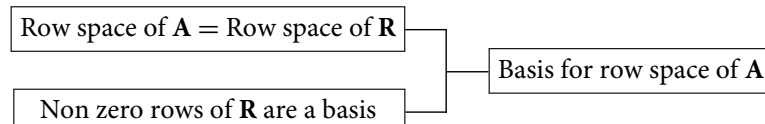
Proposition (3.26). If a matrix $\mathbf{R}$ is in row echelon form then its non-zero rows form a basis (set of axes) for the row space of matrix $\mathbf{R}$.

Proof – See Exercises 3.5. ■

By combining the above two propositions (3.25) and (3.26) we have:

Proposition (3.27). If matrix $\mathbf{A}$ is row equivalent to matrix $\mathbf{R}$ where matrix $\mathbf{R}$ is in row echelon form then the non-zero rows of matrix $\mathbf{R}$ form a basis for the row space of matrix $\mathbf{A}$.

Proof.



By the above Proposition (3.26) the basis of the row space of $\mathbf{R}$ is given by the non-zero rows of matrix $\mathbf{R}$. By the above Proposition (3.25) we have

Row space of $\mathbf{A}$ = Row space of $\mathbf{R}$ [because $\mathbf{R}$ and $\mathbf{A}$ are row equivalent]

Thus the non-zero rows of matrix $\mathbf{R}$ form a basis for the row space of matrix $\mathbf{A}$. ■



How can we use this proposition to find a basis for the row space of a given matrix?

By carrying out elementary row operations and placing the given matrix into row echelon form. We aim to achieve the following:

$$A = \begin{pmatrix} \mathbf{a}_1 \\ \vdots \\ \mathbf{a}_m \\ \mathbf{a}_{m+1} \\ \vdots \end{pmatrix} \longrightarrow R = \begin{pmatrix} \mathbf{r}_1 \\ \vdots \\ \mathbf{r}_m \\ \mathbf{0} \\ \vdots \end{pmatrix} \left. \vphantom{\begin{pmatrix} \mathbf{r}_1 \\ \vdots \\ \mathbf{r}_m \\ \mathbf{0} \\ \vdots \end{pmatrix}} \right\} \begin{array}{l} \text{These non-zero rows (vectors)} \\ \text{form a basis for the row} \\ \text{space of matrix } A. \end{array}$$

We can use MATLAB to place a matrix into row echelon form. Actually MATLAB and the command `rref` places a given matrix into *reduced* row echelon form. In the following examples we will use the reduced row echelon form but it is enough to find the row echelon form.

Example 3.28

Determine a basis for the row space of the following matrices:

(a) $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$

(b) $B = \begin{pmatrix} -1 & 3 & 9 \\ -5 & 2 & 6 \end{pmatrix}$

(c) $C = \begin{pmatrix} -3 & 6 \\ -5 & 2 \\ -2 & 7 \end{pmatrix}$

Solution

(a) Applying row operations we have

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \longrightarrow R = \begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix} \leftarrow \text{Non-zero row}$$

The vector $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ is a basis for the row space of A (Fig. 3.18).

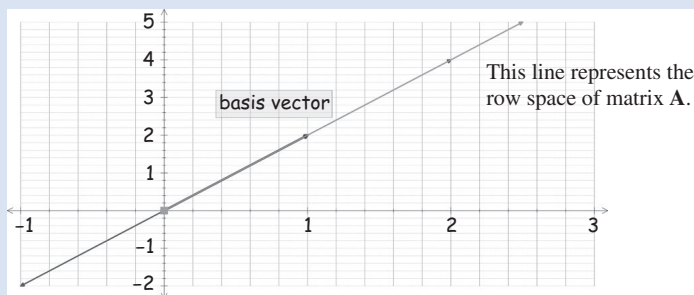


Figure 3.18

(b) Similarly, to find the reduced row echelon form of B we use MATLAB:

$$B = \begin{pmatrix} -1 & 3 & 9 \\ -5 & 2 & 6 \end{pmatrix} \longrightarrow R = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 3 \end{pmatrix}$$

(continued...)

What is a basis for the row space of matrix **B**?

The vectors $\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix} \right\}$ form a basis for the row space of matrix **B**. Hence the row space of **B** is a subspace of $\mathbb{R}^3$.

(c) By using MATLAB, the reduced row echelon form of matrix **C** is given by:

$$\mathbf{C} = \begin{pmatrix} -3 & 6 \\ -5 & 2 \\ -2 & 7 \end{pmatrix} \quad \longrightarrow \quad \mathbf{R} = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} \right\} \quad \text{Non-zero rows}$$

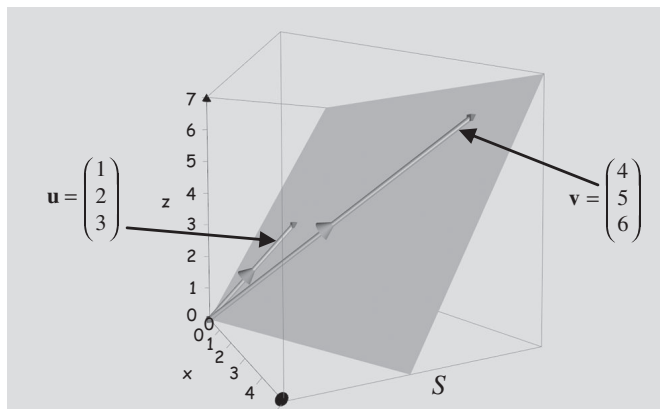
Remember, it is the non-zero rows which form a basis for the row space. Thus the vectors

$\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$ form a basis for the row space of matrix **C**. We need two vectors $\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$ to span the row space of **C** rather than the three given row vectors in matrix **C**. Remember, this basis spans the whole of $\mathbb{R}^2$, so the row space of matrix **C** is $\mathbb{R}^2$.

The above example can easily be carried out by hand calculations because of the simple integer entries for the given matrices.

3.5.4 Basis of a spanned subspace of $\mathbb{R}^n$

Let $\mathbf{u} = (1 \ 2 \ 3)^T$ and $\mathbf{v} = (4 \ 5 \ 6)^T$ be vectors in $\mathbb{R}^3$. Let S be the space spanned by these vectors which is illustrated in Fig. 3.19.



Vectors $\mathbf{u}$ and $\mathbf{v}$ span the dark shaded plane S in $\mathbb{R}^3$.

Figure 3.19

We are generally interested in finding a simple set of axes to describe this plane, or more formally, a basis for this space S .



How can we find such a basis?

Writing the vectors as rows of a matrix $A = \begin{pmatrix} \mathbf{u} \\ \mathbf{v} \end{pmatrix}$ then the *row space* of A is the vector space spanned by vectors $\mathbf{u}$ and $\mathbf{v}$. By using row operations on A , we can find a basis for the row space of a matrix A as we did in subsection 3.5.3 above.

The procedure of finding a basis for a subspace of $\mathbb{R}^n$ which is spanned by the vectors $\{\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3, \dots, \mathbf{r}_m\}$ is given by:

1. Form the matrix $A = \begin{pmatrix} \mathbf{r}_1 \\ \vdots \\ \mathbf{r}_m \end{pmatrix}$. The row space of A is the space spanned by $\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_m$.
2. Convert this matrix A into (reduced) row echelon form, R say.
3. The non-zero rows of matrix R form a basis for the vector space $\text{span}\{\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_m\}$.

Example 3.29

Determine a basis for the vector space $S = \text{span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ where

$$\mathbf{u} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \mathbf{v} = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix} \text{ and } \mathbf{w} = \begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix}$$

Solution

Let A be the matrix given by $A = \begin{pmatrix} \mathbf{u} \\ \mathbf{v} \\ \mathbf{w} \end{pmatrix}$, that is $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$. The row space of matrix A is the subspace spanned by vectors $\mathbf{u}, \mathbf{v}$ and $\mathbf{w}$. Thus

$$\text{span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\} = \text{Row space of } A$$

What do we need to find?

The basis for the row space of the matrix A .

How?

By finding the (reduced) row echelon form matrix R of A and then the non-zero rows of R form a basis (axes) for the row space of matrix A . By using MATLAB we obtain:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \quad \longrightarrow \quad \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix} = R$$

The non-zero rows of the above matrix R form a basis of the row space of A :

$$\mathbf{r}_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \text{ and } \mathbf{r}_2 = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$$

Thus a basis (axes) for the given vector space $S = \text{span}(\mathbf{u}, \mathbf{v}, \mathbf{w})$ is $B = \{\mathbf{r}_1, \mathbf{r}_2\}$ where $\mathbf{r}_1$ and $\mathbf{r}_2$ are vectors stated above. Note that S is a subspace of $\mathbb{R}^3$, but is actually a plane and can be spanned by two vectors, $\mathbf{r}_1$ and $\mathbf{r}_2$, rather than the three given vectors $\mathbf{u}, \mathbf{v}$ and $\mathbf{w}$.

Example 3.30

Determine a basis for the vector space $S = \text{span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ where

$$\mathbf{u} = \begin{pmatrix} 1 \\ 3 \\ 5 \\ 6 \end{pmatrix}, \mathbf{v} = \begin{pmatrix} 1 \\ 7 \\ 8 \\ 9 \end{pmatrix} \text{ and } \mathbf{w} = \begin{pmatrix} 3 \\ 9 \\ 15 \\ 18 \end{pmatrix}$$

Solution

By applying the above procedure we have:

Step 1 and 2: Writing the matrix $\mathbf{A}$ and evaluating its reduced row echelon form:

$$\mathbf{A} = \begin{pmatrix} \mathbf{u} \\ \mathbf{v} \\ \mathbf{w} \end{pmatrix} = \begin{pmatrix} 1 & 3 & 5 & 6 \\ 1 & 7 & 8 & 9 \\ 3 & 9 & 15 & 18 \end{pmatrix} \quad \longrightarrow \quad \mathbf{R} = \begin{pmatrix} 1 & 0 & 2.75 & 3.75 \\ 0 & 1 & 0.75 & 0.75 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$S = \text{span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\} = \text{Row space of matrix } \mathbf{A}.$$

Step 3:

What is a basis for $S = \text{span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$?

It is the non-zero rows of the reduced row echelon form matrix $\mathbf{R}$. We have

$$\mathbf{r}_1 = \begin{pmatrix} 1 \\ 0 \\ 2.75 \\ 3.75 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 11/4 \\ 15/4 \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 4 \\ 0 \\ 11 \\ 15 \end{pmatrix} \text{ and } \mathbf{r}_2 = \begin{pmatrix} 0 \\ 1 \\ 0.75 \\ 0.75 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 3/4 \\ 3/4 \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 0 \\ 4 \\ 3 \\ 3 \end{pmatrix}$$

Since the row space is the linear combination of vectors $\mathbf{r}_1$ and $\mathbf{r}_2$ so any non-zero scalar multiple of $\mathbf{r}_1$ and $\mathbf{r}_2$ is also a basis for the row space. (The basis gives the directions of the two axes for the space.) Hence, multiplying vectors $\mathbf{r}_1$ and $\mathbf{r}_2$ by 4 gives vectors:

$$\left\{ (4 \ 0 \ 11 \ 15)^T, (0 \ 4 \ 3 \ 3)^T \right\}$$

which form a basis for $S = \text{span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ and this is a subspace of $\mathbb{R}^4$. This S is a plane in $\mathbb{R}^4$ because we have two axes or two basis vectors for this space.

3.5.5 Rank of a matrix**Why is rank important in linear algebra?**

Consider the linear system $\mathbf{Ax} = \mathbf{b}$. The augmented matrix $(\mathbf{A} \mid \mathbf{b})$ in row echelon form may produce zero rows, which means $0 = 0$, but these are not important in the solution of linear equations. It is the number (rank) of the non-zero rows in row echelon form which gives the solution of a linear system of equations. We will discover in the next section that the rank of matrix $\mathbf{A}$ and of the augmented matrix $(\mathbf{A} \mid \mathbf{b})$ tell us if there are no, a unique or an infinite number of solutions.

The rank of a matrix gives the *number* of linearly *independent* rows in a matrix which means that all the rows that are linearly *dependent* are counted as one. For example, the following matrix has a rank of 1:

$$\begin{matrix} R_1 \\ R_2 \end{matrix} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \end{pmatrix} \text{ can be transformed to } \begin{matrix} R_1 \\ R_2 - 2R_1 \end{matrix} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

The second row is double the first, so carrying out row operations results in a single independent row. The rank of a matrix measures the amount of important information represented by the matrix.

An application of linear algebra is the transfer of digital data which is normally stored as a matrix. In these fields it is important that data is transferred as fast and efficiently as possible without losing any of it. The concept of a rank is critical here because a matrix with a lower rank takes up less memory and time to be transferred. Low rank matrices are much more efficient in the sense that they are much less computationally expensive to deal with.

Computer graphics rely on matrices to generate and manipulate images. The rank of the matrix tells you the dimension of the image. For example, the matrix $\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 4 & 5 & 6 \\ 2 & 2 & 2 \end{pmatrix}$ transforms a vector in 3D onto a 2D plane because matrix $\mathbf{A}$ does not have 'full rank' (the top and bottom rows are linearly dependent) as shown in Fig. 3.20.

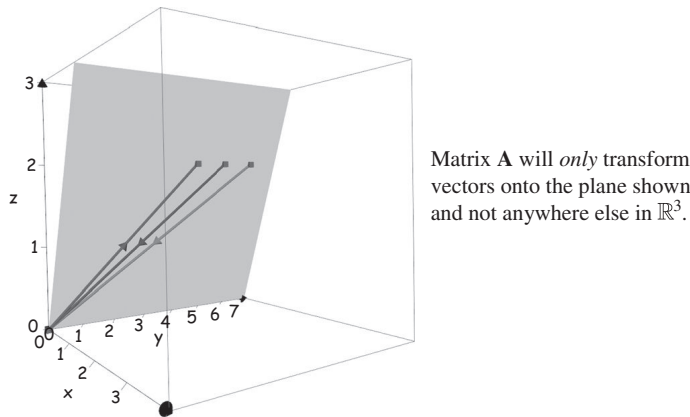


Figure 3.20

We define the rank in terms of dimension.



Can you recall what the term dimension of a vector space means?

It is the least number of axes needed to describe the vector space, or in other words, the number of vectors in the basis of a vector space.

The *dimension* of the *row space* of a matrix is called *row rank* and the *dimension* of the *column space* is called the *column rank*. Note that the row rank of a given matrix $\mathbf{A}$ is the *number* of non-zero row vectors in row echelon form matrix $\mathbf{R}$ because the non-zero rows form a basis for the row space.

$$\mathbf{A} = \begin{pmatrix} \mathbf{a}_1 \\ \vdots \\ \mathbf{a}_m \\ \mathbf{a}_{m+1} \\ \vdots \end{pmatrix} \longrightarrow \mathbf{R} = \left. \begin{pmatrix} \mathbf{r}_1 \\ \vdots \\ \mathbf{r}_m \\ \mathbf{0} \\ \vdots \end{pmatrix} \right\} m \text{ non-zero rows}$$

Row rank of matrix $\mathbf{A} = m$

The row rank of a matrix is called the **rank** of a matrix.

Definition (3.28). The **rank** of a matrix $\mathbf{A}$ is the row rank of $\mathbf{A}$.

The rank of matrix $\mathbf{A}$ is denoted by $\text{rank}(\mathbf{A})$. Thus (3.28) says

$$\text{rank}(\mathbf{A}) = \text{row rank of } \mathbf{A}$$



What is the difference between rank and dimension?

Strictly speaking, the rank is an attribute of a matrix, while dimension is an attribute of a vector space.

Example 3.31

Determine the rank of the matrices $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ given in Example 3.28.

Solution

What is $\text{rank}(\mathbf{A})$ equal to?

From the solutions to Example 3.28 we have one vector in the basis of the row space of $\mathbf{A}$, therefore the row rank is equal to 1 which means that $\text{rank}(\mathbf{A}) = 1$.

Similarly we have $\text{rank}(\mathbf{B}) = 2$ and $\text{rank}(\mathbf{C}) = 2$.

Example 3.32

Determine the rank of matrix $\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 27 & 28 & 29 & 30 & 31 & 32 \\ 15 & 16 & 17 & 18 & 19 & 20 \\ 31 & 32 & 33 & 34 & 35 & 36 \\ 45 & 46 & 47 & 48 & 49 & 50 \end{pmatrix}$.

Solution

What do we need to do?

Place the given matrix $\mathbf{A}$ into row echelon form. By using MATLAB, we can find the reduced row echelon form matrix $\mathbf{R}$:

$$\mathbf{R} = \begin{pmatrix} \boxed{1} & 0 & -1 & -2 & -3 & -4 \\ 0 & \boxed{1} & 2 & 3 & 4 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad \begin{array}{l} \leftarrow \text{Non-zero row} \\ \leftarrow \text{Non-zero row} \end{array}$$

What is the rank of the given matrix $\mathbf{A}$ equal to?

$\text{rank}(\mathbf{A}) = 2$ because we have two non-zero rows. In this case the rank tells you that two rows are important and three rows are redundant because the row space of matrix $\mathbf{A}$ can be spanned by the first two rows of matrix $\mathbf{R}$.

If matrix $\mathbf{A}$ represents a digital image then we can compress this data to a matrix with rank 2, which is more computationally efficient than matrix $\mathbf{A}$.

Proposition (3.29). Let $\mathbf{A}$ be any matrix. Then

$$\text{Row rank of matrix } \mathbf{A} = \text{Column rank of matrix } \mathbf{A}$$

This means

$$\text{rank}(\mathbf{A}) = \text{Row rank of } \mathbf{A} = \text{Column rank of } \mathbf{A}$$

Proof – Exercises 3.5. ■

The row and column rank of a matrix are equal.

We can also find a basis for the column space of a matrix by considering the columns of the reduced row echelon form matrix with leading 1's. In the above Example 3.32 the first two columns of matrix $\mathbf{R}$ have leading 1's. It can shown that the corresponding column vectors of matrix $\mathbf{A}$ form a basis for the column space of $\mathbf{A}$. Hence the following vectors are a basis:

$$\begin{pmatrix} 1 & 27 & 15 & 31 & 45 \end{pmatrix}^T \text{ and } \begin{pmatrix} 2 & 28 & 16 & 32 & 46 \end{pmatrix}^T$$

for the column space of matrix $\mathbf{A}$.

3.5.6 Rank and invertible matrices

In this subsection we discuss the relationships between rank and invertible matrices.

Proposition (3.30). Let $\mathbf{A}$ be an n by n matrix. The matrix $\mathbf{A}$ is invertible $\Leftrightarrow \text{rank}(\mathbf{A}) = n$.



What does this mean?

Matrix is invertible $\Leftrightarrow$ it has no redundant rows. We say that the matrix $\mathbf{A}$ has **full rank**.

Proof.

($\Rightarrow$). We assume matrix $\mathbf{A}$ is invertible. By (1.35):

Theorem (1.35). Let $\mathbf{A}$ be a n by n matrix, then the following are equivalent:

- (a) The matrix $\mathbf{A}$ is invertible.
- (b) The reduced row echelon form of the matrix $\mathbf{A}$ is the identity matrix $\mathbf{I}$.

The reduced row echelon form of the matrix $\mathbf{A}$ is the identity n by n matrix $\mathbf{I}$. Thus there are n non-zero rows of $\mathbf{I}$, therefore $\text{rank}(\mathbf{A}) = n$.

($\Leftarrow$). In this case, we assume that $\text{rank}(\mathbf{A}) = n$ and we need to prove that the matrix $\mathbf{A}$ is invertible. Since $\text{rank}(\mathbf{A}) = n$, the reduced row echelon form of $\mathbf{A}$ has no zero rows. By (1.39);

Proposition (1.39). $\mathbf{R}$ has at least one row of zeros $\Leftrightarrow \mathbf{A}$ is non-invertible (singular).

This means that matrix $\mathbf{A}$ must be invertible. ■

Hence if a square matrix is not of full rank then it is non-invertible.

**Summary**

(3.23) The *row space* of a matrix **A** is the vector space spanned by the row vectors of matrix **A**.

(3.25). If matrices **A** and **R** are row equivalent then their row spaces are equal.

(3.27). A basis for the row space of matrix **A** is the non-zero rows of the equivalent row echelon form matrix **R**.

The rank of a matrix is the *dimension* of the row or column space of the matrix.

**EXERCISES 3.5**

(Brief solutions at end of book. Full solutions available at <http://www.oup.co.uk/companion/singh>.)

1. For the following matrices find the row and column vectors:

$$(a) \mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \quad (b) \mathbf{B} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{pmatrix} \quad (c) \mathbf{C} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}$$

$$(d) \mathbf{D} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \quad (e) \mathbf{E} = \begin{pmatrix} -1 & 2 & 5 \\ -3 & 7 & 0 \\ -8 & 1 & 3 \end{pmatrix} \quad (f) \mathbf{F} = \begin{pmatrix} -5 & 2 & 3 \\ 7 & 1 & 0 \\ -7 & 6 & 1 \\ -2 & 5 & 2 \end{pmatrix}$$

2. Determine a basis for the row space of the matrices given in question 1. Also state the rank of the given matrix.

3. Determine a basis for the following subspace of $\mathbb{R}^n$ which are spanned by the vectors:

$$(a) \mathbf{u} = \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix}, \mathbf{v} = \begin{pmatrix} 7 \\ 2 \end{pmatrix} \quad (b) \mathbf{u} = \begin{pmatrix} -1 \\ -3 \end{pmatrix}, \mathbf{v} = \begin{pmatrix} 4 \\ 12 \end{pmatrix}$$

$$(c) \mathbf{u} = \begin{pmatrix} 3 \\ 6 \\ 5 \end{pmatrix}, \mathbf{v} = \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix}, \mathbf{w} = \begin{pmatrix} 12 \\ 15 \\ 15 \end{pmatrix}$$

$$(d) \mathbf{u} = \begin{pmatrix} -1 \\ -2 \\ -5 \end{pmatrix}, \mathbf{v} = \begin{pmatrix} 0 \\ 1 \\ 5 \end{pmatrix}, \mathbf{w} = \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix}, \mathbf{x} = \begin{pmatrix} -4 \\ 1 \\ -7 \end{pmatrix}$$

$$(e) \mathbf{u} = \begin{pmatrix} 1 \\ 2 \\ 2 \\ 2 \end{pmatrix}, \mathbf{v} = \begin{pmatrix} -1 \\ 3 \\ 5 \\ 7 \end{pmatrix}, \mathbf{w} = \begin{pmatrix} 2 \\ -1 \\ -3 \\ -5 \end{pmatrix}, \mathbf{x} = \begin{pmatrix} 0 \\ 5 \\ 7 \\ 9 \end{pmatrix}$$

4. Find a basis for the subspace of $\mathbb{R}^4$ which is given by $S = \text{span}(\mathbf{u}, \mathbf{v}, \mathbf{w}, \mathbf{x}, \mathbf{y})$ where

$$\mathbf{u} = \begin{pmatrix} -1 \\ 1 \\ -1 \\ 1 \end{pmatrix}, \mathbf{v} = \begin{pmatrix} -5 \\ -9 \\ -7 \\ -1 \end{pmatrix}, \mathbf{w} = \begin{pmatrix} 0 \\ 7 \\ 1 \\ 3 \end{pmatrix}, \mathbf{x} = \begin{pmatrix} -6 \\ -1 \\ -7 \\ 3 \end{pmatrix}, \mathbf{y} = \begin{pmatrix} 2 \\ 5 \\ 3 \\ 1 \end{pmatrix}.$$

[You can convert the decimal entries of the reduced row echelon form matrix into fractional entries by using the command `rats(rref(A))` in MATLAB. In MATLAB, the `rats()` command converts the argument into a rational number.]

5. Determine a basis for the column space of the matrices given in question 1 by taking the transpose of the given matrices. Also state the rank of the given matrix.
6. Let $\mathbf{A}$ be any matrix. Prove that $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}^T)$.
7. Let $\mathbf{A}$ be an invertible matrix. Prove that $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}^{-1})$.
8. Let $\mathbf{A}$ be a square n by n matrix. Prove the following:
 - (a) $\mathbf{A}$ has rank $n \Leftrightarrow$ the linear system $\mathbf{A}\mathbf{x} = \mathbf{0}$ has the trivial solution $\mathbf{x} = \mathbf{0}$.
 - (b) $\mathbf{A}$ has rank $n \Leftrightarrow$ the linear system $\mathbf{A}\mathbf{x} = \mathbf{b}$ has a unique solution.
9. Let $\mathbf{A}$ be a square n by n matrix whose row vectors are given by the set of vectors, $S = \{\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3, \dots, \mathbf{r}_n\}$:

$$\mathbf{A} = \begin{pmatrix} \mathbf{r}_1 \\ \vdots \\ \mathbf{r}_n \end{pmatrix}$$

Prove that matrix $\mathbf{A}$ is invertible $\Leftrightarrow S$ is a set of linearly independent vectors.

10. Let $\mathbf{A}$ be a square n by n matrix whose column vectors are given by the set of vectors; $S = \{\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3, \dots, \mathbf{c}_n\}$:

$$\mathbf{A} = (\mathbf{c}_1 \cdots \mathbf{c}_n)$$

Prove that matrix $\mathbf{A}$ is invertible $\Leftrightarrow S$ is a set of linearly independent vectors.

11. Prove that the row space of a matrix $\mathbf{A}$ is identical to the row space of $k\mathbf{A}$, where k is a non-zero scalar.
12. Let $\mathbf{A}$ be any matrix and k be a non-zero scalar. Show that $\text{rank}(k\mathbf{A}) = \text{rank}(\mathbf{A})$.
13. Prove Proposition (3.24).
14. Prove that if $\mathbf{R}$ is the reduced row echelon matrix then the non-zero rows of $\mathbf{R}$ are linearly independent.
15. Prove that if $\mathbf{A}$ is a matrix and the reduced row echelon form matrix $\mathbf{R}$ of $\mathbf{A}$ contains zero rows then the rows of $\mathbf{A}$ are linearly dependent.
16. Prove Proposition (3.26).
17. Let $\mathbf{A}$ be any matrix whose rows are given by the set of linear independent vectors $S = \{\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3, \dots, \mathbf{r}_n\}$. Prove that $\text{rank}(\mathbf{A}) = n$.
18. For the following you may assume:
The columns of matrix $\mathbf{A}$ with the corresponding leading 1's in reduced row echelon form matrix $\mathbf{R}$ are a basis for the column space. Prove Proposition (3.29).

SECTION 3.6 Linear Systems Revisited

By the end of this section you will be able to

- understand and determine the null space and nullity of a matrix
- prove some properties of null space, nullity and rank of a matrix
- determine solutions to a non-homogeneous system of linear equations

In the last section we concentrated on the row space of a matrix, but in this section we examine the part played by the *column space* of a matrix in the solution of the linear system $\mathbf{Ax} = \mathbf{b}$.

In this section we answer the critical question of linear algebra:



What conditions provide infinite, unique or no solutions to a linear system?

You will need to ensure you are familiar with the concepts of row space, column space and the rank of a matrix to analyse solutions of linear systems.

3.6.1 Null space

We consider the homogeneous system first, which is a linear system of equations written in matrix form as $\mathbf{Ax} = \mathbf{O}$. By using elementary row operations we solve the equivalent system $\mathbf{Rx} = \mathbf{O}$ where matrix $\mathbf{R}$ is row equivalent to matrix $\mathbf{A}$.

Example 3.33

Solve the homogeneous system of linear equations:

$$x + 2y + 3z = 0$$

$$4x + 5y + 6z = 0$$

$$7x + 8y + 9z = 0$$

Solution

We have the linear system $\mathbf{Ax} = \mathbf{O}$ where $\mathbf{A}$ is the coefficient matrix, $\mathbf{x}$ is the vector of unknowns and $\mathbf{O}$ is the zero vector.

How can we find the unknowns x , y and z ?

We can write out the augmented matrix and then convert this into reduced row echelon form by using hand calculations or MATLAB with command `rref(A)`:

$$(\mathbf{A} | \mathbf{O}) = \left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 4 & 5 & 6 & 0 \\ 7 & 8 & 9 & 0 \end{array} \right) \quad \longrightarrow \quad \begin{array}{ccc} x & y & z \\ \left(\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right) \end{array}$$

From the middle row of the matrix on the right hand side we have

$$y + 2z = 0 \text{ which gives } y = -2z$$

None of the rows start with z so z is a free variable.

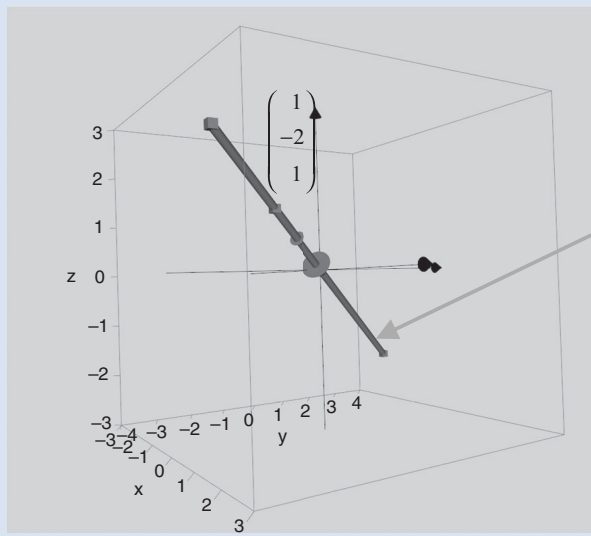
Let $z = s$ where s is any real number then $y = -2s$ and from the top row we have

$$x - z = 0 \text{ which gives } x = z = s$$

Our solution set is $x = s$, $y = -2s$ and $z = s$ which we can write in vector form as

$$\mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} s \\ -2s \\ s \end{pmatrix} = s \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \text{ where } s \text{ is a parameter}$$

This *free variable* s is any real number and the solution is the set of *all* points on the axis (thick line) as shown in Fig. 3.21:



All the points along this axis are solutions to the given linear system $\mathbf{Ax} = \mathbf{0}$.

Figure 3.21

The solution $\mathbf{x} = s \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$ is a scalar multiple of the vector $\mathbf{u} = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$, so this vector $\mathbf{u}$ is a basis vector on the axis shown for the solution space. Note that this solution space is a subspace of $\mathbb{R}^3$.

The vector or solution space, call it N , of a homogeneous system $\mathbf{Ax} = \mathbf{0}$ is called the **null space** of matrix $\mathbf{A}$. In general, we have:

Definition (3.31). Let $\mathbf{A}$ be any matrix. The set of vectors $\mathbf{x}$ in N (or $N(\mathbf{A})$) which satisfy the homogeneous system $\mathbf{Ax} = \mathbf{0}$ is called the **null space** of matrix $\mathbf{A}$. The dimension of this null space is called the **nullity** of matrix $\mathbf{A}$ and is denoted by $\text{nullity}(\mathbf{A})$.

For the above Example 3.33 the null space is the set of vectors $N = \{s\mathbf{u} \mid s \in \mathbb{R}\}$ which is shown as an axis in Fig. 3.21. In this case, $\text{nullity}(\mathbf{A}) = 1$ because we have one axis or a single vector $\mathbf{u}$, which is a basis for N . Hence the null space of matrix $\mathbf{A}$ is of dimension 1.

? How can we write the general solution to $\mathbf{Ax} = \mathbf{0}$?

$$\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

Multiplying these matrices we have:

$$\begin{array}{ccccccc} a_{11}x_1 & + & \cdots & + & a_{1n}x_n & = & 0 \\ \vdots & & & & \vdots & & \vdots \\ a_{m1}x_1 & + & \cdots & + & a_{mn}x_n & = & 0 \end{array}$$

We can write this in terms of the column vectors $\mathbf{c}_1, \mathbf{c}_2, \dots$ and $\mathbf{c}_n$ of matrix $\mathbf{A}$:

$$\begin{pmatrix} \mathbf{c}_1 \\ a_{11} \\ \vdots \\ a_{m1} \end{pmatrix} x_1 + \cdots + \begin{pmatrix} \mathbf{c}_n \\ a_{1n} \\ \vdots \\ a_{mn} \end{pmatrix} x_n = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix} \quad (*)$$

Note that the left hand side is a linear combination of the column vectors $\mathbf{c}_1, \mathbf{c}_2, \dots$ and $\mathbf{c}_n$ of matrix $\mathbf{A}$. Recall that this linear combination is the *column space* of the matrix $\mathbf{A}$. The null space consists of vectors $\mathbf{x} = (x_1 \ x_2 \ \cdots \ x_n)^T$ such that they satisfy the linear combination (*).

The null space is a non-empty set.

? How do we know it is non-empty?

Because the homogeneous system $\mathbf{Ax} = \mathbf{0}$ always has the trivial solution $\mathbf{x} = \mathbf{0}$ ($x_1 = \cdots = x_n = 0$) so we know the null space of matrix $\mathbf{A}$ is *not* empty.

Proposition (3.32). If $\mathbf{A}$ is a matrix with n columns then the null space $N(\mathbf{A})$ is a subspace of $\mathbb{R}^n$.

Proof – Exercises 3.6. ■

The null space of a matrix is a vector space.

Example 3.34

Determine the null space, nullity and rank of the matrix $\mathbf{B} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{pmatrix}$.

Solution

How do we find the null space of the given matrix $\mathbf{B}$?

The null space is the set of vectors $\mathbf{x}$ which satisfies the homogeneous system $\mathbf{Bx} = \mathbf{0}$:

$$(\mathbf{B} | \mathbf{0}) = \left(\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 0 \\ 5 & 6 & 7 & 8 & 0 \end{array} \right) \longrightarrow \left(\begin{array}{cccc|c} x & y & z & w & \\ 1 & 0 & -1 & -2 & 0 \\ 0 & 1 & 2 & 3 & 0 \end{array} \right) = (\mathbf{R} | \mathbf{0})$$

The matrix on the right hand side $\mathbf{R}$ is actually the reduced row echelon form of matrix $\mathbf{B}$.
Expanding out the bottom row gives:

$$y + 2z + 3w = 0 \text{ yields } y = -2z - 3w$$

None of the rows begin with z and w , so these are our free variables. Let $z = s$ and $w = t$ where t and s are any real numbers. We have $y = -2z - 3w = -2s - 3t$.

Expanding the top row we have

$$x - z - 2w = 0 \quad \text{gives} \quad x = z + 2w$$

Substituting $z = s$ and $w = t$ into $x = z + 2w$ gives $x = s + 2t$. The solution of $\mathbf{B}\mathbf{x} = \mathbf{0}$ is the null space of matrix $\mathbf{B}$ given by the set of vectors $\mathbf{x} = (x \ y \ z \ w)^T$ which has the entries $x = s + 2t$, $y = -2s - 3t$, $z = s$ and $w = t$:

$$\mathbf{x} = \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = \begin{pmatrix} s + 2t \\ -2s - 3t \\ s \\ t \end{pmatrix} = s \begin{pmatrix} 1 \\ -2 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 2 \\ -3 \\ 0 \\ 1 \end{pmatrix}$$

Let $\mathbf{u} = (1 \ -2 \ 1 \ 0)^T$ and $\mathbf{v} = (2 \ -3 \ 0 \ 1)^T$ then we can write the null space N as

$$N = \{s\mathbf{u} + t\mathbf{v} \mid t \in \mathbb{R}, s \in \mathbb{R}\}$$

Substituting $s = t = 0$ into $s\mathbf{u} + t\mathbf{v}$ gives the zero vector $\mathbf{0}$, which of course is in the null space N . We can substitute any real numbers for s and t to obtain an infinite number of vectors in N .

Note that N is a subspace of $\mathbb{R}^4$ because vectors $\mathbf{u}$ and $\mathbf{v}$ have four real entries.

What can we say about the vectors $\mathbf{u}$ and $\mathbf{v}$?

The vectors $\mathbf{u}$ and $\mathbf{v}$ span the null space N and they are also linearly independent.

How do we know these vectors are linearly independent?

Because $\mathbf{u}$ and $\mathbf{v}$ are not multiples of each other. This means that vectors $\mathbf{u}$ and $\mathbf{v}$ form a basis for the null space N . (Actually the null space N is a plane in $\mathbb{R}^4$ because we have two axes or basis vectors, $\mathbf{u}$ and $\mathbf{v}$.)

What is the nullity of matrix $\mathbf{B}$ equal to?

$$\text{nullity}(\mathbf{B}) = 2$$

This means the set of vectors $\mathbf{x}$ which satisfy $\mathbf{B}\mathbf{x} = \mathbf{0}$ is of dimension 2. Hence we require two axes or basis vectors to describe the null space of matrix $\mathbf{B}$.

What is the rank of the given matrix $\mathbf{B}$ equal to?

The above reduced row echelon form matrix $\mathbf{R}$ has two non-zero rows, therefore

$$\text{rank}(\mathbf{B}) = 2$$

Using the procedure outlined in this Example 3.34 gives us a basis for the null space.

Example 3.35

Determine the null space, nullity and rank of the matrix $\mathbf{C} = \begin{pmatrix} 1 & 3 \\ 5 & 15 \end{pmatrix}$.

(continued...)

Solution

The null space N is the set of vectors $\mathbf{x}$ which satisfy $\mathbf{C}\mathbf{x} = \mathbf{O}$.

How do we find this vector space?

By applying elementary row operations to convert into reduced row echelon form:

$$(\mathbf{C} | \mathbf{O}) = \left(\begin{array}{cc|c} 1 & 3 & 0 \\ 5 & 15 & 0 \end{array} \right) \quad \longrightarrow \quad (\mathbf{R} | \mathbf{O}) = \left(\begin{array}{cc|c} 1 & 3 & 0 \\ 0 & 0 & 0 \end{array} \right) \text{ non-zero row}$$

We only have *one* non-zero row in the reduced row echelon form matrix $\mathbf{R}$, therefore

$$\text{rank}(\mathbf{C}) = 1$$

Considering the equivalent homogeneous system $\mathbf{R}\mathbf{x} = \mathbf{O}$ we have

$$x + 3y = 0 \text{ implies } x = -3y$$

Let $y = s$ then $x = -3y = -3s$ where s is any real number. Hence

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -3s \\ s \end{pmatrix} = s \begin{pmatrix} -3 \\ 1 \end{pmatrix}$$

Let $\mathbf{u} = \begin{pmatrix} -3 \\ 1 \end{pmatrix}$ then the null space $N = \{s\mathbf{u} \mid s \in \mathbb{R}\}$ and the graph of this space is:

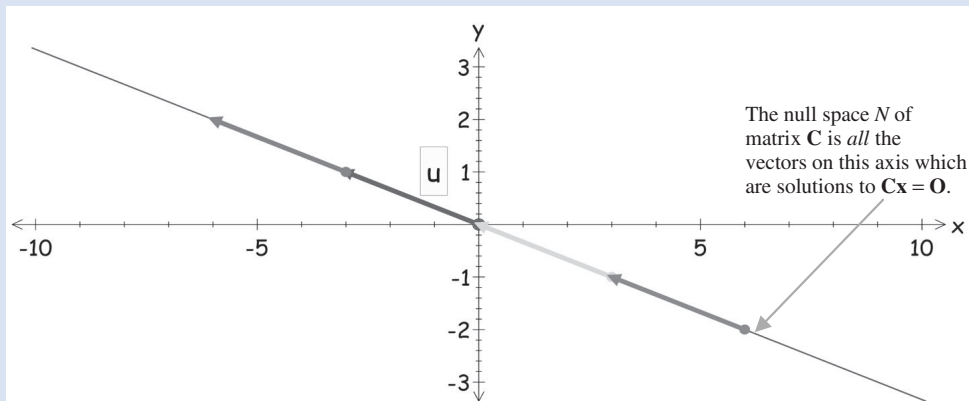


Figure 3.22

Note that the vector $\mathbf{u}$ spans the null space and is linearly independent so it is a basis for N .

What is the nullity of matrix $\mathbf{C}$ equal to?

It is the dimension of the null space N , which is 1, because it has only one basis vector $\mathbf{u}$ (or one free variable – s). We have $\text{nullity}(\mathbf{C}) = 1$.

We can also evaluate the nullity of a matrix by finding the number of free variables in the system.

Proposition (3.33). The number of free variables in the system $\mathbf{A}\mathbf{x} = \mathbf{O}$ is equal to the nullity of the matrix $\mathbf{A}$.

Proof – Exercises 3.6. ■

3.6.2 Properties of rank and nullity

In the above examples 3.33, 3.34 and 3.35 we had

$$\text{nullity}(\mathbf{A}) + \text{rank}(\mathbf{A}) = 1 + 2 = 3$$

$$\text{nullity}(\mathbf{B}) + \text{rank}(\mathbf{B}) = 2 + 2 = 4$$

$$\text{nullity}(\mathbf{C}) + \text{rank}(\mathbf{C}) = 1 + 1 = 2$$



Can you see any relationship between the nullity, rank and the number of unknowns in vector $\mathbf{x}$?

$$\text{Nullity} + \text{Rank} = \text{Number of unknowns}$$

In general, if we have a m by n matrix $\mathbf{A}$:

$$\underbrace{\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}}_{n \text{ columns}} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix} \quad [\mathbf{Ax} = \mathbf{O}]$$

Then

$$\text{nullity}(\mathbf{A}) + \text{rank}(\mathbf{A}) = n$$

Note that n is the number of columns of matrix $\mathbf{A}$ which is the total number of unknowns in the homogeneous system $\mathbf{Ax} = \mathbf{O}$. This result normally has the grand title of ‘The Dimension Theorem of Matrices’.

Theorem (3.34). The dimension theorem of matrices (rank-nullity theorem):

If $\mathbf{A}$ is a matrix with n columns (number of unknowns) then

$$\text{nullity}(\mathbf{A}) + \text{rank}(\mathbf{A}) = n$$

Proof.

Let $\mathbf{R}$ be the reduced row echelon form of matrix $\mathbf{A}$ and let it have rank r , which means that it has r non-zero rows located at the top of the matrix. By the definition of reduced row echelon form, we know that each of these r rows has a leading 1.

$$\mathbf{R} = \left(\begin{array}{cccccc} \boxed{1} & & & & & \\ & \ddots & & & & \\ & & & \boxed{1} & \cdots & \\ 0 & 0 & 0 & \cdots & \cdots & 0 \\ 0 & 0 & 0 & \cdots & \cdots & 0 \end{array} \right) \left. \begin{array}{l} \leftarrow \\ \vdots \\ \leftarrow \end{array} \right\} r \text{ non-zero rows}$$

n unknowns

There are n unknowns in total and r of these can be expressed in terms of the remaining $n - r$ unknowns. This means there are $n - r$ free variables in the system. By the above Proposition (3.33):

The number of free variables in the system $\mathbf{Ax} = \mathbf{O}$ is equal to the nullity of the matrix $\mathbf{A}$.

We have $\text{nullity}(\mathbf{A}) = n - r$ and transposing this gives us our result

$$\text{nullity}(\mathbf{A}) + r = n \quad \text{which is} \quad \text{nullity}(\mathbf{A}) + \text{rank}(\mathbf{A}) = n$$

 What is the nullity of a matrix $\mathbf{A}$ which is of full rank?

Full rank means $\text{rank}(\mathbf{A}) = n$, therefore

$$\text{nullity}(\mathbf{A}) = n - \text{rank}(\mathbf{A}) = n - n = 0$$

Hence the dimension of the null space is 0, so null space only contains the trivial solution – zero vector, $\mathbf{x} = \mathbf{O}$. In case of full rank, $\mathbf{Ax} = \mathbf{O} \Rightarrow \mathbf{x} = \mathbf{O}$ is the only solution.

Example 3.36

Determine the nullity and rank of the matrix:

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 8 & 9 & 10 & 11 & 12 & 13 & 14 \\ 15 & 16 & 17 & 18 & 19 & 20 & 21 \\ 22 & 23 & 24 & 25 & 26 & 27 & 28 \\ 29 & 30 & 31 & 32 & 33 & 34 & 35 \end{pmatrix}$$

Solution

By using MATLAB, the reduced row echelon form matrix $\mathbf{R}$ of $\mathbf{A}$ is given by

$$\mathbf{R} = \begin{pmatrix} 1 & 0 & -1 & -2 & -3 & -4 & -5 \\ 0 & 1 & 2 & 3 & 4 & 5 & 6 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad \begin{array}{l} \leftarrow \text{non-zero row} \\ \leftarrow \text{non-zero row} \end{array}$$

What is $\text{rank}(\mathbf{A})$ equal to?

Since there are two non-zero rows in matrix $\mathbf{R}$, therefore $\text{rank}(\mathbf{A}) = 2$. Note that the matrix has five rows but there are only two linearly independent rows.

What is $\text{nullity}(\mathbf{A})$ equal to?

$\text{nullity}(\mathbf{A})$ is the dimension of the null space and can be evaluated by:

Theorem (3.34) $\text{nullity}(\mathbf{A}) + \text{rank}(\mathbf{A}) = n$

Substituting $\text{rank}(\mathbf{A}) = 2$ and $n = 7$ (the number of columns of matrix $\mathbf{A}$) we have

$$\text{nullity}(\mathbf{A}) + 2 = 7 \text{ gives } \text{nullity}(\mathbf{A}) = 5$$

This means that the set of vectors $\mathbf{x}$ which satisfy $\mathbf{Ax} = \mathbf{O}$ is of dimension five. We need five axes or five basis vectors to describe the vectors $\mathbf{x}$ in the null space of matrix $\mathbf{A}$.

Sometimes it is more convenient to write $\text{nullity}(\mathbf{A}) + \text{rank}(\mathbf{A}) = n$ as

$$\text{nullity}(\mathbf{A}) = n - \text{rank}(\mathbf{A})$$

3.6.3 Non-homogeneous linear systems

Now we consider linear equations of the form $\mathbf{Ax} = \mathbf{b}$ where $\mathbf{b} \neq \mathbf{0}$ (not zero). Equations of the form $\mathbf{Ax} = \mathbf{b}$, where $\mathbf{b} \neq \mathbf{0}$, are called **non-homogeneous** linear equations. Throughout the remaining part of this section we assume $\mathbf{b} \neq \mathbf{0}$.

Example 3.37

Solve the above Example 3.35 with $\mathbf{Cx} = \mathbf{b}$ where $\mathbf{b} = (7 \ 35)^T$:

$$\begin{aligned} x + 3y &= 7 \\ 5x + 15y &= 35 \end{aligned}$$

Solution

We first write out the augmented matrix and then evaluate the reduced row echelon form:

$$\left(\begin{array}{cc|c} 1 & 3 & 7 \\ 5 & 15 & 35 \end{array} \right) \longrightarrow \left(\begin{array}{cc|c} 1 & 3 & 7 \\ 0 & 0 & 0 \end{array} \right)$$

Expanding the top row of this right hand matrix we have

$$x + 3y = 7 \text{ which gives } x = 7 - 3y$$

Let $y = s$, then we have $x = 7 - 3s = 7 - 3s$.

We have $x = 7 - 3s$ and $y = s$. The general solution $\mathbf{x}$ is

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 7 - 3s \\ s \end{pmatrix} = s \begin{pmatrix} -3 \\ 1 \end{pmatrix} + \begin{pmatrix} 7 \\ 0 \end{pmatrix}$$

The homogeneous solution $\mathbf{x}_H$ to $\mathbf{Cx} = \mathbf{0}$ in Example 3.35 was $\mathbf{x}_H = s(-3 \ 1)^T$.

What do you notice?

Solving this non-homogeneous system $\mathbf{Cx} = \mathbf{b}$ where $\mathbf{b} = (7 \ 35)^T$ gives a two-part solution:

the homogeneous solution; $\mathbf{x}_H = s(-3 \ 1)^T$ plus an extra term, $(7 \ 0)^T$, called the particular solution, which we denote by $\mathbf{x}_P$.

$\mathbf{x}_H = s \begin{pmatrix} -3 \\ 1 \end{pmatrix}$ gives us the slope $-\frac{1}{3}$, and $\begin{pmatrix} 7 \\ 0 \end{pmatrix}$ moves the line horizontally by seven units to the solution $\mathbf{x}$. We demonstrate this in Fig. 3.23.

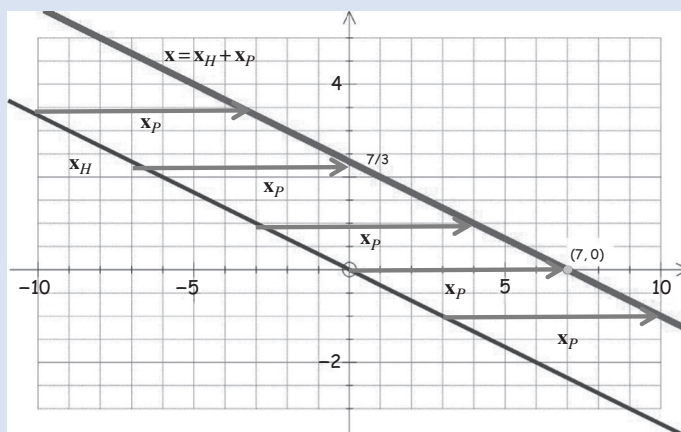


Figure 3.23

(continued...)

Hence our general solution to $\mathbf{C}\mathbf{x} = \mathbf{b}$ is $\mathbf{x} = \mathbf{x}_H + \mathbf{x}_P$ where

$$\mathbf{x}_H = s(-3 \ 1)^T \quad \text{and} \quad \mathbf{x}_P = (7 \ 0)^T$$

Notice that in the above example we have

$$\mathbf{C}\mathbf{x}_H = \mathbf{0} \text{ and } \mathbf{C}\mathbf{x}_P = \mathbf{b}$$

because $\mathbf{x}_H$ is the homogeneous solution and $\mathbf{x}_P$ is the particular solution. Combining these to solve the non-homogeneous system $\mathbf{C}\mathbf{x} = \mathbf{b}$ we have

$$\begin{aligned} \mathbf{C}\mathbf{x}_H + \mathbf{C}\mathbf{x}_P &= \mathbf{0} + \mathbf{b} \\ \mathbf{C}(\mathbf{x}_H + \mathbf{x}_P) &= \mathbf{b} \quad \Rightarrow \quad \mathbf{x} = \mathbf{x}_H + \mathbf{x}_P \end{aligned}$$

Hence the general solution of $\mathbf{C}\mathbf{x} = \mathbf{b}$ is given by:

$$\mathbf{x} = (\text{homogeneous solution}) + (\text{particular solution})$$

In general, for non-homogeneous systems we have the following:

Proposition (3.35). Let $\mathbf{x}_P$ be the particular solution of $\mathbf{A}\mathbf{x} = \mathbf{b}$ and $\mathbf{x}_H$ be the solution to the homogeneous system $\mathbf{A}\mathbf{x} = \mathbf{0}$. All the solutions of $\mathbf{A}\mathbf{x} = \mathbf{b}$ are of the form $\mathbf{x}_P + \mathbf{x}_H$.



What does this proposition mean?

The solution of $\mathbf{A}\mathbf{x} = \mathbf{b}$ consists of two parts:

$$(\text{homogeneous solution}) + (\text{particular solution})$$

Remember, the homogeneous solution $\mathbf{x}_H$ (vector) belongs to the null space of matrix $\mathbf{A}$.

Proof.

Let $\mathbf{x}$ be the solution of $\mathbf{A}\mathbf{x} = \mathbf{b}$ then

$$\begin{aligned} \mathbf{A}(\mathbf{x} - \mathbf{x}_P) &= \mathbf{A}\mathbf{x} - \mathbf{A}\mathbf{x}_P \\ &= \mathbf{b} - \mathbf{b} = \mathbf{0} \end{aligned}$$

Since we have $\mathbf{A}(\mathbf{x} - \mathbf{x}_P) = \mathbf{0}$, therefore $\mathbf{x} - \mathbf{x}_P$ is the homogeneous solution, that is $\mathbf{x}_H = \mathbf{x} - \mathbf{x}_P$. Hence we have our result $\mathbf{x} = \mathbf{x}_P + \mathbf{x}_H$.

We need to show *all* the solutions are of this format $\mathbf{x}_P + \mathbf{x}_H$.

Let $\mathbf{x}'$ be a solution of $\mathbf{A}\mathbf{x} = \mathbf{0}$, then

$$\mathbf{A}(\mathbf{x} + \mathbf{x}') = \mathbf{A}\mathbf{x} + \mathbf{A}\mathbf{x}' = \mathbf{A}\mathbf{x} + \mathbf{0} = \mathbf{b} + \mathbf{0} = \mathbf{b}$$

Hence $\mathbf{x} + \mathbf{x}'$ is a solution of $\mathbf{A}\mathbf{x} = \mathbf{b}$.

We conclude that *all* the solutions are of this form $\mathbf{x} = \mathbf{x}_P + \mathbf{x}_H$. ■

Example 3.38

Solve the non-homogeneous linear system:

$$\begin{aligned} x - 2y + 2z + 0.03t &= 1.7 \\ -y + z - 0.02t &= -1.6 \\ x - y & \quad 0.01t = 0 \end{aligned}$$

Solution

Writing the augmented matrix and evaluating the reduced row echelon form gives

$$\left(\begin{array}{cccc|c} 1 & -2 & 2 & 0.03 & 1.7 \\ 0 & -1 & 1 & -0.02 & -1.6 \\ 1 & -1 & 0 & 0.01 & 0 \end{array} \right) \longrightarrow \left(\begin{array}{cccc|c} x & y & z & t & \\ 1 & 0 & 0 & 0.07 & 4.9 \\ 0 & 1 & 0 & 0.06 & 4.9 \\ 0 & 0 & 1 & 0.04 & 3.3 \end{array} \right)$$

The general solution is

$$x + 0.07t = 4.9, \quad y + 0.06t = 4.9 \text{ and } z + 0.04t = 3.3$$

$$x = 4.9 - 0.07t, \quad y = 4.9 - 0.06t \text{ and } z = 3.3 - 0.04t$$

where t is our free variable. In vector form we have

$$\mathbf{x} = \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} = \begin{pmatrix} 4.9 - 0.07t \\ 4.9 - 0.06t \\ 3.3 - 0.04t \\ t \end{pmatrix} = \underbrace{\begin{pmatrix} 4.9 \\ 4.9 \\ 3.3 \\ 0 \end{pmatrix}}_{\text{particular sol'n}=\mathbf{x}_p} + \underbrace{\begin{pmatrix} -0.07 \\ -0.06 \\ -0.04 \\ 1 \end{pmatrix}}_{\text{homogeneous sol'n}=\mathbf{x}_H} t$$

This is an example of an **underdetermined** system: a system with *more unknowns* than equations, which gives infinitely many solutions provided it is consistent. The row echelon form matrix $\mathbf{R}$ of an underdetermined system has the following shape:

$$\begin{array}{c} \text{More unknowns} \\ \boxed{\mathbf{R}} \end{array} \left. \vphantom{\begin{array}{c} \text{More unknowns} \\ \boxed{\mathbf{R}} \end{array}} \right\} \text{Fewer equations}$$

An **overdetermined** system is a system with *more equations* than unknowns. The row echelon form matrix $\mathbf{R}$ has the following shape:

$$\begin{array}{c} \boxed{\mathbf{R}} \\ \text{Fewer unknowns} \end{array} \left. \vphantom{\begin{array}{c} \boxed{\mathbf{R}} \\ \text{Fewer unknowns} \end{array}} \right\} \text{More equations than unknowns}$$

3.6.4 Properties of non-homogeneous linear systems

Proposition (3.36). The linear system $\mathbf{Ax} = \mathbf{b}$ has a solution $\Leftrightarrow \mathbf{b}$ can be generated by the column space of matrix $\mathbf{A}$. (Fig. 3.24)



Figure 3.24

Proof – Exercises 3.6. ■

For the homogeneous system, $\mathbf{Ax} = \mathbf{0}$, we have the zero vector $\mathbf{0}$ in the column space of matrix $\mathbf{A}$. However, in the non-homogeneous case, $\mathbf{Ax} = \mathbf{b}$, we cannot guarantee that the vector $\mathbf{b}$ is in the column space of matrix $\mathbf{A}$. The vector $\mathbf{b}$ must be in the *column space* in order for $\mathbf{Ax} = \mathbf{b}$ to have a solution.

Proposition (3.37). Let $\mathbf{A}$ be a m by n matrix and $\mathbf{b} \neq \mathbf{0}$ then the linear system $\mathbf{Ax} = \mathbf{b}$ has a solution $\Leftrightarrow \text{rank}(\mathbf{A} \mid \mathbf{b}) = \text{rank}(\mathbf{A})$.

? *What does this proposition mean in simple English?*

The ranks of augmented matrix $(\mathbf{A} \mid \mathbf{b})$ and $\mathbf{A}$ must be equal for $\mathbf{Ax} = \mathbf{b}$ to have a solution. Also, the other way, if $\mathbf{Ax} = \mathbf{b}$ has a solution then the ranks of $\mathbf{A}$ and $(\mathbf{A} \mid \mathbf{b})$ are equal.

Proof.

($\Rightarrow$). Let $\mathbf{Ax} = \mathbf{b}$ have a solution.

? *How do we prove $\text{rank}(\mathbf{A} \mid \mathbf{b}) = \text{rank}(\mathbf{A})$?*

We show that the following are impossible:

- (a) $\text{rank}(\mathbf{A} \mid \mathbf{b}) > \text{rank}(\mathbf{A})$ (b) $\text{rank}(\mathbf{A} \mid \mathbf{b}) < \text{rank}(\mathbf{A})$

(a) Suppose $\text{rank}(\mathbf{A} \mid \mathbf{b}) > \text{rank}(\mathbf{A})$ where $\text{rank}(\mathbf{A}) = p$ say. Then we have p non-zero rows in the row echelon form $\mathbf{R}$ of matrix $\mathbf{A}$ but *more than* p non-zero rows in the augmented matrix $(\mathbf{R} \mid \mathbf{b}')$:

$$\begin{array}{c} p \text{ non-zero rows} \end{array} \left\{ \begin{array}{cccccc|c} x_1 & x_2 & \cdots & & & x_n & b'_1 \\ r_{11} & & & & & & \vdots \\ 0 & \ddots & & & & & b' \\ \vdots & & r_{pj} & \cdots & & & \\ \hline 0 & 0 & 0 & \cdots & \cdots & 0 & b'_{p+1} \\ \vdots & \vdots & \vdots & \cdots & \cdots & 0 & \vdots \end{array} \right\} \leftarrow (p+1) \text{ th non-zero row}$$

Expanding along the $(p+1)$ th row we have

$$0x_1 + 0x_2 + \cdots + 0x_{n-1} + 0x_n = b'_{p+1} \quad (*)$$

However, b'_{p+1} cannot equal zero.

? *Why not?*

Because by our supposition $\text{rank}(\mathbf{A} \mid \mathbf{b}) > \text{rank}(\mathbf{A}) = p$ which means $(p+1)$ th row must be non-zero, therefore $b'_{p+1} \neq 0$.

It is impossible to have the solution (*) because the left hand side is zero but the right hand is non-zero. We have a contradiction, which means that our supposition $\text{rank}(\mathbf{A} \mid \mathbf{b}) > \text{rank}(\mathbf{A})$ must be wrong.

(b) We cannot have $\text{rank}(\mathbf{A} \mid \mathbf{b}) < \text{rank}(\mathbf{A})$ or $\text{rank}(\mathbf{A}) > \text{rank}(\mathbf{A} \mid \mathbf{b})$.



Why not?

Because this would mean we have:

$$\mathbf{R} = \left(\begin{array}{cc} r_{11} & r_{1n} \\ \hline 0 \dots & 0 \end{array} \right) \left. \vphantom{\begin{array}{cc} r_{11} & r_{1n} \\ \hline 0 \dots & 0 \end{array}} \right\} \begin{array}{l} \text{Non-zero} \\ \text{rows} \end{array} \quad (\mathbf{R} \mid \mathbf{b}') = \left(\begin{array}{cc|c} r_{11} & r_{1n} & b'_1 \\ \hline 0 \dots & 0 & 0 \\ 0 \dots & 0 & 0 \end{array} \right) \left. \vphantom{\begin{array}{cc|c} r_{11} & r_{1n} & b'_1 \\ \hline 0 \dots & 0 & 0 \\ 0 \dots & 0 & 0 \end{array}} \right\} \begin{array}{l} \text{Same matrix as } \mathbf{R} \text{ on the left.} \\ \text{Non-zero} \\ \text{rows} \end{array}$$

The number of non-zero rows in matrix $\mathbf{R}$ is *greater than* the number of non-zero rows in the augmented matrix $(\mathbf{R} \mid \mathbf{b}')$. This is impossible because the same matrix $\mathbf{R}$ cannot have more non-zero rows in matrix $\mathbf{R}$ than $(\mathbf{R} \mid \mathbf{b}')$.

Since both $\text{rank}(\mathbf{A} \mid \mathbf{b}) > \text{rank}(\mathbf{A})$ and $\text{rank}(\mathbf{A} \mid \mathbf{b}) < \text{rank}(\mathbf{A})$ are false, we must have $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A} \mid \mathbf{b})$.

($\Leftarrow$). We assume $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A} \mid \mathbf{b})$. Writing this out we have

$$\text{rank} \left(\begin{array}{cccc} r_{11} & \dots & & r_{1n} \\ \hline 0 & 0 & \dots & 0 \\ \vdots & & & \vdots \end{array} \right) = \text{rank} \left(\begin{array}{cccc|c} \mathbf{c}_1 & \mathbf{c}_2 & \dots & \mathbf{c}_n & b_1 \\ r_{11} & \dots & & r_{1n} & \vdots \\ \hline 0 & 0 & \dots & 0 & 0 \\ \vdots & & & \vdots & \vdots \end{array} \right)$$

Expanding out the augmented matrix in terms of equations, we have $\mathbf{b}$ is in the column space of matrix $\mathbf{A}$. By the above Proposition (3.36) the linear system $\mathbf{Ax} = \mathbf{b}$ has a solution

The above Proposition (3.37) can be used as a test to see if there are any solutions to the given linear system. Note that the above proposition is also saying:

$$\text{rank}(\mathbf{A} \mid \mathbf{b}) \neq \text{rank}(\mathbf{A}) \Leftrightarrow \text{No solution}$$



But how do we test for a unique solution or an infinite number of solutions?

The next proposition gives the number of solutions to a non-homogeneous linear system.

Proposition (3.38). Consider the linear system $\mathbf{Ax} = \mathbf{b}$ where $\mathbf{A}$ has n columns and $\mathbf{b} \neq \mathbf{0}$. ($\mathbf{A}$ has n columns means that there are n unknowns in the system.)

- (a) $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A} \mid \mathbf{b}) = n$ (Full rank) $\Leftrightarrow$ the linear system has a *unique* solution.
- (b) $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A} \mid \mathbf{b}) < n \Leftrightarrow$ the linear system has an *infinite* number of solutions.
- (c) $\text{rank}(\mathbf{A}) \neq \text{rank}(\mathbf{A} \mid \mathbf{b}) \Leftrightarrow$ the linear system has *no* solution.

Proof – Exercises 3.6.

Example 3.39

Determine whether the following systems have infinite, unique or no solutions (you do not need to find them):

$$\text{(a)} \quad \begin{aligned} x - y - 2z - 3w &= 5 \\ -4x + 4y + 8z + 12w &= 2 \end{aligned}$$

$$\text{(b)} \quad \begin{aligned} x + 2y + 3z &= 1 \\ 4x + 5y + 6z &= 2 \\ 7x + 8y + 8z &= 3 \end{aligned}$$

Solution

- (a) Writing out the coefficient matrix, $\mathbf{A}$, augmented matrix, $(\mathbf{A} \mid \mathbf{b})$, and placing the given matrices into reduced row echelon form we have:

$$\mathbf{A} = \begin{pmatrix} 1 & -1 & -2 & -3 \\ -4 & 4 & 8 & 12 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & -1 & -2 & -3 \\ 0 & 0 & 0 & 0 \end{pmatrix} \leftarrow \text{non-zero row}$$

We have $\text{rank}(\mathbf{A}) = 1$.

$$(\mathbf{A} \mid \mathbf{b}) = \left(\begin{array}{cccc|c} 1 & -1 & -2 & -3 & 5 \\ -4 & 4 & 8 & 12 & 2 \end{array} \right) \longrightarrow \left(\begin{array}{cccc|c} 1 & -1 & -2 & -3 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{array} \right) \leftarrow \begin{array}{l} \text{non-zero row} \\ \text{non-zero row} \end{array}$$

Hence $\text{rank}(\mathbf{A} \mid \mathbf{b}) = 2$.

We have $\text{rank}(\mathbf{A}) = 1$ but $\text{rank}(\mathbf{A} \mid \mathbf{b}) = 2$, so $\text{rank}(\mathbf{A})$ does not equal $\text{rank}(\mathbf{A} \mid \mathbf{b})$. Therefore the given system has *no* solution.

- (b) Similarly we have

$$(\mathbf{A} \mid \mathbf{b}) = \left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 4 & 5 & 6 & 2 \\ 7 & 8 & 8 & 3 \end{array} \right) \longrightarrow \left(\begin{array}{ccc|c} 1 & 0 & -1 & -1/3 \\ 0 & 1 & 2 & 2/3 \\ 0 & 0 & 1 & 0 \end{array} \right) \leftarrow \begin{array}{l} \text{non-zero row} \\ \text{non-zero row} \\ \text{non-zero row} \end{array}$$

What is the rank of $\mathbf{A}$ and $(\mathbf{A} \mid \mathbf{b})$ equal to?

In both cases, we have $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A} \mid \mathbf{b}) = 3$ which means it is of full rank because the matrix $\mathbf{A}$ has three columns, so we have a unique solution.

The matrix $\mathbf{A}$ in part (b) is of full rank so by proposition (3.30):

Let $\mathbf{A}$ be a n by n matrix. The matrix $\mathbf{A}$ is invertible $\Leftrightarrow \text{rank}(\mathbf{A}) = n$.

The matrix $\mathbf{A}$ is invertible.

We can also deduce the following results:

Proposition (3.39). Let $\mathbf{A}$ be an n by n matrix then the following are equivalent:

- (a) Matrix $\mathbf{A}$ is invertible.
- (b) $\text{rank}(\mathbf{A}) = n$ (full rank)
- (c) Null space of matrix $\mathbf{A}$ only contains the zero vector.
- (d) $\text{nullity}(\mathbf{A}) = 0$
- (e) Rows of matrix $\mathbf{A}$ are linearly independent. (See question 9 of Exercises 3.5.)
- (f) Columns of matrix $\mathbf{A}$ are linearly independent. (See question 10 of Exercises 3.5.)

It can be shown that a basis for the column space of a matrix is given by the columns containing the leading 1's in reduced row echelon form:

Proposition (3.40). The columns which have leading 1's of a matrix $\mathbf{A}$ form a basis for the column space of $\mathbf{A}$.

Proof – Website.



Summary

(3.31) The set of vectors $\mathbf{x}$ which satisfy the homogeneous system $\mathbf{Ax} = \mathbf{0}$ is called the null space of $\mathbf{A}$, and the dimension of this space is called the nullity of matrix $\mathbf{A}$.

(3.33) If the matrix $\mathbf{A}$ has n columns then $\text{nullity}(\mathbf{A}) + \text{rank}(\mathbf{A}) = n$.

A non-homogeneous system $\mathbf{Ax} = \mathbf{b}$ has the general two part solution:

$$\mathbf{x} = (\text{homogeneous solution}) + (\text{particular solution})$$



EXERCISES 3.6

(Brief solutions at end of book. Full solutions available at <http://www.oup.co.uk/companion/singh>.)

1. Determine and sketch the null space of the following matrices:

$$(a) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad (b) \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \quad (c) \begin{pmatrix} 1 & 0 \\ 1 & 2 \\ 6 & 10 \end{pmatrix} \quad (d) \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}$$

2. Determine the solution of the following homogeneous systems:

$$\begin{array}{ll} x - 2y - 3z = 0 & 2x - 2y - 2z = 0 \\ (a) \quad 4x - 5y - 6z = 0 & (b) \quad 4x - 4y - 4z = 0 \\ \quad 7x - 8y - 9z = 0 & \quad 8x - 8y - 8z = 0 \\ 2x + 9y - 3z = 0 & -3x + y - z = 0 \\ (c) \quad 5x + 6y - z = 0 & (d) \quad 2x + 5y - 7z = 0 \\ \quad 9x + 8y - 9z = 0 & \quad 4x + 8y - 4z = 0 \end{array}$$

3. Determine the null space, rank and nullity of the following matrices:

$$\begin{array}{ll} (a) \mathbf{A} = \begin{pmatrix} 1 & -2 & -3 \\ 4 & -5 & -6 \\ 7 & -8 & -9 \end{pmatrix} & (b) \mathbf{B} = \begin{pmatrix} 2 & -2 & -2 \\ 4 & -4 & -4 \\ 8 & -8 & -8 \end{pmatrix} \\ (c) \mathbf{C} = \begin{pmatrix} 2 & 9 & -3 \\ 5 & 6 & -1 \\ 9 & 8 & -9 \end{pmatrix} & (d) \mathbf{D} = \begin{pmatrix} -3 & 1 & -1 \\ 2 & 5 & -7 \\ 4 & 8 & -4 \end{pmatrix} \end{array}$$

4. Determine the null space of the matrix:

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 8 & 9 & 10 & 11 & 12 & 13 & 14 \\ 15 & 16 & 17 & 18 & 19 & 20 & 21 \\ 22 & 23 & 24 & 25 & 26 & 27 & 28 \\ 29 & 30 & 31 & 32 & 33 & 34 & 35 \end{pmatrix}$$

5. Determine bases for row, column and null space of the following matrices. Also state their rank and nullity.

$$(a) \mathbf{A} = \begin{pmatrix} 1 & -4 & -9 \\ 2 & 5 & -7 \end{pmatrix} \quad (b) \mathbf{B} = \begin{pmatrix} 1 & 3 \\ 2 & 5 \\ -14 & -37 \end{pmatrix} \quad (c) \mathbf{C} = \begin{pmatrix} 1 & 3 & -9 & 5 \\ 2 & 6 & 7 & 1 \\ 1 & 3 & -8 & 1 \end{pmatrix}$$

6. Solve the following non-homogeneous system of linear equations:

$$(a) \begin{aligned} 2x + 5y + 7z + 10w &= 3 \\ x + y + 2z + 5w &= 6 \end{aligned} \quad (b) \begin{aligned} 2x - y - 4z &= 13 \\ 3x + 3y - 5z &= 13 \\ 3x - 4y + 10z &= -10 \end{aligned}$$

7. Determine whether the following systems have infinite, unique or no solutions:

$$(a) \begin{aligned} 2x + 8y &= 12 \\ 7x + 28y &= 42 \end{aligned} \quad (b) \begin{aligned} 2x - 3y - 6z + 12w &= 2 \\ 3x - 5y - 7z + 16w &= 5 \end{aligned}$$

$$(c) \begin{aligned} x + 2y + 3z &= 1 \\ 4x + 5y + 6z &= 2 \\ 7x + 8y + 9z &= 4 \end{aligned} \quad (d) \begin{aligned} 2x + 5y - 3z - 7w &= 0 \\ x + y - 4z - 8w &= 9 \\ 3x + 4y + w &= 6 \\ 5x + 21y - z + 3w &= 2 \end{aligned}$$

8. Check to see if the following vectors $\mathbf{u}$ are in the null space of corresponding matrices:

$$(a) \mathbf{u} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \mathbf{A} = \begin{pmatrix} 1 & 1 & -1 \\ 2 & -1 & 0 \\ 5 & 2 & -3 \end{pmatrix} \quad (b) \mathbf{u} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 1 & 4 & -5 \\ -7 & 5 & 2 \end{pmatrix}$$

$$(c) \mathbf{u} = \begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix}, \mathbf{C} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{pmatrix} \quad (d) \mathbf{u} = \begin{pmatrix} 1 \\ 3 \\ -4 \\ 7 \end{pmatrix}, \mathbf{D} = \begin{pmatrix} 1 & -2 & 6 & 1 \\ 3 & -6 & 7 & 8 \\ 5 & 2 & 1 & 7 \\ 1 & 6 & 3 & 2 \end{pmatrix}$$

9. Prove that a n by n matrix $\mathbf{A}$ is invertible $\Leftrightarrow \text{nullity}(\mathbf{A}) = 0$.
10. Prove that elementary row operations do *not* change the null space of a matrix.
11. Let $\mathbf{A}$ be an m by n matrix. Prove that every vector in the null space of matrix $\mathbf{A}$ is orthogonal to every vector in the row space of matrix $\mathbf{A}$.
12. Prove Proposition (3.32).
13. Prove Proposition (3.36).
14. Prove Proposition (3.38).



MISCELLANEOUS EXERCISES 3

In this exercise you may check your numerical answers using MATLAB.

- 3.1. Find a basis for the null space of $\mathbf{A}$, where

$$\mathbf{A} = \begin{bmatrix} 1 & -2 & 1 & 1 \\ -1 & 2 & 0 & 1 \\ 2 & -4 & 1 & 0 \end{bmatrix}$$

University of Western Ontario, Canada

- 3.2. Let $\mathbf{A}$ be a 7×5 matrix, and suppose that the homogeneous linear system $\mathbf{A}\mathbf{X} = \mathbf{0}$ is uniquely solvable. Answer the following questions:

- (i) What is the rank of $\mathbf{A}$?
 (ii) If the linear system $\mathbf{A}\mathbf{X} = \mathbf{B}$ is solvable, is it then uniquely solvable?

- A. $\text{rank}(\mathbf{A}) = 7$; no.
 B. $\text{rank}(\mathbf{A}) = 5$; yes.
 C. $\text{rank}(\mathbf{A}) = 7$; yes.
 D. $\text{rank}(\mathbf{A}) = 2$; yes.
 E. $\text{rank}(\mathbf{A}) = 5$; no.
 F. $\text{rank}(\mathbf{A}) = 2$; no.

University of Ottawa, Canada

- 3.3. If we denote by C_1, C_2, C_3, C_4, C_5 the columns of the matrix

$$\mathbf{A} = \begin{bmatrix} 1 & 3 & -2 & 0 & 2 \\ 2 & 6 & -5 & -2 & 4 \\ 0 & 0 & 5 & 10 & 0 \\ 2 & 6 & 0 & 8 & 4 \end{bmatrix}$$

then a basis of the column space of $\mathbf{A}$ is

- A. C_1, C_2, C_3
 B. C_1, C_3, C_4
 C. C_1, C_3
 D. C_1, C_2
 E. C_1, C_3, C_5
 F. C_1, C_4

University of Ottawa, Canada

- 3.4. Let $\mathbf{A}$ be an $m \times n$ matrix, let $\mathbf{b}$ be a vector in $\mathbb{R}^m$, and suppose that $\mathbf{v}$ is a solution of $\mathbf{A}\mathbf{x} = \mathbf{b}$.

- (a) Prove that if $\mathbf{w}$ is a solution of $\mathbf{A}\mathbf{x} = \mathbf{0}$, then $\mathbf{v} + \mathbf{w}$ is a solution of $\mathbf{A}\mathbf{x} = \mathbf{b}$.
 (b) Prove that for any solution $\mathbf{u}$ to $\mathbf{A}\mathbf{x} = \mathbf{b}$, there is a solution to $\mathbf{A}\mathbf{x} = \mathbf{0}$.

Illinois State University, USA

- 3.5. Let $\mathbf{A}$ be an $n \times n$ matrix such that $\mathbf{A}\mathbf{x} = \mathbf{b}$ has exactly one solution for each $\mathbf{b}$ in $\mathbb{R}^n$. What three other things (or facts) can you say about $\mathbf{A}$.

Illinois State University, USA

- 3.6. For each of the following situations, provide an example of the item requested, or explain why such an example could not exist.

- (a) A homogeneous linear system of equations with no solutions.
- (b) Two matrices, $\mathbf{A}$ and $\mathbf{B}$, that are not the same size but so that $\mathbf{AB} = \mathbf{BA}$.
- (c) A set of four vectors that spans $\mathbb{R}^4$ but is not a basis for $\mathbb{R}^4$.
- (d) A three-dimensional subspace of the vector space formed by the set of 2×2 matrices.

Saint Michael's College, Vermont, USA
(part question)

- 3.7. Which of the following are true?

- (a) If V is a vector space of dimension n , then every set of n linearly independent vectors in V is a spanning set of V .
- (b) P_2 contains a basis of polynomials p satisfying $p(0) = 2$.
- (c) $\{1, \sin^2(x), \cos^2(x)\}$ are linearly independent subset of $F[0, 2\pi]$.

University of Ottawa, Canada

- 3.8. Let V be a vector space. Which of the following statements are true?

- (a) If $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is a linearly independent subset of V , then also $\{\mathbf{u}, \mathbf{v}\}$ is linearly independent.
- (b) Every spanning set of V contains a basis of V .
- (c) If $\dim(V) = n$ then every set of n linearly independent vectors of V is a basis.

University of Ottawa, Canada

- 3.9. It is given that $\mathbf{A} = \begin{bmatrix} 1 & 0 & -2 & 1 & 3 \\ -1 & 1 & 5 & -1 & -3 \\ 0 & 2 & 6 & 0 & 1 \\ 1 & 1 & 1 & 1 & 4 \end{bmatrix}$ and its reduced row echelon form is

$$\text{given by } \mathbf{B} = \begin{bmatrix} 1 & 0 & -2 & 1 & 0 \\ 0 & 1 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

- (a) Find the rank of $\mathbf{A}$.
- (b) Find the nullity of $\mathbf{A}$.
- (c) Find a basis for the column space of $\mathbf{A}$.
- (d) Find a basis for the row space of $\mathbf{A}$.
- (e) Find a basis for the null space of $\mathbf{A}$.

Purdue University, USA

3.10. (a) Let $\mathbf{u}_1, \dots, \mathbf{u}_k$ be vectors in a subspace U of $\mathbb{R}^n$. Define what it means

- (i) for $\mathbf{u}_1, \dots, \mathbf{u}_k$ to be linearly independent
- (ii) for $\mathbf{u}_1, \dots, \mathbf{u}_k$ to span U

(b) Determine the null space of the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 3 & -4 \\ 2 & -1 & -1 \\ -2 & -6 & 8 \end{pmatrix}$$

(c) Determine whether the vectors

$$\mathbf{v}_1 = \begin{bmatrix} -5 \\ 4 \\ -6 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 0 \\ -2 \\ 3 \end{bmatrix}$$

in $\mathbb{R}^3$ are linearly independent or linearly dependent. If they are linearly dependent, express one of the vectors as a linear combination of the other two.

University of Southampton, UK
(part question)

3.11. (a) When is a subset V of $\mathbb{R}^n$ a *subspace*? Give two distinct examples of subspaces of $\mathbb{R}^n$.

(b) What is meant by saying that the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_s$ in $\mathbb{R}^n$ are linearly independent?

How is the linear span of the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_s$ defined?

(c) Show that the following vectors in $\mathbb{R}^3$ are linearly dependent:

$$\mathbf{v}_1 = (1, 1, 1), \mathbf{v}_2 = (2, 3, -2), \mathbf{v}_3 = (-2, -5, 10)$$

What is the dimension of the linear span of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$?

University of Sussex, UK

3.12. Consider the following matrix $\mathbf{A}$ and vector $\vec{b}$:

$$\mathbf{A} = \begin{bmatrix} 2 & 6 & -4 \\ -2 & -5 & 4 \\ 4 & 11 & -8 \end{bmatrix}, \quad \vec{b} = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

(a) By using row reduction, reduce the augmented matrix $[\mathbf{A} \quad \vec{b}]$.

(b) Using part (a) of this question, explain why $\mathbf{A}$ and $[\mathbf{A} \quad \vec{b}]$ have the same rank.

(c) Find in vector form the general (or complete) solution to $\mathbf{A}\vec{x} = \vec{b}$.
[Note that $\vec{b}$ is the vector $\mathbf{b}$].

University of New Brunswick, Canada

3.13. Compute the reduced row echelon form of the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 4 & 6 & 1 & 2 & 5 \\ 3 & 6 & 9 & 4 & 8 & 3 \end{pmatrix} \in M_{3 \times 6}(\mathbb{R})$$

Determine a basis of the null space $N(\mathbf{A})$.

RWTH Aachen University, Germany

3.14. (a) Define the terms *null-space* $N(\mathbf{A})$ of $\mathbf{A}$, *column space* $C(\mathbf{A})$ of $\mathbf{A}$, *rank* $\mathbf{A}$, *nullity* $\mathbf{A}$, where $\mathbf{A} \in M_{m \times n}(\mathbb{R})$.

(b) For the following matrix $\mathbf{A}$, find the reduced row echelon form of $\mathbf{A}$, a basis for $N(\mathbf{A})$ and rank $\mathbf{A}$ and nullity $\mathbf{A}$:

$$\mathbf{A} = \begin{bmatrix} 1 & -1 & 2 & 0 \\ 2 & -2 & 1 & 3 \\ -2 & 2 & -7 & 3 \end{bmatrix}$$

Also explain why $(1, 5, 1)^T$ and $(0, 3, 3)^T$ form a basis for $C(\mathbf{A})$.

University of Queensland, Australia

3.15. (a) Determine whether the following subsets are subspaces (giving reasons for your answers).

(i) $U = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M(2, 2) \mid a^2 = d^2 \right\}$ in $M(2, 2)$
 $[M(2, 2) \text{ is the vector space } M_{22}]$.

(ii) $V = \{p(x) \in P_n \mid p(3) = 0\}$ in P_n

(iii) $W = \{(x, y, z, t) \in \mathbb{R}^4 \mid y = z + t\}$ in $\mathbb{R}^4$

(b) Find a basis for the real vector space $\mathbb{R}^3$ containing the vector $(3, 5, -4)$.

(c) Do the following sets form a basis for V ? If not, determine whether they are linearly independent, a spanning set for V , or neither.

(i) $\{(1, 0, 1), (1, 1, 0), (0, 1, 1), (1, 1, 1)\}$ for $V = \mathbb{R}^3$.

(ii) $\{5, 2 + x - 3x^2, 4x - 1\}$ for $V = P_2$

City University, London, UK

3.16. Let V and W be two subspaces of $\mathbb{R}^n$. Define the set

$$V + W = \{\mathbf{u} \in \mathbb{R}^n \mid \mathbf{u} = \mathbf{v} + \mathbf{w} \text{ for some } \mathbf{v} \in V \text{ and some } \mathbf{w} \in W\}.$$

Prove that $V + W$ is a subspace of $\mathbb{R}^n$.

Illinois State University, USA
(part question)

3.17. For the given set S and vector $\mathbf{v}$, determine if $\mathbf{v}$ is in the span of S .

(a) $S = \left\{ \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \right\}, \mathbf{v} = \begin{bmatrix} 3 & 1 \\ -1 & 3 \end{bmatrix}$

(b) $S = \{1 + x, x + x^2, x + x^3, 1 + x + x^2 + x^3\}$, $\mathbf{v} = 2 - 3x + 4x^2 + x^3$

Illinois State University, USA

3.18. Let $\mathbf{B} = \begin{bmatrix} 1 & 2 & 2 & 8 \\ 2 & 4 & 4 & 13 \\ 1 & 1 & 1 & 5 \end{bmatrix}$.

Reduce $\mathbf{B}$ to reduced row echelon form by performing appropriate elementary row operations. Hence, or otherwise, find the following.

- (a) A basis for $\text{NULL}(\mathbf{B})$
- (b) A basis for $\text{Col}(\mathbf{B})$
- (c) The nullity of $\mathbf{B}$.
- (d) The rank of $\mathbf{B}$.

University of Sydney, Australia

3.19. A square matrix $\mathbf{A}$ is called *skew-symmetric* if $\mathbf{A}^T = -\mathbf{A}$. Prove the set of n -by- n skew-symmetric matrices is a subspace of all n -by- n matrices.

Harvey Mudd College, California, USA

3.20. Consider the vectors:

$$\begin{aligned}\mathbf{v}_1 &= (2 \ 4 \ 1 \ 1) \\ \mathbf{v}_2 &= (4 \ 7 \ 2 \ 2) \\ \mathbf{v}_3 &= (6 \ 8 \ 7 \ 5)\end{aligned}$$

- (a) Are these vectors linearly independent? Justify.
- (b) Let these vectors span a vector space V . Find the dimension of V and a set of basis vectors for V .
- (c) Does the vector $\mathbf{v} = [2 \ 1 \ 1 \ 1]$ belong to the vector space V . Justify.
- (d) Let $\mathbf{v}_1$, $\mathbf{v}_2$ and $\mathbf{v}_3$ be arranged as the first, second and third rows of a 3×4 matrix $\mathbf{A}$. Let the null space of $\mathbf{A}$ be the vector space W . Determine the dimension of W .
- (e) Find a set of basis vectors for W .

National University of Singapore

3.21. If $\mathbf{A}$ is a 5×3 matrix, show that the rows of $\mathbf{A}$ are linearly dependent.

Clark University, USA

3.22. Can a 4×10 matrix $\mathbf{A}$ have a null space of dimension 2? Why or why not?

University of South Carolina, USA
(part question)

3.23. Start with this 2 by 4 matrix

$$\mathbf{A} = \begin{bmatrix} 2 & 3 & 1 & -1 \\ 6 & 9 & 3 & -2 \end{bmatrix}$$

- (a) Find all special solutions $\mathbf{Ax} = \mathbf{0}$ and **describe the null space** of $\mathbf{A}$.

- (b) Find the complete solutions – meaning all solutions (x_1, x_2, x_3, x_4) – to

$$\mathbf{Ax} = \begin{bmatrix} 2x_1 + 3x_2 + x_3 - x_4 \\ 6x_1 + 9x_2 + 3x_3 - 2x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

- (c) When an m by n matrix $\mathbf{A}$ has rank $r = m$, the system $\mathbf{Ax} = \mathbf{b}$ can be solved for which b (best answer)? How many special solutions to $\mathbf{Ax} = \mathbf{0}$?

Massachusetts Institute of Technology, USA

- 3.24. Find a real number c such that the vectors $\begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix}, \begin{pmatrix} 2 \\ 7 \\ c \end{pmatrix}$ do not form a basis of $\mathbb{R}^3$.

University of California, Berkeley, USA
(part question)

- 3.25. Find a 3×3 matrix whose null space (kernel) is the span of the vector $(1, 2, 3)$, and whose column space (image) is the span of the vectors $(4, 5, 6)$ and $(7, 8, 9)$.

University of California, Berkeley, USA

- 3.26. (a) Show that $\{\cos(x), \cos(2x), \cos(3x)\}$ is linearly independent.
(b) Show that $(1+x)^3, (1-x)^3, (1+2x)^3, (1-2x)^3, (1+3x)^3$ are linearly dependent functions. State clearly any general fact about vector spaces that you use to justify your assertion.

Hint: First show that these functions are in the subspace spanned by $1, x, x^2, x^3$.

McGill University, Canada

- 3.27. (a) Prove or disprove the following statement:
 $\text{Span}((1, 2, -1, -2), (2, 1, 2, -1)) = \text{Span}((-1, 4, -7, -4), (8, 7, 4, -7))$
(b) If $\mathbf{u}, \mathbf{v}, \mathbf{w}$ are linearly independent vectors in $\mathbb{R}^n$ for which values of k are the vectors $k\mathbf{u} + \mathbf{v}, \mathbf{v} + k\mathbf{w}, \mathbf{w} + k\mathbf{u}$ linearly independent?

McGill University, Canada

- 3.28. True or False? Give a complete justification for your answers.

- (a) Let $\mathbf{A}$ be an $n \times n$ matrix. If there is a vector $\mathbf{y} \in \mathbb{R}^n$ so that the equation $\mathbf{Ax} = \mathbf{y}$ has more than one solution, then the columns of $\mathbf{A}$ span $\mathbb{R}^n$.
(b) The subset $H = \{(x, y) \in \mathbb{R}^2 \mid y = x^2\}$ of $\mathbb{R}^2$ is a vector subspace of $\mathbb{R}^2$.
(c) Let $\mathbf{A}$ be a 5×6 matrix. Then the null space of $\mathbf{A}$ is a vector subspace of $\mathbb{R}^5$.

University of Maryland, Baltimore County, USA

- 3.29. Suppose that $\mathbf{v}_1, \dots, \mathbf{v}_n$ are vectors in $\mathbb{R}^n$ and that $\mathbf{A}$ is an $n \times n$ matrix. If $\mathbf{Av}_1, \dots, \mathbf{Av}_n$ form a basis of $\mathbb{R}^n$, show that $\mathbf{v}_1, \dots, \mathbf{v}_n$ form a basis of $\mathbb{R}^n$ and that $\mathbf{A}$ is invertible.

University of California, Berkeley, USA